

SUPPLEMENTARY MATERIALS

NCCT-Based Radiomics and Deep Learning for mRS Prognostication After Spontaneous Intracerebral Hemorrhage: A Network Meta-Analysis

This supplementary material includes Supplementary Tables S3-S19 and Supplementary Figures S1-S18, providing search strategies, the list of excluded studies, risk-of-bias and certainty assessments, detailed data extraction tables, network meta-analysis diagnostics and consistency assessments, meta-regression analyses, and clinical decision analyses. The PRISMA-DTA and PRISMA-NMA checklists (Supplementary Tables S1 and S2) accompany the main article. The analysis code is provided in the accompanying GitHub repository.

17 Supplementary Tables · 18 Supplementary Figures · 98 pages

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PART I. SUPPLEMENTARY TABLES

Table S3. Search strategy in PubMed, Embase, Web of Science, and Cochrane Library.

Database	Search strategy	Filters and limits	Records retrieved
PubMed	("Cerebral Hemorrhage"[Mesh] OR "Intracranial Hemorrhage, Hypertensive"[Mesh] OR "Basal Ganglia Hemorrhage"[Mesh] OR intracerebral hemorrhag*[Title/Abstract] OR intracerebral haemorrhag*[Title/Abstract] OR cerebral hemorrhag*[Title/Abstract] OR cerebral haemorrhag*[Title/Abstract] OR brain hemorrhag*[Title/Abstract] OR brain haemorrhag*[Title/Abstract] OR intraparenchymal hemorrhag*[Title/Abstract] OR intraparenchymal haemorrhag*[Title/Abstract] OR parenchymal hemorrhag*[Title/Abstract] OR parenchymal haemorrhag*[Title/Abstract] OR hemorrhagic stroke*[Title/Abstract] OR haemorrhagic stroke*[Title/Abstract] OR spontaneous intracerebral hemorrhag*[Title/Abstract] OR spontaneous intracerebral haemorrhag*[Title/Abstract] OR spontaneous cerebral hemorrhag*[Title/Abstract] OR spontaneous cerebral haemorrhag*[Title/Abstract] OR primary intracerebral hemorrhag*[Title/Abstract] OR primary intracerebral haemorrhag*[Title/Abstract] OR hypertensive intracerebral hemorrhag*[Title/Abstract] OR hypertensive intracerebral haemorrhag*[Title/Abstract] OR basal ganglia hemorrhag*[Title/Abstract] OR basal ganglia haemorrhag*[Title/Abstract] OR putaminal hemorrhag*[Title/Abstract] OR putaminal haemorrhag*[Title/Abstract] OR thalamic hemorrhag*[Title/Abstract] OR thalamic haemorrhag*[Title/Abstract] OR ICH[Title/Abstract] OR SICH[Title/Abstract] OR HICH[Title/Abstract]) AND ("Artificial Intelligence"[Mesh] OR "Machine Learning"[Mesh] OR "Deep Learning"[Mesh] OR "Neural Networks, Computer"[Mesh] OR "Image Processing, Computer-Assisted"[Mesh] OR artificial intelligence[Title/Abstract] OR machine learn*[Title/Abstract] OR deep learn*[Title/Abstract] OR radiomic*[Title/Abstract] OR radiomics-based[Title/Abstract] OR radiomic signature*[Title/Abstract] OR radiomics signature*[Title/Abstract] OR radiomic score*[Title/Abstract] OR radiomics score*[Title/Abstract] OR Rad-score[Title/Abstract] OR Radscore[Title/Abstract] OR texture analys*[Title/Abstract] OR texture feature*[Title/Abstract] OR quantitative imag*[Title/Abstract] OR quantitative image feature*[Title/Abstract] OR image feature*[Title/Abstract] OR imaging feature*[Title/Abstract] OR neural network*[Title/Abstract] OR convolutional neural network*[Title/Abstract] OR CNN[Title/Abstract] OR transformer[Title/Abstract] OR vision transformer[Title/Abstract] OR ResNet[Title/Abstract] OR DenseNet[Title/Abstract] OR random forest*[Title/Abstract] OR support vector machine*[Title/Abstract] OR SVM[Title/Abstract] OR gradient boost*[Title/Abstract] OR XGBoost[Title/Abstract] OR LightGBM[Title/Abstract] OR CatBoost[Title/Abstract] OR supervised learn*[Title/Abstract] OR ensemble learn*[Title/Abstract] OR ensemble model*[Title/Abstract]) AND ("Prognosis"[Mesh] OR "Treatment Outcome"[Mesh] OR "Recovery of Function"[Mesh] OR "Functional Status"[Mesh] OR "Disability Evaluation"[Mesh] OR "Outcome Assessment, Health Care"[Mesh] OR modified Rankin[Title/Abstract] OR Rankin Scale[Title/Abstract] OR mRS[Title/Abstract] OR functional outcome*[Title/Abstract] OR functional status[Title/Abstract] OR functional prognos*[Title/Abstract] OR functional recovery[Title/Abstract] OR functional independen*[Title/Abstract] OR functional dependen*[Title/Abstract] OR poor functional outcome*[Title/Abstract] OR unfavorable functional outcome*[Title/Abstract] OR unfavourable functional outcome*[Title/Abstract] OR neurological outcome*[Title/Abstract] OR neurologic outcome*[Title/Abstract] OR clinical outcome*[Title/Abstract] OR patient outcome*[Title/Abstract] OR poor outcome*[Title/Abstract] OR unfavorable outcome*[Title/Abstract] OR unfavourable outcome*[Title/Abstract] OR disability[Title/Abstract] OR dependence[Title/Abstract] OR independence[Title/Abstract] OR prognos*[Title/Abstract] OR outcome prediction[Title/Abstract] OR outcome predict*[Title/Abstract] OR prognostic prediction[Title/Abstract])	No restrictions on date, language, or publication type were applied.	734
Embase	('intracerebral hemorrhage'/exp OR 'cerebral hemorrhage'/exp OR 'hemorrhagic stroke'/exp OR 'intracerebral hemorrhag*':ab,ti OR 'intracerebral haemorrhag*':ab,ti OR 'cerebral hemorrhag*':ab,ti OR 'cerebral haemorrhag*':ab,ti OR 'brain hemorrhag*':ab,ti OR 'brain haemorrhag*':ab,ti OR 'intraparenchymal hemorrhag*':ab,ti OR 'intraparenchymal haemorrhag*':ab,ti OR 'parenchymal hemorrhag*':ab,ti OR 'parenchymal haemorrhag*':ab,ti OR 'hemorrhagic stroke*':ab,ti	No restrictions on date, language, or publication type were applied.	3032

	OR 'haemorrhagic stroke*':ab,ti OR 'spontaneous intracerebral hemorrhag*':ab,ti OR 'spontaneous intracerebral haemorrhag*':ab,ti OR 'spontaneous cerebral hemorrhag*':ab,ti OR 'spontaneous cerebral haemorrhag*':ab,ti OR 'primary intracerebral hemorrhag*':ab,ti OR 'primary intracerebral haemorrhag*':ab,ti OR 'hypertensive intracerebral hemorrhag*':ab,ti OR 'hypertensive intracerebral haemorrhag*':ab,ti OR 'basal ganglia hemorrhag*':ab,ti OR 'basal ganglia haemorrhag*':ab,ti OR 'putaminal hemorrhag*':ab,ti OR 'putaminal haemorrhag*':ab,ti OR 'thalamic hemorrhag*':ab,ti OR 'thalamic haemorrhag*':ab,ti OR ICH:ab,ti OR SICH:ab,ti OR HICH:ab,ti) AND ('artificial intelligence'/exp OR 'machine learning'/exp OR 'deep learning'/exp OR 'artificial neural network'/exp OR 'image analysis'/exp OR 'artificial intelligence':ab,ti OR 'machine learn*':ab,ti OR 'deep learn*':ab,ti OR radiomic*':ab,ti OR 'radiomics-based':ab,ti OR 'radiomic signature*':ab,ti OR 'radiomics signature*':ab,ti OR 'radiomic score*':ab,ti OR 'radiomics score*':ab,ti OR 'rad-score':ab,ti OR radscore:ab,ti OR 'texture analys*':ab,ti OR 'texture feature*':ab,ti OR 'quantitative imag*':ab,ti OR 'quantitative image feature*':ab,ti OR 'image feature*':ab,ti OR 'imaging feature*':ab,ti OR 'neural network*':ab,ti OR 'convolutional neural network*':ab,ti OR CNN:ab,ti OR transformer:ab,ti OR 'vision transformer':ab,ti OR ResNet:ab,ti OR DenseNet:ab,ti OR 'random forest*':ab,ti OR 'support vector machine*':ab,ti OR SVM:ab,ti OR 'gradient boost*':ab,ti OR XGBoost:ab,ti OR LightGBM:ab,ti OR CatBoost:ab,ti OR 'supervised learn*':ab,ti OR 'ensemble learn*':ab,ti OR 'ensemble model*':ab,ti) AND ('prognosis'/exp OR 'treatment outcome'/exp OR 'functional status'/exp OR 'disability'/exp OR 'modified Rankin':ab,ti OR 'Rankin Scale':ab,ti OR mRS:ab,ti OR 'functional outcome*':ab,ti OR 'functional status':ab,ti OR 'functional prognos*':ab,ti OR 'functional recovery':ab,ti OR 'functional independen*':ab,ti OR 'functional dependen*':ab,ti OR 'poor functional outcome*':ab,ti OR 'unfavorable functional outcome*':ab,ti OR 'unfavourable functional outcome*':ab,ti OR 'neurological outcome*':ab,ti OR 'neurologic outcome*':ab,ti OR 'clinical outcome*':ab,ti OR 'patient outcome*':ab,ti OR 'poor outcome*':ab,ti OR 'unfavorable outcome*':ab,ti OR 'unfavourable outcome*':ab,ti OR disability:ab,ti OR dependence:ab,ti OR independence:ab,ti OR prognos*':ab,ti OR 'outcome prediction':ab,ti OR 'outcome predict*':ab,ti OR 'prognostic prediction':ab,ti)		
Web of Science	TS=("Intracerebral Hemorrhag*" OR "Intracerebral Haemorrhag*" OR "Cerebral Hemorrhag*" OR "Cerebral Haemorrhag*" OR "Brain Hemorrhag*" OR "Brain Haemorrhag*" OR "Intraparenchymal Hemorrhag*" OR "Intraparenchymal Haemorrhag*" OR "Parenchymal Hemorrhag*" OR "Parenchymal Haemorrhag*" OR "Hemorrhagic Stroke*" OR "Haemorrhagic Stroke*" OR "Spontaneous Intracerebral Hemorrhag*" OR "Spontaneous Intracerebral Haemorrhag*" OR "Spontaneous Cerebral Hemorrhag*" OR "Spontaneous Cerebral Haemorrhag*" OR "Primary Intracerebral Hemorrhag*" OR "Primary Intracerebral Haemorrhag*" OR "Hypertensive Intracerebral Hemorrhag*" OR "Hypertensive Intracerebral Haemorrhag*" OR "Basal Ganglia Hemorrhag*" OR "Basal Ganglia Haemorrhag*" OR "Putaminal Hemorrhag*" OR "Putaminal Haemorrhag*" OR "Thalamic Hemorrhag*" OR "Thalamic Haemorrhag*" OR ICH OR SICH OR HICH) AND TS=("Artificial Intelligence" OR "Machine Learn*" OR "Deep Learn*" OR Radiomic* OR "Radiomics-based" OR "Radiomic Signature*" OR "Radiomics Signature*" OR "Radiomic Score*" OR "Radiomics Score*" OR "Rad-score" OR Radscore OR "Texture Analys*" OR "Texture Feature*" OR "Quantitative Imag*" OR "Quantitative Image Feature*" OR "Image Feature*" OR "Imaging Feature*" OR "Neural Network*" OR "Convolutional Neural Network*" OR CNN OR Transformer OR "Vision Transformer" OR ResNet OR DenseNet OR "Random Forest*" OR "Support Vector Machine*" OR SVM OR "Gradient Boost*" OR XGBoost OR LightGBM OR CatBoost OR "Supervised Learn*" OR "Ensemble Learn*" OR "Ensemble Model*") AND TS=("Modified Rankin" OR "Rankin Scale" OR mRS OR "Functional Outcome*" OR "Functional Status" OR "Functional Prognos*" OR "Functional Recovery" OR "Functional Independen*" OR "Functional Dependen*" OR "Poor Functional Outcome*" OR "Unfavorable Functional Outcome*" OR "Unfavourable Functional Outcome*" OR "Neurological Outcome*" OR "Neurologic Outcome*" OR "Clinical Outcome*" OR "Patient Outcome*" OR "Poor Outcome*" OR "Unfavorable Outcome*" OR "Unfavourable Outcome*" OR Disability OR Dependence OR Independence OR Prognos* OR "Outcome Prediction" OR "Outcome Predict*" OR "Prognostic Prediction")	No restrictions on date, language, or publication type were applied.	593
Cochrane Library	([mh "Cerebral Hemorrhage"] OR [mh "Intracranial Hemorrhage, Hypertensive"] OR [mh "Basal Ganglia Hemorrhage"] OR "intracerebral hemorrhag*":ti,ab,kw OR "intracerebral haemorrhag*":ti,ab,kw OR "cerebral hemorrhag*":ti,ab,kw OR	No restrictions on date, language, or publication	32

	"cerebral haemorrhag*":ti,ab,kw OR "brain hemorrhag*":ti,ab,kw OR "brain haemorrhag*":ti,ab,kw OR "intraparenchymal hemorrhag*":ti,ab,kw OR "intraparenchymal haemorrhag*":ti,ab,kw OR "parenchymal hemorrhag*":ti,ab,kw OR "parenchymal haemorrhag*":ti,ab,kw OR "hemorrhagic stroke*":ti,ab,kw OR "haemorrhagic stroke*":ti,ab,kw OR "spontaneous intracerebral hemorrhag*":ti,ab,kw OR "spontaneous intracerebral haemorrhag*":ti,ab,kw OR "spontaneous cerebral hemorrhag*":ti,ab,kw OR "spontaneous cerebral haemorrhag*":ti,ab,kw OR "primary intracerebral hemorrhag*":ti,ab,kw OR "primary intracerebral haemorrhag*":ti,ab,kw OR "hypertensive intracerebral hemorrhag*":ti,ab,kw OR "hypertensive intracerebral haemorrhag*":ti,ab,kw OR "basal ganglia hemorrhag*":ti,ab,kw OR "basal ganglia haemorrhag*":ti,ab,kw OR "putaminal hemorrhag*":ti,ab,kw OR "putaminal haemorrhag*":ti,ab,kw OR "thalamic hemorrhag*":ti,ab,kw OR "thalamic haemorrhag*":ti,ab,kw OR ICH:ti,ab,kw OR SICH:ti,ab,kw OR HICH:ti,ab,kw) AND ([mh "Artificial Intelligence"] OR [mh "Machine Learning"] OR [mh "Deep Learning"] OR [mh "Neural Networks, Computer"] OR [mh "Image Processing, Computer-Assisted"] OR "artificial intelligence":ti,ab,kw OR "machine learn*":ti,ab,kw OR "deep learn*":ti,ab,kw OR radiomic*":ti,ab,kw OR "radiomics-based":ti,ab,kw OR "radiomic signature*":ti,ab,kw OR "radiomics signature*":ti,ab,kw OR "radiomic score*":ti,ab,kw OR "radiomics score*":ti,ab,kw OR "Rad-score":ti,ab,kw OR Radscore:ti,ab,kw OR "texture analys*":ti,ab,kw OR "texture feature*":ti,ab,kw OR "quantitative imag*":ti,ab,kw OR "quantitative image feature*":ti,ab,kw OR "image feature*":ti,ab,kw OR "imaging feature*":ti,ab,kw OR "neural network*":ti,ab,kw OR "convolutional neural network*":ti,ab,kw OR CNN:ti,ab,kw OR Transformer:ti,ab,kw OR "Vision Transformer":ti,ab,kw OR ResNet:ti,ab,kw OR DenseNet:ti,ab,kw OR "random forest*":ti,ab,kw OR "support vector machine*":ti,ab,kw OR SVM:ti,ab,kw OR "gradient boost*":ti,ab,kw OR XGBoost:ti,ab,kw OR LightGBM:ti,ab,kw OR CatBoost:ti,ab,kw OR "supervised learn*":ti,ab,kw OR "ensemble learn*":ti,ab,kw OR "ensemble model*":ti,ab,kw) AND ([mh "Prognosis"] OR [mh "Treatment Outcome"] OR [mh "Recovery of Function"] OR [mh "Functional Status"] OR [mh "Disability Evaluation"] OR [mh "Outcome Assessment, Health Care"] OR "modified Rankin":ti,ab,kw OR "Rankin Scale":ti,ab,kw OR mRS:ti,ab,kw OR "functional outcome*":ti,ab,kw OR "functional status":ti,ab,kw OR "functional prognos*":ti,ab,kw OR "functional recovery":ti,ab,kw OR "functional independen*":ti,ab,kw OR "functional dependen*":ti,ab,kw OR "poor functional outcome*":ti,ab,kw OR "unfavorable functional outcome*":ti,ab,kw OR "unfavourable functional outcome*":ti,ab,kw OR "neurological outcome*":ti,ab,kw OR "neurologic outcome*":ti,ab,kw OR "clinical outcome*":ti,ab,kw OR "patient outcome*":ti,ab,kw OR "poor outcome*":ti,ab,kw OR "unfavorable outcome*":ti,ab,kw OR "unfavourable outcome*":ti,ab,kw OR disability:ti,ab,kw OR dependence:ti,ab,kw OR independence:ti,ab,kw OR prognos*:ti,ab,kw OR "outcome prediction":ti,ab,kw OR "outcome predict*":ti,ab,kw OR "prognostic prediction":ti,ab,kw)	type were applied.	
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Table S4. Excluded Studies with Reasons for Exclusion

Excluded Reference	Detailed Explanation
There were neither discharge data, nor 90-day data, nor (6-month) data. (n = 8)	
01. Machine Learning-Based Prediction of Transition to Functional Upper Limb Recovery After Intensive Inpatient Rehabilitation in Early Subacute Stroke.	The reported functional outcome was assessed at a time point outside the prespecified discharge, 90-day and 6-month windows.
02. Multimodal machine learning for predicting postoperative functional outcomes in surgically treated supratentorial deep intracerebral hemorrhage: a prospective multicenter study.	The study did not report functional outcome at discharge, 90-days, or 6 months as required.
03. Integrating multimodal data and deep learning for functional assessment and rehabilitation prediction after cerebral hemorrhage.	No eligible discharge, 90-day, or 6-month functional outcome time point was reported.
04. Prediction of Neurological Recovery in Patients with Primary Basal Ganglia Hemorrhage Based on Non-Contrast Computed Tomography Radiomics Combined with Clinical Features.	The outcome timing did not meet the prespecified discharge, 90-day, or 6-month follow-up requirement.
05. Automated Computer Vision Methods for Image Segmentation, Stereotactic Localization, and Functional Outcome Prediction of Basal Ganglia Hemorrhages.	Functional outcome was assessed within one year after surgery rather than at discharge, 90 days, or 6 months.
06. ICH-PRNet: a cross-modal intracerebral haemorrhage prognostic prediction method using joint-attention interaction mechanism.	The report did not provide an eligible discharge, 90-day, or 6-month functional outcome time point.
07. The use of cloud based machine learning to predict outcome in intracerebral haemorrhage without explicit programming expertise.	The functional outcome time point was not clearly reported as discharge, 90 days, or 6 months.
08. Machine learning-based predictive model for post-stroke dementia.	The study did not provide the prespecified discharge, 90-day, or 6-month functional outcome assessment.
There were not at least two nodes. (n = 7)	
09. Identifying treatment-responsive patient subgroups in a neutral clinical trial of Intensive blood pressure reduction in acute intracerebral hemorrhage: A post hoc explainable machine learning analysis.	The study did not provide a direct comparison of at least two prespecified model nodes within the same eligible cohort.
10. Multimodal GPT-5 for Predicting Poor Functional Outcomes After Intracerebral Hemorrhage in the Emergency Department: Validation Study.	Only one eligible prediction node was available; no second prespecified node was reported for direct comparison.
11. Machine learning-based prediction of 6-month functional recovery in hypertensive cerebral hemorrhage: insights from XGBoost and SHAP analysis.	Different machine-learning procedures did not constitute separate prespecified input nodes; fewer than two eligible nodes were available.
12. Clinical, radiological, and radiomics feature-based explainable machine learning models for prediction of neurological deterioration and 90-day outcomes in mild intracerebral hemorrhage.	Multiple algorithms used the same clinical, radiological and radiomics input set, so they represented a single prespecified node.
13. Predicting functional outcomes of patients with spontaneous intracerebral hemorrhage based on explainable machine learning models: a multicenter retrospective study.	CNB, SVM, GNB, MLP and XGBoost were algorithmic alternatives using the same input framework, not separate prespecified nodes.
14. A machine learning model based on results of a comprehensive radiological evaluation can predict the prognosis of basal ganglia cerebral hemorrhage treated with neuroendoscopy.	The reported model/ROI alternatives did not provide at least two distinct prespecified nodes for an eligible direct comparison.
15. Prediction Model for Unfavorable Outcome in Spontaneous Intracerebral Hemorrhage Based on Machine Learning.	SVM, decision tree, ANN and logistic regression were different algorithms applied within one input node, leaving fewer than two eligible nodes.
TP, FP, FN, TN could not be extracted. (n = 4)	
16. Radiomic hemorrhage roundness predicts outcome beyond the ICH score in deep intracerebral hemorrhage with IVH.	Eligible functional outcome models were reported, but TP, FP, FN and TN could not be extracted or reliably reconstructed from the reported results.
17. Development and validation of a clinical-radiomics nomogram for predicting 180-day functional outcomes in patients with spontaneous thalamic hemorrhage.	The study reported an eligible mRS-based 6-month outcome, but the 2×2 diagnostic counts could not be extracted or reconstructed.
18. An interpretable artificial intelligence model based on CT for prognosis of intracerebral	Multiple eligible model types were compared for an mRS-based 6-month outcome,

hemorrhage: a multicenter study.	but TP, FP, FN and TN were not recoverable from the reported performance data.
19. Predicting Outcome of Patients With Cerebral Hemorrhage Using a Computed Tomography-Based Interpretable Radiomics Model: A Multicenter Study.	Eligible radiomics and combined models were reported, but the available results did not permit reconstruction of TP, FP, FN and TN.
The gold standard was not based on mRS. (n = 4)	
20. Prediction of 180-day post-admission's GOS in patients with spontaneous intracerebral hemorrhage based on explainable machine learning technology.	The reference standard was the Glasgow Outcome Scale (GOS), not the modified Rankin Scale (mRS).
21. Factors Associated with Early Functional Prognosis in Upper Limb Rehabilitation of Stroke Patients.	Functional prognosis was assessed with upper-limb functional scales rather than mRS.
22. Development of predictive model for post-stroke depression at discharge based on decision tree algorithm: A multi-center hospital-based cohort study.	The target outcome was post-stroke depression rather than an mRS-defined functional outcome.
23. An artificial intelligence-based prognostic prediction model for hemorrhagic stroke.	The discharge prognosis was defined using the Glasgow Outcome Scale (GOS), not mRS.

Only full-text excluded studies are listed below.

Table S5. Risk of bias assessment (PROBAST+AI) model development based on seven domains.

Author, year	Participants and data sources ^a	Quality			Applicability concerns			Overall judgement	
		Predictors ^b	Outcome ^c	Analysis ^d	Participants and data sources ^e	Predictors ^f	Outcome ^g	Quality ^h	Applicability concerns ⁱ
Pszczolkowski et al., 2021	H	H	U	L	L	L	L	H	L
Wang et al., 2022	L	L	L	L	L	L	L	L	L
Xia et al., 2025	L	L	L	L	L	L	L	L	L
Wang et al., 2024	H	L	U	L	L	L	L	H	L
Dierksen et al., 2024	L	L	L	L	L	L	L	L	L
Ji et al., 2026	L	L	L	L	L	L	L	L	L
Huang et al., 2023	L	L	H	L	L	L	L	H	L
Li et al., 2024	L	L	L	L	L	L	L	L	L
Huang et al., 2022	H	L	H	L	L	L	H	H	H
Zhang et al., 2026	L	L	L	L	L	L	L	L	L
Zhou et al., 2021	L	L	L	L	L	L	L	L	L
Zhao et al., 2023	L	L	U	L	L	L	L	U	L
Wu et al., 2025	L	L	U	L	L	L	L	U	L
Tran et al., 2025	H	L	H	L	L	L	H	H	H
Chen et al., 2024	L	L	L	L	L	L	L	L	L
Liu et al., 2026	U	U	U	U	U	L	L	U	U
Song et al., 2021	L	L	U	L	L	L	L	U	L
Song et al., 2024	L	L	U	L	L	L	L	U	L
Jiang et al., 2025	L	L	U	L	L	L	L	U	L
Pei et al., 2024	L	L	U	L	L	L	L	U	L

Abbreviation: PROBAST+AI, Prediction model Risk of Bias Assessment Tool + AI, L low; H high; U unclear.

Footnote: Signaling questions are rated as "yes" (Y), "probably yes" (PY), "probably no" (PN), "no" (N), "no information" (NI), and in some cases "not applicable" (NA). All signaling questions are phrased in such a way that "yes" or "probably yes" indicates a low risk of bias. Any signaling questions rated as "no" or "probably no" indicate a potential high risk of bias in that domain. If there are no "no" or "probably no" ratings, but "no information" (NI) is present, the risk of bias in that domain is classified as unclear.

a. Participants and data sources

1.1 Were appropriate data sources used?

1.2 Was an appropriate study design used?

1.3 Did the in- and exclusions of study participants result in a representative dataset?

b. Predictors

2.1 Were predictors defined and assessed in a similar way for all participants?

- 2.2 Was any pre-processing of predictors similar for all participants?
- 2.3 Were predictor assessments made without knowledge of outcome data?
- 2.4 Were the predictors included in the model available at the time the model was intended to be used?

c. Outcome

- 3.1 Were outcomes defined and assessed appropriately?
- 3.2 Were outcomes defined and assessed in a similar way for all participants?
- 3.3 Were outcome assessments made without use or knowledge of predictor data?
- 3.4 Was the time interval between predictor assessment and outcome assessment appropriate?

d. Analysis

- 4.1 Was there evidence that the sample size was reasonable?
- 4.2 Were continuous and categorical predictors handled appropriately?
- 4.3 Were participants with missing or censored data handled appropriately in the analysis?
- 4.4 If methods to address class imbalance were used, was the model or the model predictions recalibrated?
- 4.5 Were methods used to address potential model overfitting?

e. Participants and data sources

Concern that the (data of the) included participants do not match the review question or the assessor's intended use of the prediction model.

f. Predictors

Concern that the definition, pre-processing, assessment, or timing of assessment of the predictors in the model do not match the review question or the assessor's intended use.

g. Outcome

Concern that the outcome, its definition, assessment, or timing of assessment do not match the review question or the assessor's intended use.

h. Quality

Low risk: If all four domains were rated low concern regarding quality.

High risk: If at least one domain was rated high concern regarding quality .

Unclear: If at least one domain was rated unclear concern regarding quality and no domains were rated high concern.

i. Applicability concerns

Low risk: If all three domains were rated low concern for applicability.

High risk: If at least one domain was rated high concern for applicability.

Unclear: If at least one domain was rated unclear concern for applicability and no domains were rated high concern.

Table S6. Risk of bias assessment (PROBAST+AI) model evaluation based on seven domains.

Author, year	Participants and data sources ^a	Risk of bias			Applicability concerns			Overall judgement	
		Predictors ^b	Outcome ^c	Analysis ^d	Participants and data sources ^e	Predictors ^f	Outcome ^g	Risk of bias ^h	Applicability concerns ⁱ
Pszczolkowski et al., 2021	H	H	U	L	L	L	L	H	L
Wang et al., 2022	L	L	L	H	L	L	L	H	L
Xia et al., 2025	L	L	L	L	L	L	L	L	L
Wang et al., 2024	H	L	U	L	L	L	L	H	L
Dierksen et al., 2024	L	L	L	L	L	L	L	L	L
Ji et al., 2026	L	L	L	L	L	L	L	L	L
Huang et al., 2023	H	L	H	H	L	L	H	H	H
Li et al., 2024	L	L	L	L	L	L	L	L	L
Huang et al., 2022	L	L	L	L	L	L	L	L	L
Zhang et al., 2026	L	L	L	L	L	L	L	L	L
Zhou et al., 2021	L	L	L	L	L	L	L	L	L
Zhao et al., 2023	L	L	U	L	L	L	L	U	L
Wu et al., 2025	L	U	U	L	L	L	L	U	L
Tran et al., 2025	H	L	H	L	L	L	H	H	H
Chen et al., 2024	L	L	L	L	L	L	L	L	L
Liu et al., 2026	U	U	U	U	U	L	L	U	U
Song et al., 2021	L	L	U	L	L	L	L	U	L
Song et al., 2024	L	L	U	H	L	L	L	H	L
Jiang et al., 2025	L	L	U	L	L	L	L	U	L
Pei et al., 2024	L	L	U	H	L	L	L	H	L

Abbreviation: PROBAST+AI, Prediction model Risk of Bias Assessment Tool + AI, L low; H high; U unclear.

Footnote: Signaling questions are rated as "yes" (Y), "probably yes" (PY), "probably no" (PN), "no" (N), "no information" (NI), and in some cases "not applicable" (NA). All signaling questions are phrased in such a way that "yes" or "probably yes" indicates a low risk of bias. Any signaling questions rated as "no" or "probably no" indicate a potential high risk of bias in that domain. If there are no "no" or "probably no" ratings, but "no information" (NI) is present, the risk of bias in that domain is classified as unclear.

a. Participants and data sources

1.1 Were appropriate data sources used?

1.2 Was an appropriate study design used?

1.3 Did the in- and exclusions of study participants result in a representative dataset?

b. Predictors

2.1 Were predictors defined and assessed in a similar way for all participants?

- 2.2 Was any pre-processing of predictors similar for all participants?
- 2.3 Were predictor assessments made without knowledge of outcome data?
- 2.4 Were the predictors included in the model available at the time the model was intended to be used?

c. Outcome

- 3.1 Were outcomes defined and assessed appropriately?
- 3.2 Were outcomes defined and assessed in a similar way for all participants?
- 3.3 Were outcome assessments made without use or knowledge of predictor data?
- 3.4 Was the time interval between predictor assessment and outcome assessment appropriate?

d. Analysis

- 4.1 Was model evaluation based on only apparent performance avoided?
- 4.2 Was there evidence that the sample size was reasonable?
- 4.3 Were participants with missing or censored data handled appropriately in the analysis?
- 4.4 If methods to address class imbalance were used, was the evaluation done in a dataset without imbalance correction?
- 4.5 If data splitting was done to create training and test datasets, was there evidence that data leakage was avoided?
- 4.6 If resampling methods were used to evaluate model performance, were all model development steps replicated in the resampling process?
- 4.7 Was the predictive performance of the model evaluated appropriately, e.g., calibration, discrimination, and net benefit?

e. Participants and data sources

Concern that the (data of the) included participants do not match the review question or the assessor's intended use of the prediction model.

f. Predictors

Concern that the definition, pre-processing, assessment, or timing of assessment of the predictors in the model do not match the review question or the assessor's intended use.

g. Outcome

Concern that the outcome, its definition, assessment, or timing of assessment do not match the review question or the assessor's intended use.

h. Risk of bias

Low risk: If all four domains were rated low risk of bias.

High risk: If at least one domain was rated high risk of bias.

Unclear: If at least one domain was rated unclear risk of bias and no domains were rated high risk of bias.

i. Applicability concerns

Low risk: If all three domains were rated low concern for applicability.

High risk: If at least one domain was rated high concern for applicability.

Unclear: If at least one domain was rated unclear concern for applicability and no domains were rated high concern.

Table S7. GRADE certainty assessment for diagnostic test accuracy outcomes across DTA-NMA nodes.

Index test	Summary accuracy (95% CrI)	Participants (studies)	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Certainty of evidence
Conventional clinical-CT models	Sensitivity: 0.75 (0.69 to 0.80)	1,819 (14 studies)	No downgrade	Downgrade 1	No downgrade	No downgrade	No downgrade	⊕⊕⊕○ Moderate
	Specificity: 0.78 (0.72 to 0.83)		No downgrade	Downgrade 1	No downgrade	No downgrade	No downgrade	⊕⊕⊕○ Moderate
CT-based radiomics-only ML models	Sensitivity: 0.73 (0.67 to 0.78)	2,012 (12 studies)	No downgrade	Downgrade 1	No downgrade	No downgrade	Downgrade 1	⊕⊕○○ Low
	Specificity: 0.69 (0.61 to 0.76)		No downgrade	Downgrade 1	No downgrade	No downgrade	Downgrade 1	⊕⊕○○ Low
Clinical-CT-radiomics ML models	Sensitivity: 0.76 (0.71 to 0.81)	1,454 (12 studies)	No downgrade	Downgrade 1	No downgrade	No downgrade	No downgrade	⊕⊕⊕○ Moderate
	Specificity: 0.81 (0.75 to 0.86)		No downgrade	Downgrade 1	No downgrade	No downgrade	No downgrade	⊕⊕⊕○ Moderate
CT-based DL-only models	Sensitivity: 0.78 (0.72 to 0.83)	1,127 (4 studies)	No downgrade	Downgrade 1	No downgrade	No downgrade	No downgrade	⊕⊕⊕○ Moderate
	Specificity: 0.66 (0.57 to 0.74)		No downgrade	Downgrade 1	No downgrade	No downgrade	No downgrade	⊕⊕⊕○ Moderate
Clinical-CT-DL fusion models	Sensitivity: 0.82 (0.75 to 0.88)	482 (3 studies)	No downgrade	Downgrade 1	No downgrade	No downgrade	No downgrade	⊕⊕⊕○ Moderate
	Specificity: 0.83 (0.75 to 0.89)		No downgrade	Downgrade 1	No downgrade	No downgrade	No downgrade	⊕⊕⊕○ Moderate

Node definitions: index-test labels correspond directly to the diagnostic NMA nodes.

Certainty of evidence levels: High = no downgrade; Moderate = downgraded one level (e.g., Downgrade 1 on one domain); Low = downgraded two levels (e.g., Downgrade 1 on two domains or Downgrade 2 on one domain); Very low = downgraded three or more levels.

Detailed GRADE assessment criteria (each domain is rated 'No downgrade', 'Downgrade 1' [one level], or 'Downgrade 2' [two levels]):

a. Risk of bias: rated on the basis of PROBAST/QUADAS-2 style signalling questions covering participant selection, index test, reference standard, and flow/timing.

b. Inconsistency: derived from the width of the posterior prediction interval (PI) relative to the 95% credible interval and the magnitude of between-study variance (τ); downgrade when the PI is materially wider than the CrI, indicating substantial unexplained heterogeneity across studies.

c. Indirectness: assigned when the evidence differs from the review question in population, index-test definition/version, or reference standard.

d. Imprecision: downgraded when the 95% CrI is wide (width > 0.20) or crosses the prespecified minimum acceptable sensitivity/specificity threshold.

e. Publication bias: based on the Deeks funnel-plot asymmetry signal when estimable ($p < 0.10$), combined with visual inspection and study-distribution judgement when evidence is sparse.

GRADE certainty rating and specific reasons for downgrading (listed per index test and outcome):

Conventional clinical-CT models — Sensitivity (certainty: Moderate): Inconsistency: Downgrade 1 (prediction interval materially wider than the credible interval).

Conventional clinical-CT models — Specificity (certainty: Moderate): Inconsistency: Downgrade 1 (prediction interval materially wider than the credible interval).

CT-based radiomics-only ML models — Sensitivity (certainty: Low): Inconsistency: Downgrade 1 (prediction interval materially wider than the credible interval). Publication bias: Downgrade 1 (Deeks' test $p = 0.01$; threshold $p < 0.10$).

CT-based radiomics-only ML models — Specificity (certainty: Low): Inconsistency: Downgrade 1 (prediction interval materially wider than the credible interval). Publication bias: Downgrade 1 (Deeks' test $p = 0.01$; threshold $p < 0.10$).

Clinical-CT-radiomics ML models — Sensitivity (certainty: Moderate): Inconsistency: Downgrade 1 (prediction interval materially wider than the credible interval).

Clinical-CT-radiomics ML models — Specificity (certainty: Moderate): Inconsistency: Downgrade 1 (prediction interval materially wider than the credible interval).

CT-based DL-only models — Sensitivity (certainty: Moderate): Inconsistency: Downgrade 1 (prediction interval materially wider than the credible interval).

CT-based DL-only models — Specificity (certainty: Moderate): Inconsistency: Downgrade 1 (prediction interval materially wider than the credible interval).

Clinical-CT-DL fusion models — Sensitivity (certainty: Moderate): Inconsistency: Downgrade 1 (prediction interval materially wider than the credible interval).

Clinical-CT-DL fusion models — Specificity (certainty: Moderate): Inconsistency: Downgrade 1 (prediction interval materially wider than the credible interval).

Table S8. Methodological Details of the Bayesian Hierarchical Network Meta-Analysis of Diagnostic Test Accuracy (NMA-DTA).

Statistical software	Main platform	R 4.3.2 (https://www.r-project.org)
	MCMC engine	JAGS 4.3.1, interfaced via the R2jags package
	Visualisation packages	ggplot2, bayesplot, igraph, ggridges, ellipse, and related extensions
	Random seed	set.seed(2024) to ensure full reproducibility
Layer 1 — Data Layer (Likelihood)		
Likelihood	True positives (TP)	Modelled as a binomial count, parameterised by the algorithm-specific sensitivity and the number of disease-positive subjects in each study
	True negatives (TN)	Modelled as a binomial count, parameterised by the algorithm-specific specificity and the number of disease-negative subjects in each study
Link function	Logit transformation	Maps sensitivity and specificity from the probability scale to an unconstrained real-valued space, facilitating linear decomposition of effects
Effect decomposition	Global summary effect	Algorithm-level fixed effect representing the pooled logit-sensitivity and logit-specificity across all studies
	Study baseline effect	Study-level random effect capturing the overall diagnostic performance shift attributable to each study's population and setting
	Multi-arm data structure	Each study may evaluate several algorithm categories and contributes one observation row per algorithm; the study-level random effects are shared across all algorithms evaluated within the same study, linking the multi-arm evidence in the network [6]
Derived indices	Full diagnostic accuracy set	Posterior distributions of sensitivity (Se), specificity (Sp), positive likelihood ratio (LR+), negative likelihood ratio (LR-), and diagnostic odds ratio (DOR) are derived from the fixed-effect posterior samples
Layer 2 — Study-Level Random Effects (Scale Mixture of Normals)		
Motivation	Limitation of standard models	The conventional bivariate normal random-effects model assumes all studies arise from a single normal distribution; in the presence of outlying studies, this can cause summary estimates to be unduly influenced by a small number of extreme observations
Methodological source	Reference	Scale Mixture of Normals proposed by Verde (2010) [4]
	Per-study mixture weight	A continuous weight parameter is introduced for each study; marginally, this is equivalent to replacing the bivariate normal distribution with a bivariate t-distribution, granting the model the ability to down-weight outlying studies automatically
Core mechanism	Weight prior	Mixture weight follows a Gamma distribution whose shape and rate parameters are both equal to $v/2$, where v is the degrees-of-freedom parameter
	Degrees-of-freedom parameter v	Controls the heaviness of the tail of the between-study distribution. Smaller v indicates stronger departure from normality and greater support for the robust model over the standard normal alternative. Prior: Exponential distribution with mean = 1
Outlier criterion	Decision threshold	A study whose posterior mean mixture weight falls below 0.75 is classified as a high-heterogeneity outlying study
Layer 3 — Hyperprior Layer		
Covariance structure	Prior for the precision matrix	Wishart distribution with scale matrix equal to the 2×2 identity matrix and degrees of freedom $df = 5$, providing a weakly informative hyperprior on the between-study precision matrix that avoids imposing directional constraints on the Se–Sp relationship [4,5]
	Covariance matrix	Obtained by inverting the precision matrix; used to extract the between-study standard deviations $\tau(\text{Se})$ and $\tau(\text{Sp})$, and their correlation coefficient ρ
	log-DOR heterogeneity $\tau(\log\text{DOR})$	Derived from the between-study covariance matrix of the logit-transformed sensitivity and specificity as $\tau(\log\text{DOR}) = \sqrt{\tau^2(\text{Se}) + \tau^2(\text{Sp}) + 2 \cdot \text{Cov}(\text{Se}, \text{Sp})}$; monitored within the model and recomputed post-MCMC in R from posterior samples of the covariance matrix, with small negative values arising from numerical rounding truncated to zero [5]
Fixed-effect priors	Algorithm fixed effects	Weakly informative normal prior: mean = 0, standard deviation = 2 [4]; on the probability scale this spans approximately 99% of the (0, 1) interval, imposing no directional constraint
	Meta-regression coefficients	Same weakly informative normal prior as the fixed effects: mean = 0, standard deviation = 2

Design-specific effects (DBT model)	Variance hyperprior	In the design-by-treatment (inconsistency) model, the standard deviations of the design-specific deviations for logit-sensitivity and logit-specificity are assigned Uniform(0, 2) weakly informative hyperpriors [8]
III. MCMC Configuration		
Chains and iterations	Number of parallel chains	4 chains, each initialised with independently drawn random starting values
	Burn-in (warm-up)	40,000 iterations per chain, discarded to eliminate the influence of initial values
	Sampling iterations	60,000 iterations per chain after burn-in (100,000 total per chain including burn-in)
Effective samples	Thinning factor	5 (every 5th post-burn-in sample retained) to reduce within-chain autocorrelation
	Per chain	12,000 retained posterior samples per chain, i.e. (100,000 – 40,000) / 5
	Total across all chains	48,000 posterior samples pooled across the four chains used for inference
Initialisation	Initial value strategy	Initial values generated automatically by JAGS by sampling from the prior distributions; each chain was assigned a distinct random-number generator with a chain-specific reproducible seed derived from the master analysis seed
IV. Convergence Diagnostics		
Visual diagnostics	MCMC trace plots	Inspect whether all chains are well-mixed, trajectories are stationary, and no systematic drift or inter-chain separation is present
	Posterior density overlay plots	Inspect whether density curves from all chains overlap closely and are unimodal
Quantitative diagnostics	Brooks–Gelman–Rubin statistic ($\hat{R}$)	Convergence criterion: $\hat{R} < 1.05$, indicating that between-chain variance has contracted sufficiently relative to within-chain variance
	Effective sample size (ESS)	Criterion: ESS > 1,000 for each key parameter, ensuring adequate precision of posterior estimates
Model fit	Deviance Information Criterion (DIC)	Evaluates overall model fit quality; lower DIC indicates a better balance between goodness-of-fit and model complexity
V. Post-Processing & Derived Statistics		
Ranking analysis	SUCRA computation	Based on the posterior rank distribution of each algorithm across all MCMC iterations, the cumulative ranking probability curve is computed and integrated; SUCRA = 100% indicates the algorithm always ranks first, 0% always last [5]
Pairwise comparisons	League table construction	Upper triangle: posterior difference in sensitivity (column algorithm – row algorithm); lower triangle: corresponding difference in specificity; diagonal: posterior mean DOR with 95% CrI for each algorithm; all estimates reported as posterior mean (95% CrI)
	Relative diagnostic odds ratio (RDOR)	Ratio of two algorithms' DOR values, computed iteration-wise from posterior samples; RDOR > 1 with 95% CrI lower bound > 1 indicates the column algorithm significantly outperforms the row algorithm overall
Heterogeneity reporting	Format for τ and τ^2	Reported as posterior mean (95% CrI) separately for the sensitivity scale $\tau(\text{Se})$, specificity scale $\tau(\text{Sp})$, and log-DOR scale $\tau(\log\text{DOR})$
Small-study effects	Deeks' funnel plot asymmetry test	Horizontal axis: log-DOR; vertical axis: $1/\sqrt{\text{ESS}}$, where the effective sample size $\text{ESS} = 4 \cdot (\text{TP} + \text{FN}) \cdot (\text{FP} + \text{TN}) / (\text{TP} + \text{FP} + \text{FN} + \text{TN})$ with a 0.5 continuity correction applied to zero cells; a linear regression of log-DOR on $1/\sqrt{\text{ESS}}$ weighted by ESS tests whether the slope departs significantly from zero, performed for the whole network and separately for each algorithm with at least 3 studies. $P < .10$ indicates significant asymmetry, suggesting potential small-study effects [7]
VI. Inconsistency Assessment and Additional Analyses		
Inconsistency	Global: design-by-treatment interaction model	The consistency model was compared with a design-by-treatment (unconstrained) model that adds design-specific effects to both logit-sensitivity and logit-specificity. $\Delta\text{DIC} = \text{DIC}(\text{inconsistency model}) - \text{DIC}(\text{consistency model})$; $ \Delta\text{DIC} \leq 2$ indicates equivalent fit compatible with consistency, $\Delta\text{DIC} > 2$ favours the consistency model, and $\Delta\text{DIC} < -2$ suggests potential global inconsistency [9]
	Local: node splitting	For each comparison informed by both direct and indirect evidence, direct and indirect estimates were contrasted on the logit scale for sensitivity and specificity (inconsistency factor = direct – indirect); a two-sided posterior probability, $2 \times \min[P(\text{IF} > 0), P(\text{IF} < 0)]$, below .05 was considered evidence of local inconsistency [8]
Transitivity	Covariate distribution comparison	The distribution of prespecified study-level characteristics (centre composition, input-data modality, geographical region, validation setting, and molecular reference standard) was compared across the direct comparisons of the network; disproportionate composition

Meta-regression	Covariate specification	across comparisons would signal potential violation of the transitivity assumption Each prespecified covariate was added separately to the network model on the logit scale, with distinct regression coefficients for sensitivity and specificity; continuous covariates were mean-centred and categorical covariates entered as indicator variables with the reference level fixed at 0; coefficients were summarised as posterior mean (95% CrI)
Uncertainty	95% prediction intervals	95% prediction intervals for sensitivity, specificity, and DOR, representing the expected performance of a future study, were derived from the posterior between-study covariance matrix; in the summary ROC plane, the 95% prediction region is displayed as an ellipse around the summary operating point [4]
Prior sensitivity	Robustness assessment	Not formally assessed in this report; priors were prespecified as weakly informative (see priors above), and the scale-mixture formulation limits sensitivity to the between-study distribution.

Abbreviations: CrI, credible interval; DIC, deviance information criterion; DOR, diagnostic odds ratio; ESS, effective sample size; MCMC, Markov chain Monte Carlo; RDOR, relative diagnostic odds ratio; SUCRA, surface under the cumulative ranking curve; DBT, design-by-treatment; PI, prediction interval; IF, inconsistency factor. Reference numbers correspond to those in the main text.

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ANN, artificial neural networks; DL, deep learning; ML, machine learning.

Table S9. Supplementary baseline clinical and imaging characteristics of patients and cohorts.

Author, year	Node(s)	Recruit. period	Treatment	Age	Male	GCS/NIHSS	Onset-to-CT	ICH vol.	ICH site	IVH	PHE	Follow-up
Pszczolkowski et al., 2021	A, B, C	NR	TXA/placebo	Tr: P 73.10 ± 12.80; G 64.08 ± 13.52 y; Te: P 72.55 ± 12.30; G 65.29 ± 12.75 y	Tr: P 319/643 (49.6%); G 353/559 (63.1%); Te: P 128/258 (49.6%); G 164/259 (63.3%)	NR	Tr: P 1.92 (1.33–2.75), G 1.95 (1.37–2.83) h; Te: P 1.89 (1.33–2.93), G 2.00 (1.38–3.05) h	Tr: P 18.78 (7.71–42.14), G 7.50 (3.36–14.83) mL; Te: P 20.20 (9.79–40.49), G 7.27 (3.51–14.51) mL	NR	NR	NR	1,719/1,732 (99.2%)
Wang et al., 2022	A, D, E	Tr: 2016–2020; Ext.; Te: 2017–2020	Mixed	Tr: 53.39 ± 12.29 y; Ext.; Te: 48.5 ± 10.8 y	Tr: 169/219 (77.17%); Ext.; Te: 33/35 (94.29%)	GCS—Tr: 8.7 (6.2–8.9); Ext.; Te: 8.6 (6.3–8.88)	NR	NR	Pons	NR	NR	Tr: 219/219 (100%); Ext.; Te: 35/35 (100%)
Xia et al., 2025	A, B, C	2014–2021	NR	Tr:60.50 [52.00–73.00] y; IV:59.00 [50.00–69.00] y; EV:NR	Tr:169 (57.9%); IV:199 (68.2%); EV:65 (60.7%)	GCS Tr:12.00 [8.00–14.00]; IV:12.00 [9.00–14.00]; EV:NR	Baseline CT ≤24 h post-onset	Tr:20.65 [6.35–49.95] mL; IV:22.91 [8.19–53.98] mL; EV:NR	Deep or lobar Tr:Deep 220 (75.3%); IV:Deep 203 (69.5%); EV:NR	Tr:97 (33.2%); IV:96 (32.9%); EV:NR	Tr:90 (30.8%); IV:104 (35.6%); EV:NR	NR

Author, year	Node(s)	Recruit. period	Treatment	Age	Male	GCS/NIHSS	Onset-to-CT	ICH vol.	ICH site	IVH	PHE	Follow-up
Wang et al., 2024	A, D, E	2016–2020	NR	Ctr 1:60 [51–70] y; Ctr 2:59 [55–71] y; Ctr 3:59 [53–70] y	Ctr 1:599 (60%); Ctr 2:17 (59%); Ctr 3:35 (61%)	Ctr 1:GCS 10 ± 4; Ctr 2:GCS 10 ± 3; Ctr 3:GCS 10 ± 4	Ctr 1:3.0 [1.5–4.0] h; Ctr 2:4.0 [3.0–4.5] h; Ctr 3:2.0 [1.0–3.0] h	Ctr 1:30.3 [13.7–70.9] mm ³ ; Ctr 2:47.3 [16.1–68.9] mm ³ ; Ctr 3:39.1 [14.5–66.9] mm ³	Ctr 1: Deep 906 (90%); Lobar 106 (10%); Ctr 2: Deep 27 (93%); Lobar 2 (7%); Ctr 3: Deep 45 (79%); Lobar 12 (21%)	Ctr 1:508 (50%); Ctr 2:12 (41%); Ctr 3:28 (49%)	Ctr 1:10.9 [4.7–22.5] mm ³ ; Ctr 2:11.9 [4.6–27.2] mm ³ ; Ctr 3:16.3 [8.0–34.0] mm ³	NR
Dierksen et al., 2024	A, B, C	NR	Intensive BP reduction vs standard care	Tr: 62.4 ± 13.1 y; Ext.; Te: 62.2 ± 13.2 y	Tr: 299 (60%); Ext.; Te: 222 (63%)	Tr: GCS 14 (12–15); NIHSS 12 (5–19); Ext.; Te: GCS 14 (12–15); NIHSS 12 (5–19)	Admission CT; time NR	Tr: 13.8 ± 12.8 mL; Ext.; Te: 13.3 ± 11.6 mL	Tr: BG 281 (56%), Thal 155 (31%), lobar 63 (13%); Ext.; Te: BG 200 (57%), Thal 116 (33%), lobar 36 (10%)	Tr: 366 (73%); Ext.; Te: 254 (72%)	Tr: 2.2 ± 3.1 mL; Ext.; Te: 1.9 ± 2.0 mL	NR
Ji et al., 2026	D, E	Ctr A: 2019–2022; Ctr B: 2021–2023	NR	Ctr A: 57.0 (49.0–67.0) y; Ctr B: 58.0 (51.0–67.0) y	Ctr A: 250/391 (63.9%); Ctr B: 77/117 (65.8%)	GCS: A G 13.0 (12.0–15.0), P 6.0 (5.0–8.2); B G 14.0 (10.0–15.0), P 7.0 (6.8–9.0). NIHSS: A G 10.0 (6.0–15.5), P 27.0 (21.8–30.0); B G 10.0 (2.0–15.0), P 25.0 (20.0–25.0)	NR	NR	NR	NR	NR	Tr/IV:193/391; EV:60/117

Author, year	Node(s)	Recruit. period	Treatment	Age	Male	GCS/NIHSS	Onset-to-CT	ICH vol.	ICH site	IVH	PHE	Follow-up
Huang et al., 2023	A, B, C	2016–2021	NR	Tr: 60 (52, 70) y; IV: 58 (51, 70) y; EV: 59 (53, 70) y	Tr: 465 (57%); IV: 135 (68%); EV: 52 (61%)	GCS <9: 399 (36.3%); GCS ≥9: 699 (63.7%)	≤6 h; Tr: 3.0 (2.0–4.0) h; IV: 2.8 (1.0–4.0) h; EV: 2.5 (1.0–4.0) h	Tr: 29.8 (13.3–69.8) mL; IV: 32.9 (15.9–76.2) mL; EV: 43.8 (14.7–67.2) mL	Deep 977 (89.0%); Lobar 121 (11.0%)	546 (49.7%)	Tr: 9.2 (4.1–20.3) mL; IV: 16.7 (7.6–31.3) mL; EV: 16.9 (7.3–31.7) mL	NR
Li et al., 2024	A, B, C	2017–2020	CT-guided stereotactic hematoma evacuation	Tr: 56.72 ± 11.64 y; IV: 58.94 ± 13.44 y	Tr: 104/138 (75.4%); IV: 19/36 (52.8%)	Tr: GCS 10.45 ± 2.81; IV: GCS 10.08 ± 2.97	NR	Tr: 32.18 ± 12.56 mL; IV: 36.54 ± 12.74 mL	Tr: Basal ganglia 112/138 (81.2%), non-basal ganglia 26/138 (18.8%); IV: Basal ganglia 26/36 (72.2%), non-basal ganglia 10/36 (27.8%)	Tr: 73/138 (52.9%); IV: 23/36 (63.9%)	NR	268/268 (100%)
Huang et al., 2022	A, B, C	2016–2021	NR	Tr: 60 (52, 70) y; IV: 58 (51, 70) y; EV: 59 (53, 70) y	Tr: 465 (57%); IV: 135 (68%); EV: 52 (61%)	Tr: GCS <9, 301 (37%); ≥9, 513 (63%); IV: GCS <9, 68 (34%); ≥9, 130 (66%); EV: GCS <9, 30 (35%); ≥9, 56 (65%)	Tr: 3.0 (2.0, 4.0) h; IV: 2.8 (1.0, 4.0) h; EV: 2.5 (1.0, 4.0) h	Tr: 29.8 (13.3, 69.8); IV: 32.9 (15.9, 76.2); EV: 43.8 (14.7, 67.2)	Tr: Deep 727 (89%), Lobar 87 (11%); IV: Deep 178 (90%), Lobar 20 (10%); EV: Deep 72 (84%), Lobar 14 (16%)	Tr: 409 (50%); IV: 97 (49%); EV: 40 (47%)	Tr: 9.2 (4.1, 20.3) mL; IV: 16.7 (7.6, 31.3) mL; EV: 16.9 (7.3, 31.7) mL	1085/1098 (98.8%)

Author, year	Node(s)	Recruit. period	Treatment	Age	Male	GCS/NIHSS	Onset-to-CT	ICH vol.	ICH site	IVH	PHE	Follow-up
Zhang et al., 2026	A, B, C	2021–2023	Conservative/surgical	Tr: 18–65: 156 (54.4%); 66–79: 114 (39.7%); ≥80: 17 (5.9%); IV: 18–65: 79 (64.2%); 66–79: 37 (30.1%); ≥80: 7 (5.7%)	Tr: 179 (62.4%); IV: 89 (72.4%)	GCS Tr: 14.00 (9.00–15.00); IV: 13.00 (10.00–15.00)	≤24 h	Tr: 7 (4–11) mL; IV: 7 (4–10) mL	Thal	Tr: 212 (73.9%); IV: 84 (68.3%)	NR	NR
Zhou et al., 2021	A, B, C	2014–2021	Std SICH care; craniotomy/MIS as indicated	Deriv: 61.41 ± 13.18 y; Val: 60.89 ± 10.12 y	Deriv: 219 (62.2%); Val: 86 (60.6%)	GCS Deriv: 13 [10–14]; Val: 13 [11–14]	Deriv: 3.28 ± 2.46 h; Val: 3.07 ± 1.62 h	Deriv: 17.48 ± 13.51 mL; Val: 18.86 ± 11.29 mL	Deriv: Deep 280 (79.5%), Lobar 72 (20.5%); Val: Deep 112 (78.9%), Lobar 30 (21.1%)	Deriv: 98 (27.8%); Val: 40 (28.2%)	NR	NR
Zhao et al., 2023	A, D, E	H: 2016–2019; Y: 2018–2021	AHA/ASA-guided care	Tr: 61.1 ± 14.5 y; IV: 58.2 ± 15.3 y; EV: 68.5 ± 13.2 y	Tr: 341 (75.8%); IV: 40 (80.0%); EV: 69 (65.0%)	Tr: GCS 15.0 (10.0–15.0); IV: 15.0 (9.8–15.0); EV: 14.0 (10.3–15.0)	Tr: 8.0 (4.0–20.0) h; IV: 11.3 (4.8–18.5) h; EV: 4.0 (2.0–6.0) h	Tr: 18.9 (8.8–44.4) mL; IV: 21.5 (10.8–47.1) mL; EV: 13.8 (6.4–38.0) mL	Tr: Deep 268 (59.6%); Lobar 128 (28.4%); BS 23 (5.1%); CB 31 (6.9%); IV: Deep 38 (76.0%); Lobar 8 (16.0%); BS 3 (6.0%); CB 1 (2.0%); EV: Deep 62 (59.6%); Lobar 29 (27.9%); BS 11 (10.6%); CB 2 (1.9%)	Tr: 202/450 (44.9%); IV: 24/50 (48.0%); EV: 32/104 (30.8%)	Tr: 12.5 (5.6–23.0) mL; IV: 11.9 (5.9–19.6) mL; EV: 7.4 (3.5–19.6) mL	Tr: 450/450 (100%); IV: 50/50 (100%); EV: 104/104 (100%)

Author, year	Node(s))	Recruit. period	Treatment	Age	Male	GCS/NIHSS	Onset-to-CT	ICH vol.	ICH site	IVH	PHE	Follow-up
Wu et al., 2025	A, D, E	2008–2021	NR	61.5 ± 14.2 y	1221/1853 (65.9%)	GCS 11.1 ± 4.2; NIHSS NR	NR	NR	NR	NR	NR	1853/1853 (100%)

Author, year	Node(s)	Recruit. period	Treatment	Age	Male	GCS/NIHSS	Onset-to-CT	ICH vol.	ICH site	IVH	PHE	Follow-up
Tran et al., 2025	B, D	NR	ATACH-2: intensive vs standard BP lowering; Yale: NR	Tr/CV: 62.1 ± 12.9 y; EV: 69.6 ± 14.4 y	Tr/CV: 528 (60.9%); EV: 354 (54.9%)	GCS — Tr/CV: 3–11: 127 (14.7%); 12–14: 242 (27.9%); 15: 497 (57.4%).; EV: 3–11: 179 (27.8%); 12–14: 168 (26.1%); 15: 270 (41.9%); unknown 28 (4.3%). NIHSS — Tr/CV: 0–4: 137 (15.8%); 5–9: 226 (26.1%); 10–14: 235 (27.1%); 15–19: 159 (18.4%); 20–25: 77 (8.9%); >25: 27 (3.1%); unknown 5 (0.6%).; EV: 0–4: 181 (28.1%); 5–9: 110 (17.1%); 10–14: 76 (11.8%); 15–19: 86 (13.3%); 20–25: 67 (10.4%); >25: 38 (5.9%); unknown 87 (12.5%)	NR	Tr/CV: 13.1 ± 12.6 mL; EV: 18.7 ± 20.7 mL	NR	NR	NR	645/645 (100%)

Author, year	Node(s)	Recruit. period	Treatment	Age	Male	GCS/NIHSS	Onset-to-CT	ICH vol.	ICH site	IVH	PHE	Follow-up
Chen et al., 2024	A, D	MGH:1994–2019; Yale: NR	NR; small proportion received comfort care only	Tr: 72.0 [62.0–80.0] y; Val.: 72.0 [63.0–81.0] y; Te: 69.7 [61.0–79.0] y; EV: 68.0 [59.0–77.0] y	Tr: 229 (46.1%); Val.: 73 (44.2%); Te: 95 (43.2%); EV: 81 (55.5%)	GCS — Tr: 15 [12–15]; Val.: 15 [14–15]; Te: 15 [13–15]; EV: 15 [11.5–15]	Admission NCCT; time NR	Tr: 11.6 [3.2–27.5] mL; Val.: 10.4 [2.9–26.9] mL; Te: 11.6 [3.3–27.8] mL; EV: NR	Tr: Inf. 51 (10.3%), lobar 234 (47.1%), deep 211 (42.5%); Val.: 17 (10.3%), 69 (41.8%), 79 (47.9%); Te: 25 (11.4%), 96 (43.6%), 99 (45.0%); EV: NR	Tr: 160 (32.2%); Val.: 47 (28.5%); Te: 72 (32.7%); EV: NR	NR	Exact 6-/12-month completion rate NR; all included patients had ≥1 follow-up
Liu et al., 2026	A, B, C	Ctr 1: 2016–2020; Ctr 2/3: 2021–2021	NR	59 (51–69) y	1235/2136 (58%)	GCS: low <9: 647/2136 (30%); high ≥9: 1489/2136 (70%)	5.0 (3.0–13.0) h	29.8 (13.2–60.9) mL	Deep: 1803/2136 (85%); Lobar: 333/2136 (15%)	1000/2136 (47%)	12.2 (5.3–23.9) mL	NR
Song et al., 2021	A, B, C	2014–2020	N/A	Tr: 61.0 ± 13.7; IV: 60.7 ± 13.2; EV: 66.4 ± 12.2 y	Tr: 153/244 (62.7%); IV: 66/104 (63.5%); EV: 57/87 (65.5%)	Tr: 13.0 [10.0–14.0]; IV: 13.0 [10.3–14.0]; EV: 13.0 [10.0–14.0]	Tr: 2.5 [2.0–4.0] h; IV: 3.0 [2.0–4.0] h; EV: 3.0 [2.0–7.0] h	Tr: 11.0 [5.9–19.8] mL; IV: 14.8 [7.4–24.8] mL; EV: 12.1 [5.4–22.7] mL	Deep: 186/244 (76.2%); 82/104 (78.8%); 58/87 (66.7%); Lobar: 39/244 (16.0%); 14/104 (13.5%); 22/87 (25.3%); BS/CB NR	Tr: 80/244 (32.8%); IV: 26/104 (25.0%); EV: 26/87 (29.9%)	NR	435/435 (100%)

Author, year	Node(s)	Recruit. period	Treatment	Age	Male	GCS/NIHSS	Onset-to-CT	ICH vol.	ICH site	IVH	PHE	Follow-up
Song et al., 2024	A, B, C	2019–2022	Conservative	P 61.92 ± 13.29; G 59.96 ± 13.52 y	P 56 (63.6%); G 62 (79.5%)	NR	P 6.00 (4.00, 8.00); G 7.00 (5.00, 12.25) h	P 11.50 (7.80, 22.82); G 9.35 (3.90, 15.22) mL	P Basal ganglia 60 (68.2%), Thalamic 18 (20.5%), Lobar 10 (11.4%); G Basal ganglia 41 (52.6%), Thalamic 18 (23.1%), Lobar 19 (24.4%)	P 17 (19.3%); G 5 (6.4%)	NR	166/166 (100%)
Jiang et al., 2025	A, B, C	2022–2025	Surgical	Tr: 58.85 ± 14.28 y; EV: 59.23 ± 13.34 y	Tr: 115/154 (74.7%); EV: 49/66 (74.2%)	Tr: GCS 9.00 (6.25–13.00); EV: GCS 8.00 (6.00–12.00)	NR	Tr: 50.90 (39.08–76.92) mL; EV: 46.45 (28.47–70.50) mL	Tr: lobar 37 (24.0%), deep 117 (76.0%); EV: lobar 17 (25.8%), deep 49 (74.2%)	Tr: 92 (59.7%); EV: 35 (53.0%)	NR	NR
Pei et al., 2024	A, B, C	2020–2023	NR	Tr: 63.12 ± 14.74 y; Val: 65.21 ± 15.57 y	Tr: 81/115 (70.4%); Val: 29/46 (63.0%)	Tr: GCS 12.0 (9.0–13.0); Val: GCS 12.0 (9.5–13.0)	Tr: 9.0 (5.0–23.0) h; Val: 8.0 (6.0–21.0) h	Tr: 9.9 (5.1–21.25) mL; Val: 12.1 (6.3–32.1) mL	Deep/middle/r ight; location rec.	Tr: 39/115 (33.9%); Val: 19/46 (41.3%)	NR	NR

Notes: Data are mean ± SD, median (IQR), or n (%) as reported.

Tr = training; IV = internal validation; EV = external validation; Te = test; Val = validation; Deriv = derivation; Ctr = center; P/G = poor/good outcome; CV = cross-validation; y = years; h = hours; mL = millilitres; mm³ = cubic millimetres; BP = blood pressure; TXA = tranexamic acid; SICH = spontaneous ICH; MIS = minimally invasive surgery; AHA/ASA = American Heart Association/American Stroke Association; BG = basal ganglia; BS = brainstem; CB = cerebellum; Thal = thalamus; Inf. = infratentorial; GCS = Glasgow Coma Scale; NIHSS = National Institutes of Health Stroke Scale; ICH = intracerebral hemorrhage; IVH = intraventricular hemorrhage; PHE = perihematoma edema; NR = not reported; N/A = not applicable. Follow-up = assessed/expected participants (%).

Model nodes: A. Conventional clinical-CT models; B. CT-based radiomics-only ML models; C. Clinical-CT-radiomics ML models; D. CT-based DL-only models; E. Clinical-CT-DL fusion models.

Table S10. Diagnostic performance of included models for poor functional outcome at 90 days.

Author, year	Node(s)(A–E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
Pszczolkowski et al., 2021	A	Clinical-CT model/Elastic-net GLM	Multi-modality	Age, gender, onset-to-CT time, baseline ICH volume, antiplatelet use, ultra-early haematoma growth	70% training + 30% testing; 10-fold CV within training set	Int. val.	mRS 4–6	517	258	160	48	98	211	0.789 (0.750–0.828)
	B	NCCT radiomics/Elastic-net GLM	Single-modality	Baseline NCCT radiomics features, including ICH radiomics, PHE volume and IVH volume		Int. val.	mRS 4–6	517	258	180	67	78	192	0.783 (0.744–0.823)
	C	NCCT radiomics + Clinical-CT model/Elastic-net GLM	Multi-modality	NCCT radiomics + age, gender, onset-to-CT time, baseline ICH volume, antiplatelet use, ultra-early haematoma growth		Int. val.	mRS 4–6	517	258	179	45	79	214	0.818 (0.781–0.854)
Wang et al., 2022	A	Clinical-CT model / Logistic regression	Multi-modality	GCS score; hemorrhage volume; platelet; need for mechanical ventilation	Train. cohort n=219; independent testing cohort n=35	Ext. val.	mRS 4–6	35	8	7	8	1	19	0.875 (0.761–0.966)
	D	DL model / ResNet18	Single-modality	CT images; combined ROI (resized-ROI + crop-ROI) of hemorrhage/perihematomal region; 512-dimensional DL features		Ext. val.	mRS 4–6	35	8	8	13	0	14	0.759 (0.609–0.885)
	E	Combined model / LASSO + Logistic regression	Multi-modality	DL score, GCS score, need for mechanical ventilation, platelet		Ext. val.	mRS 4–6	35	8	7	6	1	21	0.894 (0.791–0.972)

Author, year	Node(s)(A–E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
Xia et al., 2025	A	Clinical-semantic model (CSM) / Logistic regression	Multi-modality	Age; GCS; IVH; PHE; MLS	Train. 292; internal testing 292; external validation 107	Ext. val.	mRS 4–6	107	69	47	13	22	25	0.781 (0.678–0.884)
	B	Radiomics model (RM, VOI40%) / Logistic regression	Single-modality	15 intra-/peri-hematoma CT radiomics features (VOI40%)		Ext. val.	mRS 4–6	107	69	41	9	28	29	0.793 (0.709–0.876)
	C	Clinical semantic-radiomics model (CSRM, VOI40%) / Logistic regression	Multi-modality	Age; GCS; IVH; PHE; MLS + VOI40% Rad-score		Ext. val.	mRS 4–6	107	69	48	11	21	27	0.901 (0.840–0.962)
Wang et al., 2024	A	Clinical-CT model / Logistic regression	Multi-modality	Age; IVH; ICH volume; PHE volume; admission GCS	Random 7:3 split; training n=768, test n=330; 5-fold CV in training set	Int. val.	mRS 4–6	330	212	161	29	51	89	0.830 (NR)
	D	DL model / ResNet50 + Logistic regression	Single-modality	15 selected deep-learning features extracted from hematoma and PHE ROIs on NCCT		Int. val.	mRS 4–6	330	212	159	25	53	93	0.850 (NR)
	E	Combined model / Logistic regression	Multi-modality	Age; IVH; ICH volume; PHE volume; admission GCS + Deep Score		Int. val.	mRS 4–6	330	212	174	18	38	100	0.890 (NR)
Dierksen et al., 2024	A	Clinical-CT model; SVM_rad + noFS	Multi-modality	Age, sex, race, GCS, NIHSS, BP, treatment, laboratory variables, medical history, ICH location/side, hydrocephalus, pineal/septum shift, hematoma volume, PHE volume, IVH	500 training/cross-validation + 352 independent test	Int. val.	mRS 4–6	352	134	100	39	34	179	0.830 (0.780–0.870)

Author, year	Node(s)(A–E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
	B	ICH + PHE Radiomics model; ElNet + RIDGE	Single-modality	ICH and PHE radiomic features, including shape, first-order intensity, texture-matrix features, wavelet-derived features, and Laplacian-of-Gaussian (LoG)-derived features from hematoma and perihematoma edema regions.		Int. val.	mRS 4–6	352	134	107	97	27	121	0.740 (0.700–0.790)
	C	ICH/PHE + Clinical model; SVM_rad + PCA	Multi-modality	ICH + PHE radiomics (shape, first-order, texture, wavelet, LoG) + age, sex, race, GCS, NIHSS, BP, treatment, laboratory variables, medical history, ICH location/side, hydrocephalus, pineal/septum shift, hematoma volume, PHE volume, IVH		Int. val.	mRS 4–6	352	134	93	37	41	181	0.850 (0.810–0.890)
Ji et al., 2026	D	Baseline-Image (3D CNN)	Single-modality	3D NCCT images	Five-fold cross-validation + independent external test	Ext. val.	mRS 3–6	117	60	46	24	14	33	0.737 (0.637–0.821)
	E	Image + Text multimodal model (3D CNN + BioClinicalBERT + dual-stream attention)	Multi-modality	3D NCCT, age, sex, hypertension, diabetes, smoking, alcohol, onset time, GCS, NIHSS		Ext. val.	mRS 3–6	117	60	48	9	12	48	0.846 (0.769–0.919)
Huang et al., 2023	A	Clinical logistic regression	Multi-modality	Age, GCS, glycemia, ICH volume, hematoma location	Temporal split + external-center validation	Ext. val.	mRS 4–6	86	49	43	10	6	27	0.881 (0.808–0.955)

Author, year	Node(s)(A–E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
	B	Rad-score logistic regression	Single-modality	7 PHE radiomics features		Ext. val.	mRS 4–6	86	49	34	7	15	30	0.805 (0.714–0.896)
	C	COMB clinical-radiomics logistic regression	Multi-modality	Age, GCS, glycemia, ICH volume, hematoma location, PHE Rad-score		Ext. val.	mRS 4–6	86	49	44	8	5	29	0.905 (0.839–0.970)
Li et al., 2024	A	Clinical-CT model / LightGBM	Multi-modality	Admission GCS score; maximum transverse diameter of hematoma; hematoma volume; hematoma location	Random hold-out (8:2); 5-fold CV for tuning	Int. val.	mRS 4–6	54	36	31	0	5	18	0.922 (0.845–0.999)
	B	Radiomics model / LightGBM	Single-modality	23 radiomics features: shape features, first-order histogram features, and texture features including GLCM, GLRLM, GLSZM, NGTDM		Int. val.	mRS 4–6	54	36	29	6	7	12	0.770 (0.633–0.906)
	C	Clinical-radiomics nomogram / Nomogram	Multi-modality	Admission GCS score; maximum transverse diameter of hematoma; hematoma volume; hematoma location, 23 radiomics features / Rad-score		Int. val.	mRS 4–6	54	36	32	1	4	17	0.932 (0.864–0.999)
Huang et al., 2022	A	Clinical-CT model / multivariable logistic regression	Multi-modality	Age, GCS, ICH volume, hemorrhage location	Temporal split + independent external validation	Ext. val.	mRS 4–6	86	49	43	11	6	26	0.881 (0.808–0.955)
	B	Edema radiomics model / multivariable logistic	Single-modality	PHE radiomics features (Rad-score)		Ext. val.	mRS 4–6	86	49	34	7	15	30	0.805 (0.714–0.896)

Author, year	Node(s)(A-E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
Tran et al., 2025		regression												
	C	Combined model / multivariable logistic regression	Multi-modality	Age, GCS; ICH volume; hemorrhage location; hematoma-edema Rad-score		Ext. val.	mRS 4–6	86	49	43	8	6	29	0.893 (0.825–0.961)
	B	Extra Trees Classifier (ExtraTrees)	Single-modality	1693 hematoma radiomics → NMF 128	ATACH-2 n=866, stratified 5-fold CV → Yale n=645 independent test	Ext. val.	mRS 4–6	645	301	224	146	77	198	0.724 (0.685–0.762)
Liu et al., 2026	D	Extra Trees Classifier (ExtraTrees)	Single-modality	nnU-Net + VAE-GAN latent features → NMF 128 each		Ext. val.	mRS 4–6	645	301	246	169	55	175	0.750 (0.710–0.785)
	A	Clinical-CT model(G2)/ Logistic regression	Multi-modality	Clinical features (age, sex, SBP, GCS, laboratory data, CT radiological features)		Ext. val.	mRS 4–6	35	17	15	4	2	14	0.879 (0.749–1.000)
	B	Radiomics model (Rad.G2) / Logistic regression	Single-modality	G2 ICH, PHE handcrafted radiomics (8 features)		Ext. val.	mRS 4–6	35	17	7	0	10	18	0.807 (0.661–0.954)
	C	Combined model (COMB.G2) / Logistic regression	Multi-modality	Clinical features + NCCT radiomics features	Train.: Center 1; Int. val.: Center 1; Ext. val.: Center 2+3	Ext. val.	mRS 4–6	35	17	14	2	3	16	0.915 (0.805–1.000)
	A	Clinical-CT model(G3)/ Logistic regression	Multi-modality	Clinical features (age, sex, SBP, GCS, laboratory data, CT radiological features)z		Ext. val.	mRS 4–6	15	3	3	1	0	11	0.944 (0.823 – 1.000)
	B	Radiomics model (Rad.G3) / Logistic regression	Single-modality	G3 ICH + PHE handcrafted radiomics (8 features)		Ext. val.	mRS 4–6	15	3	2	0	1	12	0.972 (0.895 – 1.000)
	C	Combined model (COMB.G3) / Logistic regression	Multi-modality	Clinical features + NCCT radiomics features		Ext. val.	mRS 4–6	15	3	2	1	1	11	0.944 (0.823 – 1.000)

Author, year	Node(s)(A–E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
	A	Clinical-CT model(G4)/ Logistic regression	Multi-modality	Clinical features (age, sex, SBP, GCS, laboratory data, CT radiological features)z		Ext. val.	mRS 4–6	15	10	10	3	0	2	0.820 (0.587 – 1.000)
	B	Radiomics model (Rad.G4) / Logistic regression	Single-modality	G4 ICH + PHE handcrafted radiomics (8 features)		Ext. val.	mRS 4–6	15	10	10	4	0	1	0.700 (0.324 – 1.000)
	C	Combined model (COMB.G4) / Logistic regression	Multi-modality	Clinical features + NCCT radiomics features		Ext. val.	mRS 4–6	15	10	7	3	3	2	0.800 (0.560 – 1.000)
Song et al., 2021	A	Clinical-CT model / Multivariate logistic regression	Multi-modality	Midline shift; CT onset-to-scan time; GCS; serum glucose; uric acid	Train 244; Int. val. 104; Ext. val. 87	Ext. val.	mRS 4–6	87	53	36	12	17	22	0.700 (0.590–0.820)
	C	Clinical-radiomics nomogram / Multivariate logistic regression	Multi-modality	Clinical predictors + hematoma Rad-score		Ext. val.	mRS 4–6	87	53	45	14	8	20	0.730 (0.620–0.840)
Song et al., 2024	A	Clinical-CT model (logistic regression)	Multi-modality	Gender, baseline ICH volume, CT onset-to-scan time, IVH, AST/ALT	Train 132; Int. val. 34	Int. val.	mRS 3–6	34	16	12	4	4	14	0.760 (NR)
	B	Perihematomal radiomics model (Random Forest)	Single-modality	NCCT perihematomal tissue radiomics features		Int. val.	mRS 3–6	34	16	10	2	6	16	0.830 (0.809–0.854)
	C	Clinical-radiomics nomogram (Random Forest)	Multi-modality	Clinical predictors + hematoma Rad-score		Int. val.	mRS 3–6	34	16	10	1	6	17	0.840 (NR)
Jiang et al., 2025	A	Clinical-CT model /	Multi-modality	Age; hydrocephalus; brain herniation;	Center-based external	Ext. val.	mRS 4–6	66	43	28	2	15	21	0.830 (0.718–0.911)

Author, year	Node(s)(A–E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
		Multivariate logistic regression		location; ICH volume; surgery duration	validation									
	B	Radscore model / Elastic net regression with 10-fold cross-validation	Single-modality	Flatness; Least axis length; VolumeCC		Ext. val.	mRS 4–6	66	43	30	5	13	18	0.743 (0.621–0.843)
	C	Combined model / Multivariate logistic regression	Multi-modality	Age; hydrocephalus; brain herniation; hematoma location; hematoma volume; surgical duration; Radscore (Flatness, Least axis length, VolumeCC)		Ext. val.	mRS 4–6	66	43	30	3	13	20	0.865 (0.758–0.936)

Notes: Input modality: Single-modality = CT image information only; Multi-modality = CT information combined with clinical variables. Algorithm/Model corresponds to the reported model producing the listed TP, FP, FN, and TN. When multiple algorithms were reported for a single node, the authors' designated primary model is presented.

External validation results were used when available; internal testing sets from three-way splits and training-plus-validation samples from k-fold or leave-one-out cross-validation were classified as internal validation. The mRS cut-off is reported as published (mRS 3-6 preferred); only 90-d/3-mo results are included. Pred. cut-off, if reported, is the classification threshold. AUC is reported as published; NR = not reported.

TP = true positive; FP = false positive; FN = false negative; TN = true negative; AUC = area under the receiver operating characteristic curve; CI = confidence interval; CV = cross-validation; ML = machine learning; DL = deep learning; NCCT = non-contrast CT; ICH = intracerebral hemorrhage; IVH = intraventricular hemorrhage; PHE = perihematoma edema; GCS = Glasgow Coma Scale; NIHSS = National Institutes of Health Stroke Scale.

Model nodes: A. Conventional clinical-CT models; B. CT-based radiomics-only ML models; C. Clinical-CT-radiomics ML models; D. CT-based DL-only models; E. Clinical-CT-DL fusion models.

Table S11. Diagnostic performance of included models for poor functional outcome at 6 months.

Author, year	Node(s)(A–E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
Zhang et al., 2026	A	Clinical-CT model / Logistic regression	Multi-modality	Age; GCS; rehabilitation therapy; mGS	Random 7:3 hold-out	Int. val.	mRS 4–6	123	45	34	14	11	64	0.839 (0.766–0.912)
	B	Radiomics model / Logistic regression	Single-modality	CT radiomics Rad-score		Int. val.	mRS 4–6	123	45	36	24	9	54	0.796 (0.716–0.873)
	C	Clinical-radiomics nomogram / Multivariable logistic regression	Multi-modality	Age; GCS; rehabilitation therapy; mGS; Rad-score		Int. val.	mRS 4–6	123	45	35	12	10	66	0.889 (0.831–0.946)
Zhou et al., 2021	A	Clinical-CT model / Multivariable logistic regression	Multi-modality	GCS; HE; hematoma volume	Center-based derivation/external validation	Int. val.	mRS 3–6	30	18	12	3	6	9	0.782 (NR)
	B	Rad-score / LASSO regression	Single-modality	NCCT radiomics features		Int. val.	mRS 3–6	30	18	11	2	7	10	0.806 (NR)
	C	Combined model / Multivariable logistic regression	Multi-modality	GCS; HE; hematoma volume; Rad-score		Int. val.	mRS 3–6	30	18	14	3	4	9	0.843 (NR)
Chen et al., 2024	A	ICH score; FUNC score	Multi-modality	Age; GCS; ICH volume; ICH location; IVH; pre-ICH cognitive status	Train/Val/Test = 497/165/220; Ext. Val.=146	Int. val.	mRS 3–6	190	67	24	17	43	106	0.698 (0.637–0.758)
	D	fICHnet (ResNet34 + RNN; DenseNet121 + RNN)	Single-modality	Admission NCCT images		Int. val.	mRS 3–6	146	84	65	26	19	36	0.753 (0.694–0.810)

Notes: Only studies meeting the full-text eligibility criteria are included. Each Model ID occupies one row. Models directly compared within a study share the same Study ID and cohort; studies with a single model node or without obtainable/reconstructable TP, FP, FN, and TN are excluded.

Input modality: Single-modality = CT image information only; Multi-modality = CT information combined with clinical variables. Algorithm/Model corresponds to the reported model producing the listed TP, FP, FN, and TN. When multiple algorithms were reported for a single node, the authors' designated primary model is presented.

External validation results were used when available; internal testing sets from three-way splits and training-plus-validation samples from k-fold or leave-one-out cross-validation were classified as internal validation. The mRS cut-off is reported as published (mRS 3-6 preferred); only 120-d/6-mo results are included. Pred. cut-off, if reported, is the classification threshold. AUC is reported as published; NR = not reported.

TP = true positive; FP = false positive; FN = false negative; TN = true negative; AUC = area under the receiver operating characteristic curve; CI = confidence interval; CV = cross-validation; ML = machine learning; DL = deep learning; NCCT = non-contrast CT; ICH = intracerebral hemorrhage; IVH = intraventricular hemorrhage; PHE = perihematomal edema; GCS = Glasgow Coma Scale; NIHSS = National Institutes of Health Stroke Scale.

Model nodes: A. Conventional clinical-CT models; B. CT-based radiomics-only ML models; C. Clinical-CT-radiomics ML models; D. CT-based DL-only models; E. Clinical-CT-DL fusion models.

Table S12. Diagnostic performance of included models for poor functional outcome at discharge.

Author, year	Node(s)(A–E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
Zhao et al., 2023	A	XGBoost model / XGBoost	Multi-modality	Gender; age; GCS; onset-to-CT time; ICH location; 9 NCCT markers; ICH/IVH/PHE size	H: 450 train + 50 internal test; Y: 104 external test	Ext. val.	mRS 4–6	104	38	33	8	5	58	0.871 (0.869–0.873)
	D	Convolution model / 3D CNN	Single-modality	3D NCCT images		Ext. val.	mRS 4–6	104	38	31	10	7	56	0.829 (0.827–0.831)
	E	Fusion model / 3D CNN + XGBoost fusion	Multi-modality	3D NCCT images + gender; age; GCS score; interval from symptom onset to admission CT; ICH location; 9 NCCT markers; ICH size; IVH size; PHE size		Ext. val.	mRS 4–6	104	38	34	6	4	60	0.905 (0.903–0.907)

Author, year	Node(s)(A–E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
Wu et al., 2025	A	Clinical-CT model / Random Forest	Single-modality	19 clinical variables: age, sex, smoking, alcohol consumption, hypertension, diabetes mellitus, ESRD, GCS, SBP, DBP, heart rate, hemoglobin, lymphocyte percentage, platelet count, BUN, creatinine, glucose, PT, APTT	Train 670 (5-fold CV); Test 288	Int. val.	mRS 3–6	288	229	161	17	68	42	0.794 (NR)
	D	Imaging model / 3D ResNet-50 + SE + SA	Single-modality	Initial NCCT images	Train 895 (80/20 train-validation); Test 288	Int. val.	mRS 3–6	288	229	190	18	39	41	0.876 (NR)
	E	Hybrid model / Random Forest	Multi-modality	19 clinical variables + Model-I predicted outcome probability (r) from NCCT	Train 670 (5-fold CV); Test 288	Int. val.	mRS 3–6	288	229	185	12	44	47	0.873 (NR)
Pei et al., 2024	A	Clinical-CT model/Logistic regression	Multi-modality	Sex; IVH; GCS score; ICH volume	Train/validation split (7:3)	Int. val.	mRS 3–6	46	21	13	4	8	21	0.711 (0.551–0.871)
	B	Radiomics model/Logistic regression	Single-modality	NCCT hematoma radiomics features (8 selected features)		Int. val.	mRS 3–6	46	21	17	4	4	21	0.869 (0.764–0.975)

Author, year	Node(s)(A–E)	Model/Alg.	Input modality	Input features	Data split method	Validation level	mRS cut-off	N	Poor outcome	TP	FP	FN	TN	AUC(95% CI)
	C	Combined model/Logistic regression	Multi-modality	Sex; IVH; GCS score; ICH volume; Radiomics score		Int. val.	mRS 3–6	46	21	16	1	5	24	0.915 (0.836–0.994)

Notes: Only studies meeting the full-text eligibility criteria are included. Each Model ID occupies one row. Models directly compared within a study share the same Study ID and cohort; studies with a single model node or without obtainable/reconstructable TP, FP, FN, and TN are excluded.

Input modality: Single-modality = CT image information only; Multi-modality = CT information combined with clinical variables. Algorithm/Model corresponds to the reported model producing the listed TP, FP, FN, and TN. When multiple algorithms were reported for a single node, the authors' designated primary model is presented.

External validation results were used when available; internal testing sets from three-way splits and training-plus-validation samples from k-fold or leave-one-out cross-validation were classified as internal validation. The mRS cut-off is reported as published (mRS 3-6 preferred); discharge outcomes refer only to in-hospital assessment at discharge (not 90-day or 120-day/6-month timepoints). Pred. cut-off, if reported, is the classification threshold. AUC is reported as published; NR = not reported.

TP = true positive; FP = false positive; FN = false negative; TN = true negative; AUC = area under the receiver operating characteristic curve; CI = confidence interval; CV = cross-validation; ML = machine learning; DL = deep learning; NCCT = non-contrast CT; ICH = intracerebral hemorrhage; IVH = intraventricular hemorrhage; PHE = perihematomal edema; GCS = Glasgow Coma Scale; NIHSS = National Institutes of Health Stroke Scale.

Model nodes: A. Conventional clinical-CT models; B. CT-based radiomics-only ML models; C. Clinical-CT-radiomics ML models; D. CT-based DL-only models; E. Clinical-CT-DL fusion models.

Table S13. CT acquisition and image-preprocessing characteristics.

Author, year	CT timing	CT type	Scanner/model	Inter-center protocol	Contrast	Tube voltage	Tube current	Slice thickness	Slice interval	Matrix	Recon. kernel	Preprocessing
Pszczolkowski et al., 2021	Baseline NCCT before randomisation; ≤ 8 h from onset	NCC T	NR	124 centers; local protocols	No contrast	NR	NR	<2 mm; ≥ 2 to <4 mm; ≥ 4 mm	NR	NR	NR	1-mm isotropic resampling; NN/cubic interpolation; LoG + sym8 Wavelet; 50/100 gray levels
Wang et al., 2022	Baseline CT; exact onset-to-CT NR	NCC T	Siemens SOMATOM Definition; Siemens SOMATOM PLUS 4; Philips iCT 256	Heterogeneous scanner/acquisition protocols	No contrast	120 kV	240 or 630 mA; 333 mA	1.2–7.2 mm; 1.0–5.0 mm	NR	512×512	NR	Resampling to $0.5 \times 0.5 \times 5$ mm; third-order spline interpolation; ROI resize/crop; z-score normalization
Xia et al., 2025	≤ 24 h	NCC T	NR	Standardized CT protocol across two centers; detailed acquisition parameters NR	No contrast	NR	NR	5 or 6 mm	NR	NR	NR	WW100 HU; WL50 HU
Wang et al., 2024	Center 1: 3.0 (1.5–4.0) h; Center 2: 4.0 (3.0–4.5) h; Center 3: 2.0 (1.0–3.0) h; all OIT <6 h	NCC T	NR	NR	No contrast	NR	NR	NR	NR	NR	NR	WW 75–85 HU; WL 35–45 HU; input $224 \times 224 \times 3$
Dierksen et al., 2024	Admission	NCCT	NR	Multi-center; heterogeneous CT acquisition; exact protocols NR	No contrast	NR	NR	Variable; exact acquisition thickness NR	NR	Variable; exact matrix NR	NR	sitkLinear resampling to $1 \times 1 \times 1$ mm; HU range PHE 0–40, ICH 0–100; bin width 2; pad distance 10; Original +

Author, year	CT timing	CT type	Scanner/mode I	Inter-center protocol	Contrast	Tube voltage	Tube current	Slice thickness	Slice interval	Matrix	Recon. kernel	Preprocessing
Ji et al., 2026												Wavelet + LoG (σ 2/4/6 mm)
	Admission	NCC T	Siemens/SOM ATOM Definition; GE/BrightSpeed, Optima CT520 series; Philips/Ingenuity CT, Incisive CT, iCT 256	Multi-center; standardized preprocessing	No contrast	120 kV	400–460 mA	5 mm	5 mm	NR	GE: Standard ; Philips: UB; Siemens: H30	Resize to 128×512×512; WW 100 HU, WL 50 HU; nnU-Netv2 skull stripping
Huang et al., 2023	Baseline NCCT within 6 h after symptom onset	NCC T	NR	No restrictions on scanner manufacturer, settings, or slice thickness	No contrast	120 kV	148–440 mA	1–5 mm	NR	NR	NR	Resample 1×1×1 mm ³ , bin width 25, median imputation, Z- score
Li et al., 2024	Preoperative	NCC T	GE Revolution Apex 128-row CT	NR	No contrast	120 kV	400 mA	5 mm	NR	512 × 512	NR	NR
Huang et al., 2022	Baseline	NCC T	NR	No restrictions on scanner manufacturer/s ettings; multicenter heterogeneous acquisition	No contrast	120 kV	148–440 mA	1–5 mm	NR	NR	NR	Resample 1×1×1 mm ³ ; bin width 25; array shift 1000; Z-score
Zhang et al., 2026	Baseline CT within 24 h of symptom onset	NCC T	Siemens SOMATOM Definition Edge, 64-slice	NR	No contrast	120 kV	240 mA	5.0 mm	NR	NR	NR	WL 40/WW 80 HU; no resampling/pre processing
Zhou et al., 2021	Baseline/admis sion NCCT ≤24 h after onset; Derivation: 3.28 ± 2.46 h; Validation: 3.07 ± 1.62 h	NCC T	Philips Ingenuity; Siemens Somatom Force; Canon Aquilion ONE	Standard departmental protocols; 3 scanners; image normalization applied	No contrast	120 kV	250–300 mA	1 mm or 5 mm	NR	512 × 512	NR	Resample 1×1×1 mm ³ ; 256 bins; WL/WW 50/110 HU
Zhao et al., 2023	Admission/day of admission;	NCC T	H: GE Discovery	Different scanners;	No contrast	H: 120 kV; Y:	H: 250 mAs; Y:	H: 5 mm; Y: 5 mm	NR	H: 512 × 512; Y: 512 × 512	NR	3D input 256×256×128;

Author, year	CT timing	CT type	Scanner/mode I	Inter-center protocol	Contrast	Tube voltage	Tube current	Slice thickness	Slice interval	Matrix	Recon. kernel	Preprocessing
Wu et al., 2025	initial CT		CT750; Y: SIEMENS SOMATOM Definition Flash	acquisition parameters differed in tube current/FOV; no harmonization protocol reported		120 kV	400 mAs					TTA rotation/scaling /cropping; prediction averaging
	NR	NCC T	NR	NR	No contrast	NR	NR	NR	NR	NR	NR	NR
Tran et al., 2025	Admission	NCC T	NR	ATACH-2: Multi-center, protocol NR; Yale: Single- center, protocol NR	No contrast	NR	NR	ATACH-2: 5.3 ± 1.8 mm; Yale: 4.8 ± 0.7 mm	NR	ATACH-2: 418×418– 512×734; Yale: 472×472– 1024×1024	NR	Intensity normalization; isotropic resampling; ROI cropping; 128 ³ /64 ³
Chen et al., 2024	Admission	NCC T	MGH: GE (82.2%), Siemens (6.6%), Philips (3.4%), Toshiba (1.7%), Mediso (0.1%), Canon (0.1%), unknown (5.9%); 36 scanners. Yale: GE Discovery CT750 HD	Multi-center; scanner protocols differed by center	No contrast	NR	NR	MGH: ≤5 mm included; thinner slices downsamp led to 5 mm. Yale: NR	NR	MGH: 512×512 for 99.7%; Yale: 512×512	NR	Skull stripping; HU 0–100 normalization; crop; resize 128×128×18
	Baseline NCCT within 72 h of onset	NCC T	Center 1: SIEMENS Sensation 64; SIEMENS SOMATOM Definition Flash; Center 2/3: Philips ICT 256; NMS NeuViz Glory	Multi-center; slice thickness grouped as <2 mm, 2–<4 mm, ≥4 mm for harmonization	No	120 kV	148–440 mA	1–8 mm	NR	NR	NR	Resampling 1×1×1 mm ³ ; bin width 25; array shift 1000; ComBat harmonization
Song et al.,	Baseline NCCT	NCC	Center 1/2:	Two centers	NR	120 kV	250–300	1 mm or 5	NR	512 × 512	NR	Resampling

Author, year	CT timing	CT type	Scanner/mode I	Inter-center protocol	Contrast	Tube voltage	Tube current	Slice thickness	Slice interval	Matrix	Recon. kernel	Preprocessing
2021	within 24 h of ictus	T	Lightspeed VCT (GE); Ingenuity (Philips); Aquilion ONE (Canon)				mA	mm				1×1×1 mm ³ ; 256 gray-level bins; WL 50 HU/WW 110 HU
Song et al., 2024	Baseline NCCT within 24 h after onset	NCC T	NR	Single center	NR	NR	NR	NR	NR	NR	NR	NR
Jiang et al., 2025	Baseline	NCC T	Philips Medical Systems; Canon Medical Systems	Two centers; standardized acquisition	NR	120 kV	250–300 mA	1 mm	NR	512 × 512	NR	Resampling to 1×1×1 mm ³ ; 256-bin discretization; WW 90 HU/WL 40 HU
Pei et al., 2024	Baseline	NCC T	Canon Aquilion ONE; GE Optima CT540; Siemens SOMATOM Definition	Single center; three scanners	NR	120 kV	Automate d modulation	6 mm	6 mm	512 × 512	NR	Brain window; 1 × 1 × 1 mm resampling; wavelet + LoG

Notes: Only studies meeting the full-text eligibility criteria are included. Acquisition, segmentation, and development details are reported separately by cohort or center. When scanners, slice thicknesses, or centers differ, items are separated by semicolons. NR = not reported; N/A = not applicable.

CT = computed tomography; NCCT = non-contrast CT; kV = kilovolts; mA = milliamperes; mAs = milliampere-seconds; mm = millimetres; HU = Hounsfield units; WL = window level; WW = window width; LoG = Laplacian of Gaussian; NR = not reported; N/A = not applicable.

Model nodes: A. Conventional clinical-CT models; B. CT-based radiomics-only ML models; C. Clinical-CT-radiomics ML models; D. CT-based DL-only models; E. Clinical-CT-DL fusion models.

Table S14. Segmentation and model-development characteristics.

Author, year	ROI definition	Seg. method	Seg. software	Rater/blinding /ICC	Radiomics SW/version	Initial/final features	Feature selection	DL input	Network/fusion	Harmonization
Pszczolkowski et al., 2021	Baseline ICH ROI; PHE + IVH volumes added	Semi-automated + manual editing	ITK-SNAP 3.6.0	3 raters; blinded to clinical data; ICC NR	MATLAB R2019b; radiomics package + in-house code	754 / 218	Correlation filtering ($ r > 0.9$) + elastic-net; 10-fold CV	N/A	N/A	ComBat; slice-thickness batches; age/sex covariates
Wang et al., 2022	Hematoma ROI for radiomics; resized-ROI + crop-ROI for DL	Manual	NR	NR	PyRadiomics; version NR	1051/8	LASSO	Two-channel 3D CT ROI, $90 \times 90 \times 36 \times 2$	ResNet18; DL score + clinical variables, LASSO + logistic regression	Resampling + z-score normalization
Xia et al., 2025	3D hematoma VOI; VOI40% = hematoma + 40% perihematoma expansion	Semi-automatic; CNN segmentation + manual correction; morphological dilation	Dr. Wise Hemorrhagic Stroke Analyzer; Deepwise platform	2 blinded neuroradiologists; resegmentation if VOI change $> 20\%$; ICC ≥ 0.75	PyRadiomics-based Deepwise platform; version NR	1,743 / 15	LASSO	N/A	N/A	N/A
Wang et al., 2024	Hematoma + PHE ROIs; axial slice-by-slice; ROI $< 125 \text{ mm}^3$ excluded	Manual, slice-by-slice; expert review	ITK-SNAP v3.8.0	3 radiologists; reviewed by 7-year-experience radiologist; Blind NR; ICC NR	NR	2048 / 15	ANOVA + LASSO	Hematoma + PHE ROI slices, $224 \times 224 \times 3$	ImageNet-pretrained ResNet50 $\rightarrow$ Deep Score; Merge = Deep Score + clinical/CT features $\rightarrow$ LR	N/A
Dierksen et al., 2024	Separate 3D ICH and PHE ROIs on admission NCCT	Manual slice-by-slice segmentation; separate ICH/PHE masks	MRICron + 3D Slicer	Trained research associates; neuroradiologist review; blind NR; ICC: ICH 0.94/0.92, PHE 0.94/0.93	PyRadiomics v2.2.0	2260 (1130 ICH + 1130 PHE) / 30	ICC filtering (> 0.8 lower 95% CI); B: RIDGE; C: PCA; repeated 5-fold CV + Bayesian optimization	N/A	N/A	1-mm isotropic resampling; statistical harmonization NR
Ji et al., 2026	Whole-brain 3D CT after skull stripping	Automated nnU-Netv2 + manual review	nnU-Netv2; 3D Slicer	Experienced clinicians; blind NR; ICC N/A	NR	NR	N/A	Raw 3D NCCT, $1 \times 128 \times 512 \times$	3D CNN; 3D CNN + BioClinicalBE	N/A

								512	RT + dual-stream cross-modal attention	
Huang et al., 2023	3D whole PHE VOI; VOI <0.125 mL excluded	Manual 3D segmentation; >20% modification resegmented	ITK-SNAP v3.8.0	3 neuroradiologists; final review by 1; blinded; ICC NR	AI Kit v3.2.2	107/7	Wilcoxon test, multivariable logistic regression	N/A	N/A	Median imputation, Z-score
Li et al., 2024	3D whole hematoma ROI on preoperative NCCT	Manual, slice-by-slice	ITK-SNAP 3.8	2 radiologists reviewed; 1 senior radiologist resolved disagreement; 1 radiologist (10 y) delineated; Blind NR; ICC NR	PyRadiomics v3.0	1,834/23	Mann–Whitney U → Spearman >0.9 → LASSO (10-fold CV)	N/A	N/A	N/A
Huang et al., 2022	3D hematoma + PHE VOIs; exclude <125 mm ³	Manual	ITK-SNAP v3.8.0	3 neuroradiologists, blinded; >20% disagreement redefined; ICC NR	A.K. v3.2.2 (PyRadiomics-based)	214 initial; Edema 7; Hematoma 9; Hematoma-Edema 12	Wilcoxon → multivariable LR	N/A	N/A	Z-score; IBSI
Zhang et al., 2025	3D whole hematoma ROI	Semi-automatic; threshold 40–100 HU + manual delineation	3D Slicer v5.6.2	NR	3D Slicer Radiomics plugin	107/6	LASSO (10-fold CV) → VIF → Rad-score	N/A	N/A	N/A
Zhou et al., 2021	3D whole-hematoma ROI on axial NCCT	Manual	ITK-SNAP v3.8.0	50 cases resegmented; inter-/intra-observer ICC>0.8: 91.6%/93.5%; outcome-blinded	PyRadiomics v3.0.1	107/10	Univariable P<0.05 → LASSO, 10-fold CV	N/A	N/A	Zero-variance removal; median imputation; zero-mean normalization
Zhao et al., 2023	ICH/IVH/PHE manually delineated; DL used 3D	Manual	ITK-SNAP 3.8	2 neuroradiologists (6/15 y); blinded to	NR	NR	N/A	3D NCCT, 256×256×128	Multi-scale 3D Convolution; residual/skip + separable	N/A

Wu et al., 2025	NCCT			clinical/lab data; 20 cases; Dice: ICH 0.89, IVH 0.87, PHE 0.83				convolution; Fusion = Convolution + XGBoost via MLP		
	NR	NR	NR	NR	NR	NR	N/A	3D initial NCCT images	3D ResNet-50 + SE + SA; hybrid: DL probability (r) + 19 clinical variables → RF	N/A
Tran et al., 2025	3D hematoma ROI; dilated hematoma + perilesional tissue	Automatic	nnU-Net	Manual ground-truth masks; rater/blinding NR; Dice 0.88 ± 0.05 (CV), 0.77 ± 0.19 (external test)	PyRadiomics, NR	B: 1693/128; D: 256 (128 nnU-Net + 128 VAE-GAN)	NMF	nnU-Net: 128 ³ /64 ³ NCCT; VAE-GAN: 64 ³ dilated ROI	nnU-Net + VAE-GAN → NMF → ExtraTrees	Intensity normalization + isotropic resampling; harmonization NR
Chen et al., 2024	NR	NR	NR	NR	N/A	NR	N/A	Admission NCCT images	fICHnet (CNN-GRU survival model; ResNet34+RN N/DenseNet121+RNN)	N/A
Liu et al., 2026	Whole hematoma + PHE VOI	Manual segmentation + radiologist correction	ITK-SNAP v3.8.0	7 radiologists; reviewed by 1 radiologist; blinded to outcome; ICC NR	Artificial Intelligence Kit (A.K.) v3.2.2	G1:12; G2:8; G3:3; G4:8 / NR	Mann–Whitney U → logistic regression → stepwise regression	N/A	N/A	ComBat harmonization
Song et al., 2021	3D whole hematoma VOI on baseline NCCT	Manual hematoma delineation; semi-automatic VOI generation	Artificial Intelligence Kit (AK) v3.2.2	2 radiologists; 50 cases reproducibility; inter-/intra-observer ICC >0.8 for most features	Artificial Intelligence Kit (AK) v3.2.2, GE Healthcare	396/12	mRMR → LASSO (10-fold CV)	N/A	N/A	Feature-domain harmonization; median imputation; standardization
Song et al., 2024	Hematoma; PHE rings at 5, 10, 15 mm	Semi-automatic; manual removal of ventricle/skull	3D Slicer v4.11.0	2 radiologists blinded to outcome; 1 radiologist review;	3D Slicer radiomics	944/19	Correlation test → Mann–Whitney U test →	N/A	N/A	NR

Jiang et al., 2025	3D whole-hematoma ROI	Semi-automatic	ITK-SNAP 3.8.0	ICC >0.75	LASSO					
				2 radiologists (≥5 y/2 y); outcome-blinded; 30 cases; intra/inter ICC≥0.75: 104/107, 100/107	Python 3.2	107/3	ICC≥0.75 → elastic net, 10-fold CV	N/A	N/A	Standardized acquisition + preprocessing; ComBat NR
Pei et al., 2024	Whole hematoma VOI	Manual	3D Slicer 5.0.3	Two radiologists; ICC ≥0.75	PyRadiomics via 3D Slicer	1,130/8	mRMR + LASSO	N/A	N/A	NR

Notes: Table S14 reports ROI definition, segmentation, radiomics, and deep-learning development for the same cohorts. Feature counts are initial/final values where reported. ROI = region of interest; VOI = volume of interest; Seg. = segmentation; SW = software; ICC = intraclass correlation coefficient; DL = deep learning; CNN = convolutional neural network; CV = cross-validation; LASSO = least absolute shrinkage and selection operator; NMF = non-negative matrix factorization; mRMR = minimum redundancy maximum relevance; VIF = variance inflation factor; LR = logistic regression; TTA = test-time augmentation; VAE-GAN = variational autoencoder-generative adversarial network; IBSI = Image Biomarker Standardisation Initiative; ICH = intracerebral hemorrhage; IVH = intraventricular hemorrhage; PHE = perihematoma edema; NR = not reported; N/A = not applicable. Model nodes: A. Conventional clinical-CT models; B. CT-based radiomics-only ML models; C. Clinical-CT-radiomics ML models; D. CT-based DL-only models; E. Clinical-CT-DL fusion models.

Table S15. Meta-regression analyses of all study-level covariates.

Covariates	Categories	Rows (n)	Logit (Sensitivity) β (95% CrI) vs. reference	Bayesian P	Logit (Specificity) β (95% CrI) vs. reference	Bayesian P	Log (DOR) β (95% CrI) vs. reference	Bayesian P
Input data modality	Multimodal	29	Reference		Reference		Reference	
	Unimodal	16	0.77 (-1.50, 3.03)	0.51	0.49 (-1.80, 2.76)	0.67	1.25 (-1.99, 4.48)	0.44
Centers	Multicenter	39	Reference		Reference		Reference	
	Single center	6	0.19 (-0.74, 1.08)	0.66	0.83 (-0.21, 1.89)	0.11	1.02 (-0.28, 2.28)	0.11
Region	Asia	37	Reference		Reference		Reference	
	Non-Asia	8	-0.12 (-0.71, 0.45)	0.66	-0.24 (-0.97, 0.51)	0.49	-0.37 (-1.16, 0.45)	0.34
Validation type	External validation	30	Reference		Reference		Reference	
	Internal validation	15	-0.14 (-0.65, 0.38)	0.57	0.43 (-0.19, 1.08)	0.16	0.29 (-0.44, 1.09)	0.43
Reference standard	mRS 3	3	Reference		Reference		Reference	
	mRS 4	42	0.60 (-0.38, 1.71)	0.21	-0.05 (-1.15, 1.42)	0.94	0.54 (-0.86, 2.52)	0.47
Publication year	Continuous (per 1-year)	45	-0.05 (-0.23, 0.11)	0.51	0.09 (-0.12, 0.32)	0.38	0.04 (-0.21, 0.29)	0.74
Age	Continuous (per 1-unit)	45	-0.03 (-0.09, 0.03)	0.29	-0.02 (-0.10, 0.05)	0.57	-0.05 (-0.13, 0.02)	0.17
Male proportion	Continuous (per 1-unit)	45	0.01 (-0.03, 0.05)	0.61	-0.01 (-0.05, 0.04)	0.65	0.00 (-0.05, 0.05)	0.92
ICH volume	Continuous (per 1-unit)	40	0.01 (-0.01, 0.03)	0.33	0.02 (-0.01, 0.05)	0.28	0.03 (-0.00, 0.06)	0.09

Note: β = difference in logit-transformed sensitivity (or specificity) relative to the reference category ($\beta = 0$ for the reference; shown as “Reference”). β relative to the reference category; continuous covariates: β per 1-unit increase. Positive β indicates higher, negative β indicates lower performance vs. the reference. For the DOR scale, $\beta = \beta \text{ logit(Se)} + \beta \text{ logit(Sp)}$, derived per posterior draw (no separate DOR model); ROR = $\exp(\beta \text{ logDOR})$. CrI, 95% credible interval; n, number of arm-level rows (one per data row). Bayesian P = $2 \times \min(\Pr(\beta \leq 0), \Pr(\beta \geq 0))$, a two-sided posterior tail probability (not a frequentist P value). Model: Bayesian bivariate random-effects meta-regression (arm-level GLMM, fitted in JAGS). Maximum R-hat = 1.00. For the continuous covariates (publication year, age, male proportion, and ICH volume), β is the change per 1-unit increase in the covariate and no reference category applies (the “vs. reference” header concerns categorical covariates only)

Table S16. Between-study variance (τ^2) explained by each covariate in the meta-regression analyses.

Covariate	Metric	τ^2 without covariate (95% CrI)	τ^2 with covariate (95% CrI)	R^2 (95% CrI)
Input data modality	τ^2 logit(Sensitivity)	0.14 (0.07, 0.34)	0.14 (0.07, 0.34)	0.00 (-2.22, 0.69)
	τ^2 logit(Specificity)	0.24 (0.10, 0.64)	0.24 (0.10, 0.64)	0.01 (-2.64, 0.73)
	τ^2 log(DOR)	0.30 (0.14, 0.77)	0.30 (0.14, 0.76)	0.00 (-2.31, 0.70)
Centers	τ^2 logit(Sensitivity)	0.14 (0.07, 0.34)	0.15 (0.07, 0.37)	-0.04 (-2.37, 0.68)
	τ^2 logit(Specificity)	0.24 (0.10, 0.64)	0.22 (0.09, 0.59)	0.09 (-2.35, 0.75)
	τ^2 log(DOR)	0.30 (0.14, 0.77)	0.28 (0.13, 0.68)	0.09 (-1.98, 0.72)
Region	τ^2 logit(Sensitivity)	0.14 (0.07, 0.34)	0.15 (0.07, 0.37)	-0.05 (-2.44, 0.67)
	τ^2 logit(Specificity)	0.24 (0.10, 0.64)	0.25 (0.10, 0.67)	-0.02 (-2.79, 0.72)
	τ^2 log(DOR)	0.30 (0.14, 0.77)	0.30 (0.14, 0.77)	0.01 (-2.38, 0.70)
Validation type	τ^2 logit(Sensitivity)	0.14 (0.07, 0.34)	0.14 (0.07, 0.36)	-0.02 (-2.33, 0.69)
	τ^2 logit(Specificity)	0.24 (0.10, 0.64)	0.23 (0.10, 0.62)	0.05 (-2.54, 0.74)
	τ^2 log(DOR)	0.30 (0.14, 0.77)	0.32 (0.14, 0.81)	-0.05 (-2.57, 0.69)
Reference standard	τ^2 logit(Sensitivity)	0.14 (0.07, 0.34)	0.14 (0.07, 0.35)	-0.00 (-2.23, 0.69)
	τ^2 logit(Specificity)	0.24 (0.10, 0.64)	0.24 (0.10, 0.66)	-0.01 (-2.75, 0.73)
	τ^2 log(DOR)	0.30 (0.14, 0.77)	0.33 (0.15, 0.86)	-0.08 (-2.73, 0.68)
Publication year	τ^2 logit(Sensitivity)	0.14 (0.07, 0.34)	0.14 (0.07, 0.35)	-0.00 (-2.29, 0.69)
	τ^2 logit(Specificity)	0.24 (0.10, 0.64)	0.24 (0.10, 0.65)	-0.00 (-2.69, 0.73)
	τ^2 log(DOR)	0.30 (0.14, 0.77)	0.32 (0.14, 0.80)	-0.04 (-2.53, 0.69)
Age	τ^2 logit(Sensitivity)	0.14 (0.07, 0.34)	0.15 (0.07, 0.37)	-0.04 (-2.39, 0.68)
	τ^2 logit(Specificity)	0.24 (0.10, 0.64)	0.25 (0.11, 0.68)	-0.04 (-2.84, 0.71)
	τ^2 log(DOR)	0.30 (0.14, 0.77)	0.29 (0.13, 0.73)	0.05 (-2.21, 0.72)
Male proportion	τ^2 logit(Sensitivity)	0.14 (0.07, 0.34)	0.15 (0.07, 0.36)	-0.03 (-2.40, 0.68)
	τ^2 logit(Specificity)	0.24 (0.10, 0.64)	0.25 (0.10, 0.67)	-0.02 (-2.75, 0.72)
	τ^2 log(DOR)	0.30 (0.14, 0.77)	0.32 (0.14, 0.82)	-0.04 (-2.54, 0.69)
ICH volume	τ^2 logit(Sensitivity)	0.15 (0.07, 0.36)	0.15 (0.07, 0.38)	-0.02 (-2.42, 0.70)
	τ^2 logit(Specificity)	0.27 (0.11, 0.74)	0.27 (0.11, 0.76)	0.00 (-2.94, 0.75)
	τ^2 log(DOR)	0.34 (0.15, 0.89)	0.30 (0.13, 0.81)	0.12 (-2.17, 0.75)

Note: τ^2 = between-study random-effects variance on the logit scale, computed per posterior iteration as $\Sigma = \text{solve}(\Omega)$ (Ω = Wishart precision matrix), the same scale as the primary analysis. $R^2 = 1 - \tau^2(\text{with covariate})/\tau^2(\text{without covariate})$, the proportion of between-study variance explained by the covariate; posterior median with 95% CrI (negative values indicate the covariate did not reduce the estimated variance in that iteration). Each meta-regression included a single covariate: for the 45-arm analyses the covariate-free (null) model is shared across covariates, so the “without covariate” τ^2 values are identical across those covariates and are shown per row for readability; the ICH-volume analysis was restricted to the 40 arms with available ICH volume data, for which the null model was refitted on the same subset (hence its baseline values differ slightly).

How to read this table: τ^2 (between-study variance, logit scale) reflects how much results differ across studies — the larger τ^2 , the stronger the between-study heterogeneity. “Without covariate” is the total between-study variance when the covariate is ignored; “with covariate” is what remains after accounting for it. If the latter is clearly smaller, the covariate explains part of the heterogeneity. R^2 is the proportion of between-study variance explained by the covariate (e.g., 0.26 = about 26%); values near 0 or negative mean the covariate did not explain the variance (negative values are posterior noise, not a reverse effect). The three rows refer to sensitivity, specificity, and the diagnostic odds ratio (DOR) scale, respectively.

Table S17. MCMC Convergence Diagnostics.

Parameter	Mean	SD	2.5%	97.5%	R-hat	ESS
Se — Conventional clinical-CT models	0.746	0.026	0.692	0.796	1.0011	14,000
Se — CT-based radiomics-only ML models	0.729	0.028	0.672	0.782	1.0011	14,000
Se — Clinical-CT-radiomics ML models	0.765	0.027	0.710	0.815	1.0012	13,000
Se — CT-based DL-only models	0.779	0.029	0.719	0.832	1.0010	35,000
Se — Clinical-CT-DL fusion models	0.820	0.031	0.753	0.876	1.0010	30,000
Sp — Conventional clinical-CT models	0.780	0.030	0.716	0.834	1.0010	30,000
Sp — CT-based radiomics-only ML models	0.690	0.037	0.614	0.758	1.0011	26,000
Sp — Clinical-CT-radiomics ML models	0.809	0.029	0.747	0.860	1.0010	48,000
Sp — CT-based DL-only models	0.655	0.043	0.567	0.737	1.0012	11,000
Sp — Clinical-CT-DL fusion models	0.831	0.037	0.751	0.895	1.0011	15,000
DOR — Conventional clinical-CT models	10.773	2.264	7.038	15.901	1.0011	14,000
DOR — CT-based radiomics-only ML models	6.176	1.286	4.047	9.056	1.0011	17,000
DOR — Clinical-CT-radiomics ML models	14.293	3.208	9.030	21.638	1.0011	17,000
DOR — CT-based DL-only models	6.988	1.720	4.255	10.946	1.0011	16,000
DOR — Clinical-CT-DL fusion models	24.477	8.263	12.375	44.334	1.0011	21,000
τ_{Se}	0.387	0.085	0.258	0.586	1.0010	36,000
τ_{Sp}	0.509	0.124	0.319	0.803	1.0010	37,000
ρ (Se-Sp)	-0.204	0.269	-0.674	0.356	1.0011	20,000
v	2.844	1.298	1.060	6.070	1.0013	6,700
$\tau \log DOR$	0.570	0.129	0.372	0.870	1.0001	26,239

Note: CrI, credible interval; SD, standard deviation; ESS, effective sample size. R-hat < 1.05 indicates good convergence; ESS > 1000 is considered adequate. Maximum R-hat = 1.001; minimum ESS = 6700. MCMC settings: 100000 iterations, 40000 burn-in, 4 chains, thin = 5.

Table S18. Clinical decision table across assumed prevalence scenarios.

Index test	Prevalence (%)	Sensitivity, % (95% CrI)	Specificity, % (95% CrI)	PPV, % (95% CrI)	NPV, % (95% CrI)	Positive post-test probability (%)	Negative post-test probability (%)	TP per 1000	FN per 1000	TN per 1000	FP per 1000
Conventional clinical-CT models	5.0	74.6 (69.2 to 79.6)	78.0 (71.6 to 83.4)	15.3 (12.2 to 19.1)	98.3 (98.0 to 98.6)	15.3 (12.2 to 19.1)	1.7 (1.4 to 2.0)	37	13	741	209
	10.0	74.6 (69.2 to 79.6)	78.0 (71.6 to 83.4)	27.6 (22.6 to 33.3)	96.5 (95.8 to 97.2)	27.6 (22.6 to 33.3)	3.5 (2.8 to 4.2)	75	25	702	198
	15.0	74.6 (69.2 to 79.6)	78.0 (71.6 to 83.4)	37.7 (31.7 to 44.3)	94.6 (93.5 to 95.6)	37.7 (31.7 to 44.3)	5.4 (4.4 to 6.5)	112	38	663	187
	20.0	74.6 (69.2 to 79.6)	78.0 (71.6 to 83.4)	46.1 (39.7 to 52.9)	92.5 (91.0 to 93.9)	46.1 (39.7 to 52.9)	7.5 (6.1 to 9.0)	149	51	624	176
	25.0	74.6 (69.2 to 79.6)	78.0 (71.6 to 83.4)	53.2 (46.7 to 60.0)	90.2 (88.3 to 92.0)	53.2 (46.7 to 60.0)	9.8 (8.0 to 11.7)	187	63	585	165
	30.0	74.6 (69.2 to 79.6)	78.0 (71.6 to 83.4)	59.4 (53.0 to 65.8)	87.8 (85.5 to 89.9)	59.4 (53.0 to 65.8)	12.2 (10.1 to 14.5)	224	76	546	154
	35.0	74.6 (69.2 to 79.6)	78.0 (71.6 to 83.4)	64.7 (58.6 to 70.8)	85.1 (82.4 to 87.6)	64.7 (58.6 to 70.8)	14.9 (12.4 to 17.6)	261	89	507	143
	40.0	74.6 (69.2 to 79.6)	78.0 (71.6 to 83.4)	69.4 (63.7 to 75.0)	82.2 (79.1 to 85.1)	69.4 (63.7 to 75.0)	17.8 (14.9 to 20.9)	298	102	468	132
	45.0	74.6 (69.2 to 79.6)	78.0 (71.6 to 83.4)	73.6 (68.3 to 78.6)	79.0 (75.5 to 82.4)	73.6 (68.3 to 78.6)	21.0 (17.6 to 24.5)	336	114	429	121
	50.0	74.6 (69.2 to 79.6)	78.0 (71.6 to 83.4)	77.3 (72.5 to 81.8)	75.5 (71.6 to 79.3)	77.3 (72.5 to 81.8)	24.5 (20.7 to 28.4)	373	127	390	110
CT-based radiomics-only ML models	5.0	72.9 (67.2 to 78.2)	69.0 (61.4 to 75.8)	11.1 (9.0 to 13.7)	98.0 (97.5 to 98.4)	11.1 (9.0 to 13.7)	2.0 (1.6 to 2.5)	36	14	655	295
	10.0	72.9 (67.2 to 78.2)	69.0 (61.4 to 75.8)	20.9 (17.3 to 25.1)	95.8 (94.9 to 96.6)	20.9 (17.3 to 25.1)	4.2 (3.4 to 5.1)	73	27	621	279
	15.0	72.9 (67.2 to 78.2)	69.0 (61.4 to 75.8)	29.5 (25.0 to 34.7)	93.5 (92.2 to 94.7)	29.5 (25.0 to 34.7)	6.5 (5.3 to 7.8)	109	41	586	264
	20.0	72.9 (67.2 to 78.2)	69.0 (61.4 to 75.8)	37.2 (32.0 to 43.0)	91.1 (89.3 to 92.7)	37.2 (32.0 to 43.0)	8.9 (7.3 to 10.7)	146	54	552	248
	25.0	72.9 (67.2 to 78.2)	69.0 (61.4 to 75.8)	44.1 (38.6 to 50.1)	88.4 (86.2 to 90.5)	44.1 (38.6 to 50.1)	11.6 (9.5 to 13.8)	182	68	517	233
	30.0	72.9 (67.2 to 78.2)	69.0 (61.4 to 75.8)	50.3 (44.7 to 56.4)	85.6 (82.9 to 88.1)	50.3 (44.7 to 56.4)	14.4 (11.9 to 17.1)	219	81	483	217
	35.0	72.9 (67.2 to 78.2)	69.0 (61.4 to 75.8)	56.0 (50.4 to 61.9)	82.6 (79.4 to 85.5)	56.0 (50.4 to 61.9)	17.4 (14.5 to 20.6)	255	95	448	202
	40.0	72.9 (67.2 to 78.2)	69.0 (61.4 to 75.8)	61.1 (55.7 to 66.8)	79.3 (75.7 to 82.7)	61.1 (55.7 to 66.8)	20.7 (17.3 to 24.3)	292	108	414	186
	45.0	72.9 (67.2 to 78.2)	69.0 (61.4 to 75.8)	65.9 (60.7 to 71.1)	75.7 (71.7 to 79.6)	65.9 (60.7 to 71.1)	24.3 (20.4 to 28.3)	328	122	379	171

	50.0	72.9 (67.2 to 78.2)	69.0 (61.4 to 75.8)	70.2 (65.3 to 75.1)	71.8 (67.5 to 76.1)	70.2 (65.3 to 75.1)	28.2 (23.9 to 32.5)	365	135	345	155
Clinical-CT-radiomics ML models	5.0	76.5 (71.0 to 81.5)	80.9 (74.7 to 86.0)	17.6 (13.8 to 22.3)	98.5 (98.1 to 98.8)	17.6 (13.8 to 22.3)	1.5 (1.2 to 1.9)	38	12	768	182
	10.0	76.5 (71.0 to 81.5)	80.9 (74.7 to 86.0)	31.1 (25.2 to 37.7)	96.9 (96.2 to 97.5)	31.1 (25.2 to 37.7)	3.1 (2.5 to 3.8)	76	24	728	172
	15.0	76.5 (71.0 to 81.5)	80.9 (74.7 to 86.0)	41.6 (34.8 to 49.0)	95.1 (94.0 to 96.1)	41.6 (34.8 to 49.0)	4.9 (3.9 to 6.0)	115	35	687	163
	20.0	76.5 (71.0 to 81.5)	80.9 (74.7 to 86.0)	50.2 (43.1 to 57.7)	93.2 (91.7 to 94.6)	50.2 (43.1 to 57.7)	6.8 (5.4 to 8.3)	153	47	647	153
	25.0	76.5 (71.0 to 81.5)	80.9 (74.7 to 86.0)	57.3 (50.2 to 64.5)	91.2 (89.3 to 92.9)	57.3 (50.2 to 64.5)	8.8 (7.1 to 10.7)	191	59	606	144
	30.0	76.5 (71.0 to 81.5)	80.9 (74.7 to 86.0)	63.3 (56.5 to 70.0)	88.9 (86.6 to 91.1)	63.3 (56.5 to 70.0)	11.1 (8.9 to 13.4)	229	71	566	134
	35.0	76.5 (71.0 to 81.5)	80.9 (74.7 to 86.0)	68.4 (62.0 to 74.6)	86.5 (83.8 to 89.0)	68.4 (62.0 to 74.6)	13.5 (11.0 to 16.2)	268	82	526	124
	40.0	76.5 (71.0 to 81.5)	80.9 (74.7 to 86.0)	72.8 (66.9 to 78.4)	83.8 (80.6 to 86.8)	72.8 (66.9 to 78.4)	16.2 (13.2 to 19.4)	306	94	485	115
	45.0	76.5 (71.0 to 81.5)	80.9 (74.7 to 86.0)	76.7 (71.3 to 81.7)	80.8 (77.2 to 84.2)	76.7 (71.3 to 81.7)	19.2 (15.8 to 22.8)	344	106	445	105
	50.0	76.5 (71.0 to 81.5)	80.9 (74.7 to 86.0)	80.0 (75.2 to 84.5)	77.5 (73.5 to 81.4)	80.0 (75.2 to 84.5)	22.5 (18.6 to 26.5)	382	118	404	96
CT-based DL-only models	5.0	77.9 (71.9 to 83.2)	65.5 (56.7 to 73.7)	10.8 (8.6 to 13.5)	98.3 (97.7 to 98.7)	10.8 (8.6 to 13.5)	1.7 (1.3 to 2.3)	39	11	622	328
	10.0	77.9 (71.9 to 83.2)	65.5 (56.7 to 73.7)	20.3 (16.6 to 24.8)	96.4 (95.3 to 97.3)	20.3 (16.6 to 24.8)	3.6 (2.7 to 4.7)	78	22	589	311
	15.0	77.9 (71.9 to 83.2)	65.5 (56.7 to 73.7)	28.7 (24.1 to 34.4)	94.4 (92.8 to 95.7)	28.7 (24.1 to 34.4)	5.6 (4.3 to 7.2)	117	33	557	293
	20.0	77.9 (71.9 to 83.2)	65.5 (56.7 to 73.7)	36.3 (31.0 to 42.6)	92.2 (90.1 to 94.0)	36.3 (31.0 to 42.6)	7.8 (6.0 to 9.9)	156	44	524	276
	25.0	77.9 (71.9 to 83.2)	65.5 (56.7 to 73.7)	43.1 (37.4 to 49.7)	89.9 (87.2 to 92.2)	43.1 (37.4 to 49.7)	10.1 (7.8 to 12.8)	195	55	491	259
	30.0	77.9 (71.9 to 83.2)	65.5 (56.7 to 73.7)	49.4 (43.5 to 56.0)	87.4 (84.2 to 90.2)	49.4 (43.5 to 56.0)	12.6 (9.8 to 15.8)	234	66	458	242
	35.0	77.9 (71.9 to 83.2)	65.5 (56.7 to 73.7)	55.0 (49.2 to 61.5)	84.6 (80.9 to 88.0)	55.0 (49.2 to 61.5)	15.4 (12.0 to 19.1)	273	77	426	224
	40.0	77.9 (71.9 to 83.2)	65.5 (56.7 to 73.7)	60.2 (54.5 to 66.4)	81.6 (77.4 to 85.6)	60.2 (54.5 to 66.4)	18.4 (14.4 to 22.6)	312	88	393	207
	45.0	77.9 (71.9 to 83.2)	65.5 (56.7 to 73.7)	65.0 (59.5 to 70.8)	78.4 (73.6 to 82.8)	65.0 (59.5 to 70.8)	21.6 (17.2 to 26.4)	351	99	360	190
	50.0	77.9 (71.9 to 83.2)	65.5 (56.7 to 73.7)	69.4 (64.2 to 74.8)	74.8 (69.5 to 79.8)	69.4 (64.2 to 74.8)	25.2 (20.2 to 30.5)	390	110	327	173

Clinical-CT-DL fusion models	5.0	82.0 (75.3 to 87.6)	83.1 (75.1 to 89.5)	20.9 (14.8 to 29.1)	98.9 (98.5 to 99.2)	20.9 (14.8 to 29.1)	1.1 (0.8 to 1.5)	41	9	789	161
	10.0	82.0 (75.3 to 87.6)	83.1 (75.1 to 89.5)	35.7 (26.8 to 46.4)	97.6 (96.8 to 98.4)	35.7 (26.8 to 46.4)	2.4 (1.6 to 3.2)	82	18	748	152
	15.0	82.0 (75.3 to 87.6)	83.1 (75.1 to 89.5)	46.7 (36.7 to 57.9)	96.3 (95.0 to 97.4)	46.7 (36.7 to 57.9)	3.7 (2.6 to 5.0)	123	27	706	144
	20.0	82.0 (75.3 to 87.6)	83.1 (75.1 to 89.5)	55.3 (45.1 to 66.1)	94.9 (93.0 to 96.4)	55.3 (45.1 to 66.1)	5.1 (3.6 to 7.0)	164	36	665	135
	25.0	82.0 (75.3 to 87.6)	83.1 (75.1 to 89.5)	62.2 (52.3 to 72.2)	93.3 (90.9 to 95.3)	62.2 (52.3 to 72.2)	6.7 (4.7 to 9.1)	205	45	623	127
	30.0	82.0 (75.3 to 87.6)	83.1 (75.1 to 89.5)	67.8 (58.5 to 77.0)	91.5 (88.6 to 94.0)	67.8 (58.5 to 77.0)	8.5 (6.0 to 11.4)	246	54	581	119
	35.0	82.0 (75.3 to 87.6)	83.1 (75.1 to 89.5)	72.5 (63.9 to 80.8)	89.6 (86.1 to 92.6)	72.5 (63.9 to 80.8)	10.4 (7.4 to 13.9)	287	63	540	110
	40.0	82.0 (75.3 to 87.6)	83.1 (75.1 to 89.5)	76.5 (68.7 to 83.9)	87.4 (83.4 to 91.0)	76.5 (68.7 to 83.9)	12.6 (9.0 to 16.6)	328	72	498	102
	45.0	82.0 (75.3 to 87.6)	83.1 (75.1 to 89.5)	80.0 (72.9 to 86.4)	85.0 (80.3 to 89.1)	80.0 (72.9 to 86.4)	15.0 (10.9 to 19.7)	369	81	457	93
	50.0	82.0 (75.3 to 87.6)	83.1 (75.1 to 89.5)	83.0 (76.7 to 88.6)	82.3 (77.0 to 87.0)	83.0 (76.7 to 88.6)	17.7 (13.0 to 23.0)	410	90	415	85

CrI, credible interval; PPV, positive predictive value; NPV, negative predictive value. Positive post-test probability equals PPV. Negative post-test probability is the probability of disease after a negative test, calculated as 1 minus NPV. Absolute results are standardized to 1,000 tested persons at each assumed prevalence.

Table S19. Sensitivity analyses of diagnostic-node rankings by outcome

Diagnostic node	90 days	6 months	Discharge	External validation studies only	Excluding outlier studies	mRS >3 studies only	Excluding high-risk studies	Excluding overlapping cohorts
Sensitivity								
Conventional clinical-CT models	#4 (28.0%)	#4 (13.0%)	#5 (10.0%)	#3 (57.0%)	#4 (25.0%)	#4 (26.0%)	#3 (59.0%)	#4 (20.4%)
CT-based radiomics-only ML models	#5 (8.0%)	#3 (36.0%)	#2 (65.0%)	#5 (2.0%)	#5 (7.0%)	#5 (8.0%)	#5 (26.0%)	#5 (13.5%)
Clinical-CT-radiomics ML models	#3 (53.0%)	#2 (51.0%)	#4 (47.0%)	#1 (79.0%)	#3 (56.0%)	#3 (55.0%)	#1 (61.0%)	#3 (51.7%)
CT-based DL-only models	#2 (66.0%)	#1 (100.0%)	#1 (67.0%)	#4 (48.0%)	#2 (66.0%)	#2 (66.0%)	#4 (43.0%)	#2 (68.5%)
Clinical-CT-DL fusion models	#1 (96.0%)	—	#3 (61.0%)	#2 (64.0%)	#1 (96.0%)	#1 (95.0%)	#1 (61.0%)	#1 (96.0%)
Specificity								
Conventional clinical-CT models	#3 (54.0%)	#2 (79.0%)	#4 (34.0%)	#4 (36.0%)	#3 (54.0%)	#3 (57.0%)	#2 (70.0%)	#3 (55.0%)
CT-based radiomics-only ML models	#4 (22.0%)	#3 (34.0%)	#3 (39.0%)	#2 (57.0%)	#4 (22.0%)	#4 (21.0%)	#4 (23.0%)	#4 (21.6%)
Clinical-CT-radiomics ML models	#2 (80.0%)	#1 (86.0%)	#1 (91.0%)	#3 (53.0%)	#2 (80.0%)	#2 (82.0%)	#1 (80.0%)	#2 (80.1%)
CT-based DL-only models	#5 (3.0%)	#4 (1.0%)	#5 (19.0%)	#5 (6.0%)	#5 (3.0%)	#5 (4.0%)	#5 (7.0%)	#5 (3.4%)
Clinical-CT-DL fusion models	#1 (91.0%)	—	#2 (67.0%)	#1 (98.0%)	#1 (91.0%)	#1 (85.0%)	#2 (70.0%)	#1 (89.9%)
Diagnostic odds ratio (DOR)								
Conventional clinical-CT models	#3 (51.0%)	#3 (42.0%)	#5 (10.0%)	#3 (49.0%)	#3 (50.0%)	#3 (51.0%)	#3 (69.0%)	#3 (50.8%)
CT-based radiomics-only ML models	#5 (7.0%)	#4 (17.0%)	#3 (50.0%)	#5 (11.0%)	#5 (7.0%)	#5 (6.0%)	#4 (21.0%)	#5 (6.7%)
Clinical-CT-radiomics ML models	#2 (75.0%)	#1 (82.0%)	#1 (84.0%)	#2 (72.0%)	#2 (76.0%)	#2 (76.0%)	#1 (78.0%)	#2 (75.5%)
CT-based DL-only models	#4 (19.0%)	#2 (59.0%)	#4 (38.0%)	#4 (22.0%)	#4 (19.0%)	#4 (20.0%)	#5 (10.0%)	#4 (18.9%)
Clinical-CT-DL fusion models	#1 (98.0%)	—	#2 (68.0%)	#1 (95.0%)	#1 (98.0%)	#1 (97.0%)	#2 (72.0%)	#1 (98.1%)

Abbreviations: DOR, diagnostic odds ratio; PROBAST+AI, Prediction model Risk of Bias Assessment Tool for artificial intelligence; SUCRA, surface under the cumulative ranking curve.

Note: Rank #1 indicates the highest SUCRA value, and values in parentheses represent SUCRA values (%). An em dash (—) indicates that no SUCRA rank was estimable for the corresponding diagnostic node in the respective sensitivity analysis. For sensitivity analyses, “outlier studies” were defined as studies identified as potential outliers using the scale-mixture weight criterion ($w < 0.75$). “High-risk studies” were defined as studies rated as high risk in at least one of the final two columns of the PROBAST+AI model evaluation (validation-set) assessment table. For the overlapping-cohort sensitivity analysis, one of two companion reports of the same cohort was excluded (Huang et al., 2022), and only one of the three external validation cohorts of Liu et al. (2026) was retained (Group 2).

PART II. SUPPLEMENTARY FIGURES

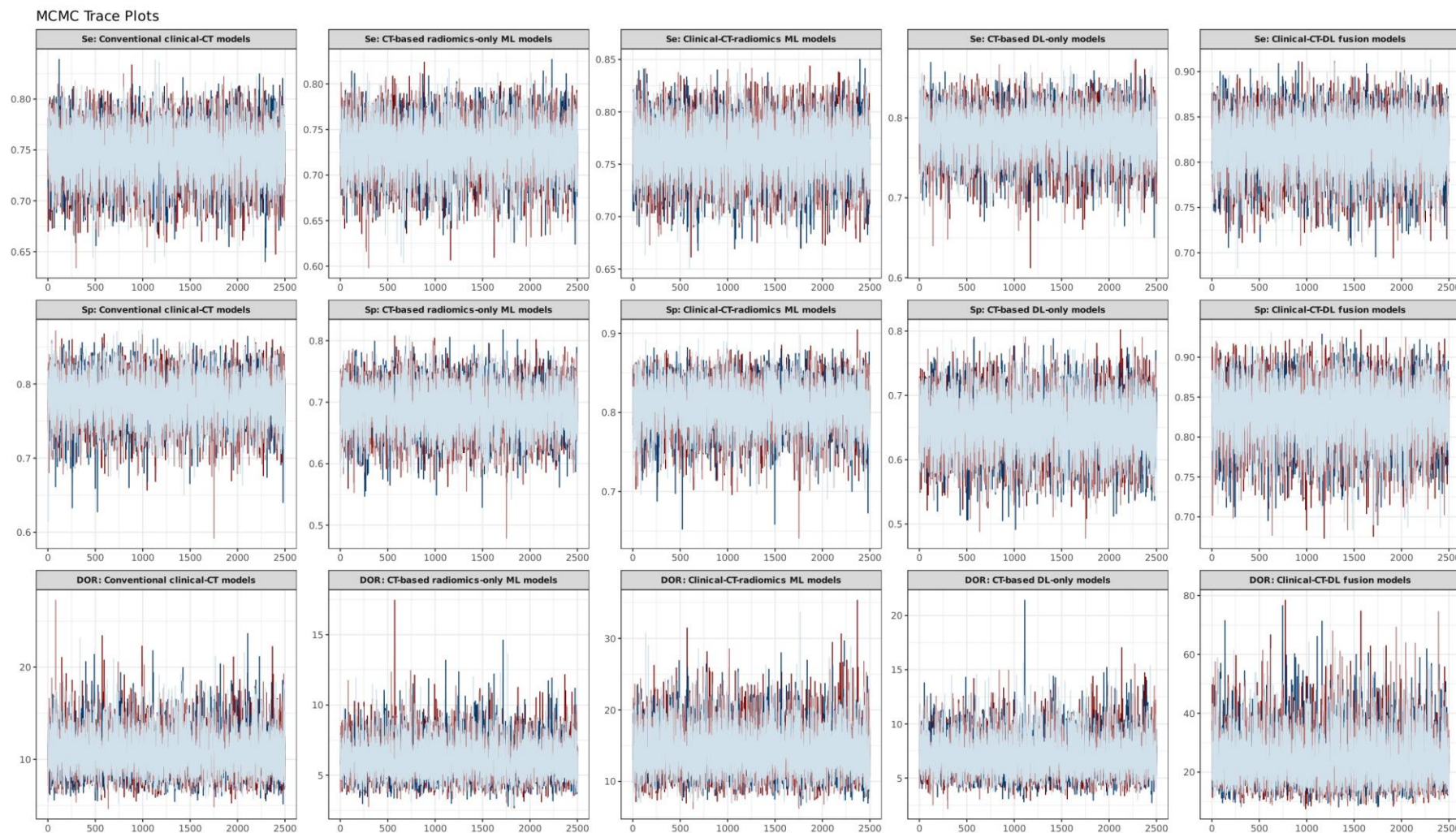


Figure S1. Trace plots of the Markov chain Monte Carlo simulations for the 90-day analysis.

Note: Trace plots of the Markov chain Monte Carlo simulations for sensitivity, specificity, and diagnostic odds ratio (DOR) across the five model categories (90-day analysis). Red and blue lines represent two parallel chains per parameter.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; MCMC, Markov chain Monte Carlo; ML, machine learning.

Posterior Density — Sensitivity & Specificity

Filled curves = posterior density (wider = more uncertainty) dashed lines = posterior means

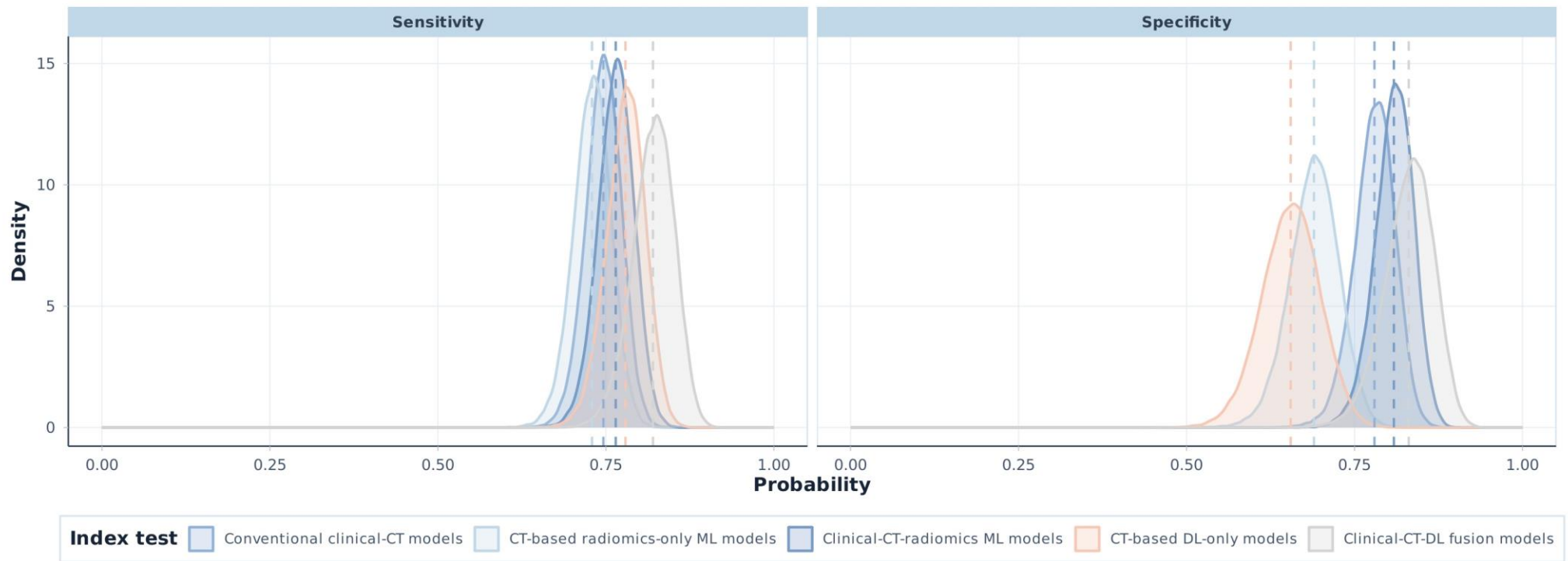


Figure S2. Posterior density plots of sensitivity and specificity for the evaluated model categories (90-day analysis).

Note: Posterior density plots of sensitivity and specificity for the five model categories (90-day analysis). Dashed vertical lines indicate posterior means.

Abbreviations: CT, computed tomography; DL, deep learning; ML, machine learning.

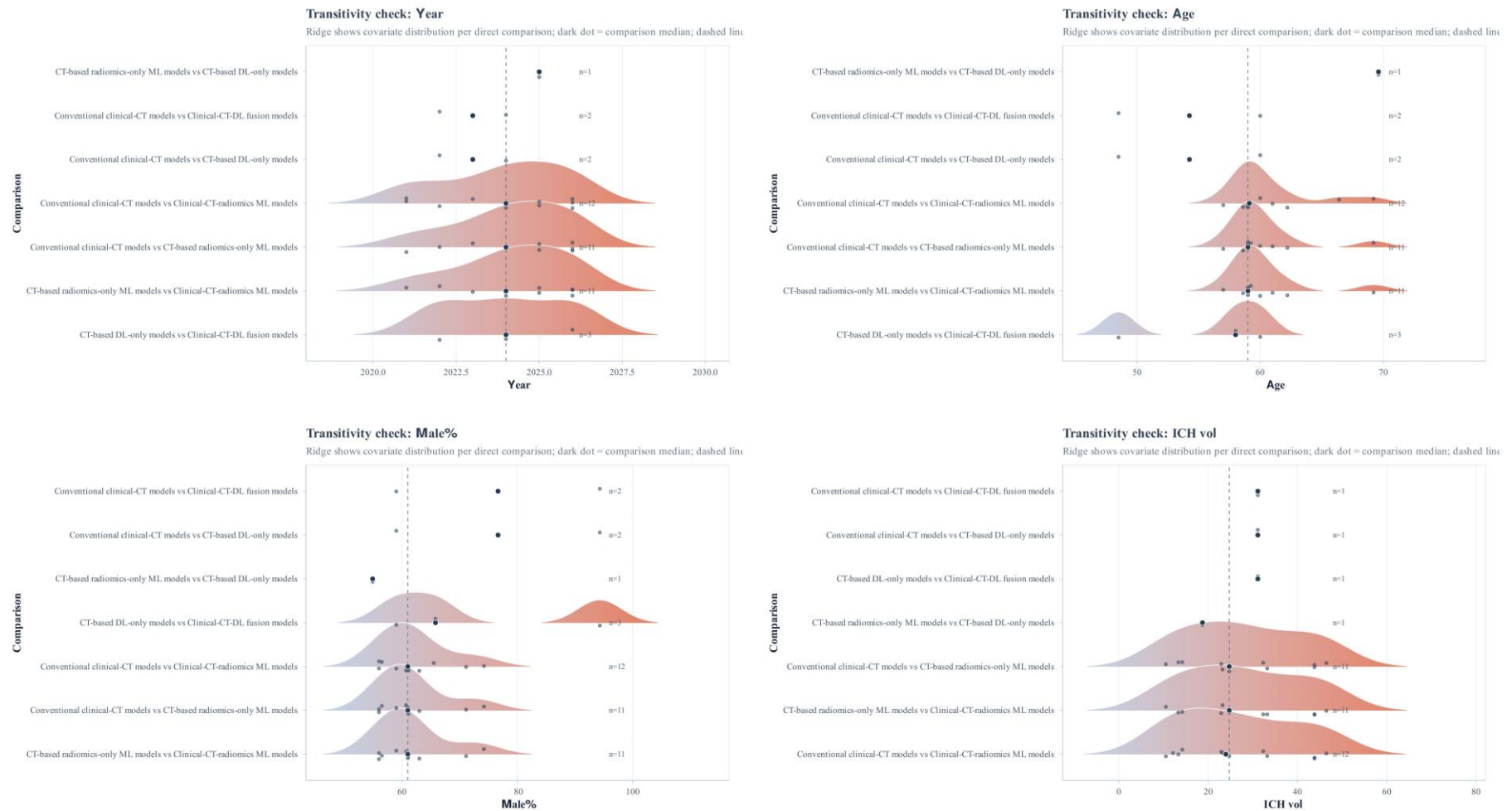


Figure S3A. Transitivity checks across direct comparisons for continuous study-level covariates (publication year, age, male proportion, and ICH volume).

Note: Ridge plots show the distribution of continuous study-level covariates within each direct comparison. Blue and red ridges represent the two compared model categories; dots represent individual studies; dark vertical lines indicate overall medians.

Abbreviations: CT, computed tomography; DL, deep learning; ICH, intracerebral hemorrhage; ML, machine learning.

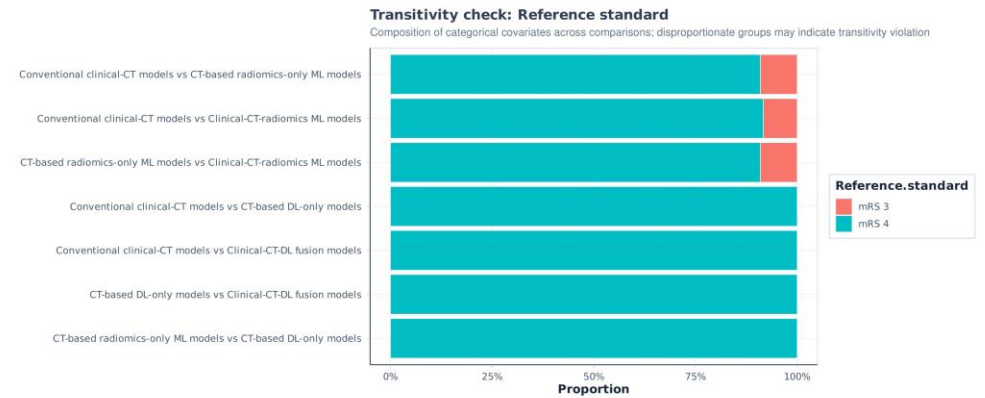
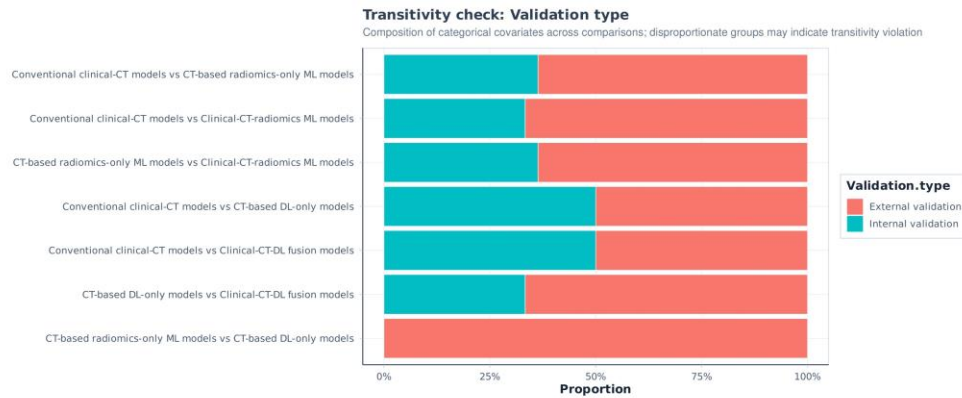
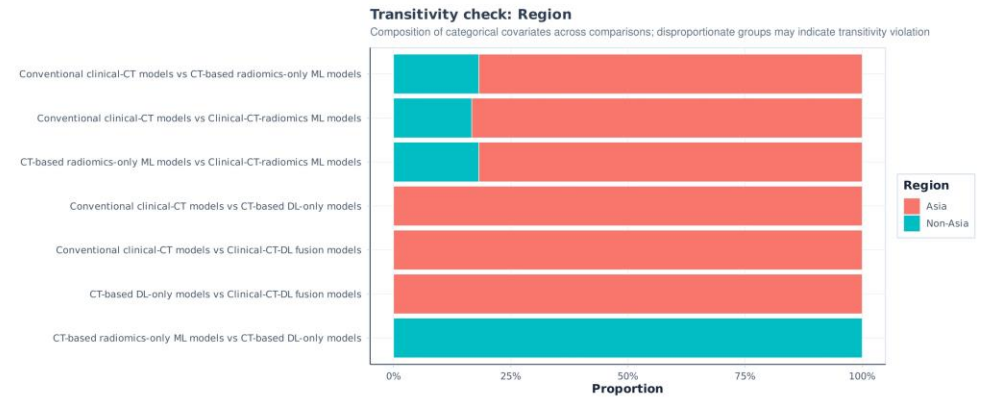
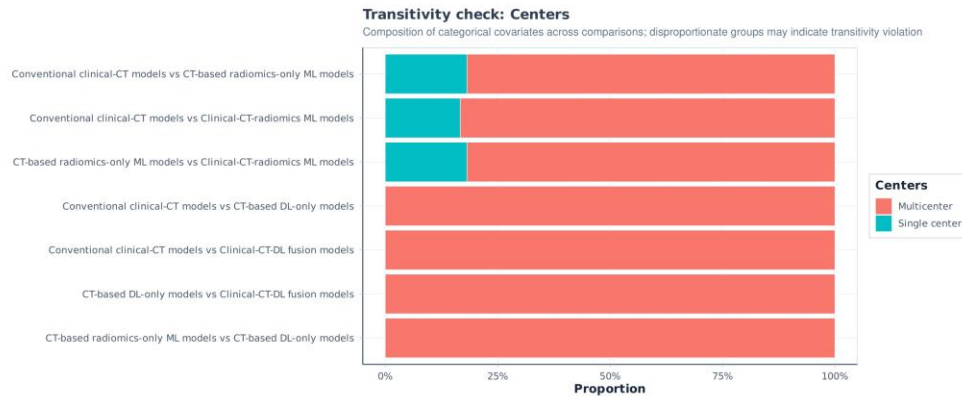


Figure S3B. Transitivity checks across direct comparisons for categorical study-level covariates (center, region, validation type, and reference standard).

Note: Stacked bar charts show the composition of categorical study-level covariates within each direct comparison. Disproportionate category distributions may indicate potential transitivity violations.

Abbreviations: CT, computed tomography; DL, deep learning; ML, machine learning.

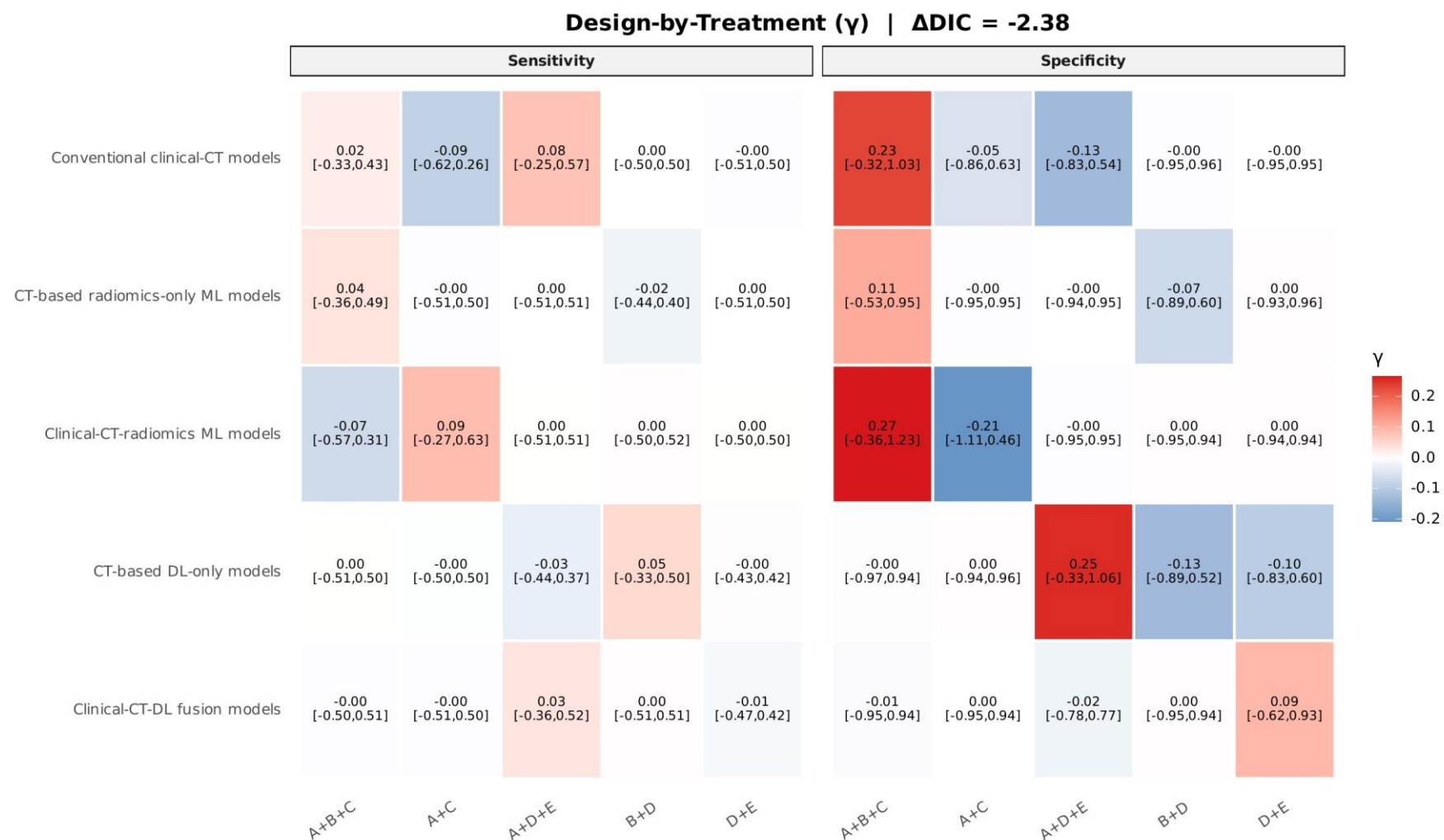


Figure S4. Design-by-treatment interaction assessment of global inconsistency (Δ DIC) for sensitivity and specificity.

Note: Heatmap of the design-by-treatment interaction model assessing global inconsistency for sensitivity and specificity. Cells show the interaction parameter (γ) with 95% credible intervals for each comparison loop; Δ DIC is shown in the figure.

Abbreviations: CrI, credible interval; CT, computed tomography; DIC, deviance information criterion; DL, deep learning; ML, machine learning.

Node Splitting Consistency Test

Overlapping CrIs = consistent; non-overlapping = potential inconsistency

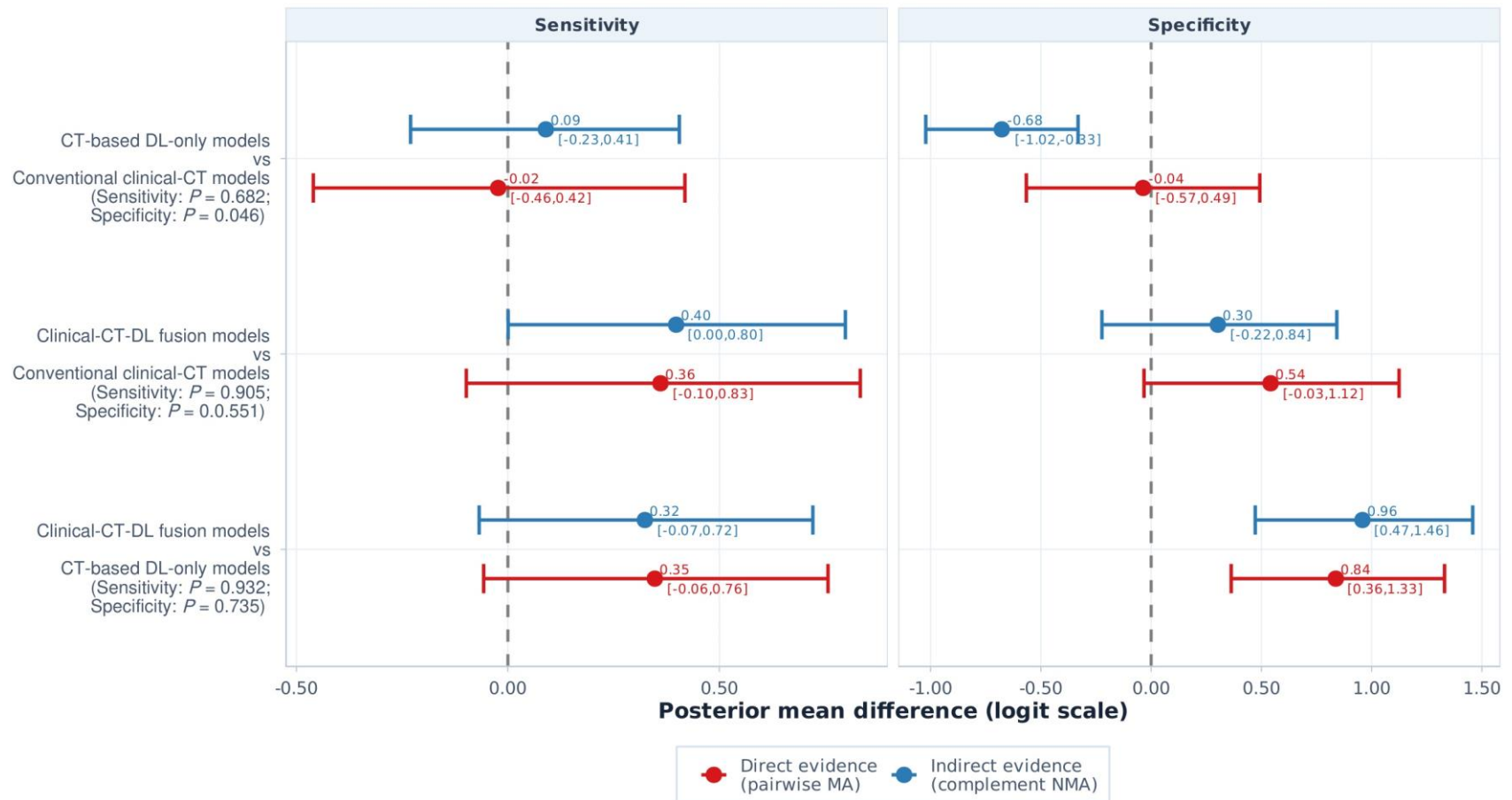


Figure S5. Node-splitting consistency tests for sensitivity and specificity (90-day analysis).

Note: Node-splitting consistency tests comparing direct and indirect evidence for sensitivity and specificity. Overlapping credible intervals indicate consistency; P values are annotated for each comparison.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; ML, machine learning.

Conventional clinical-CT models - per-study estimates & 95% CI

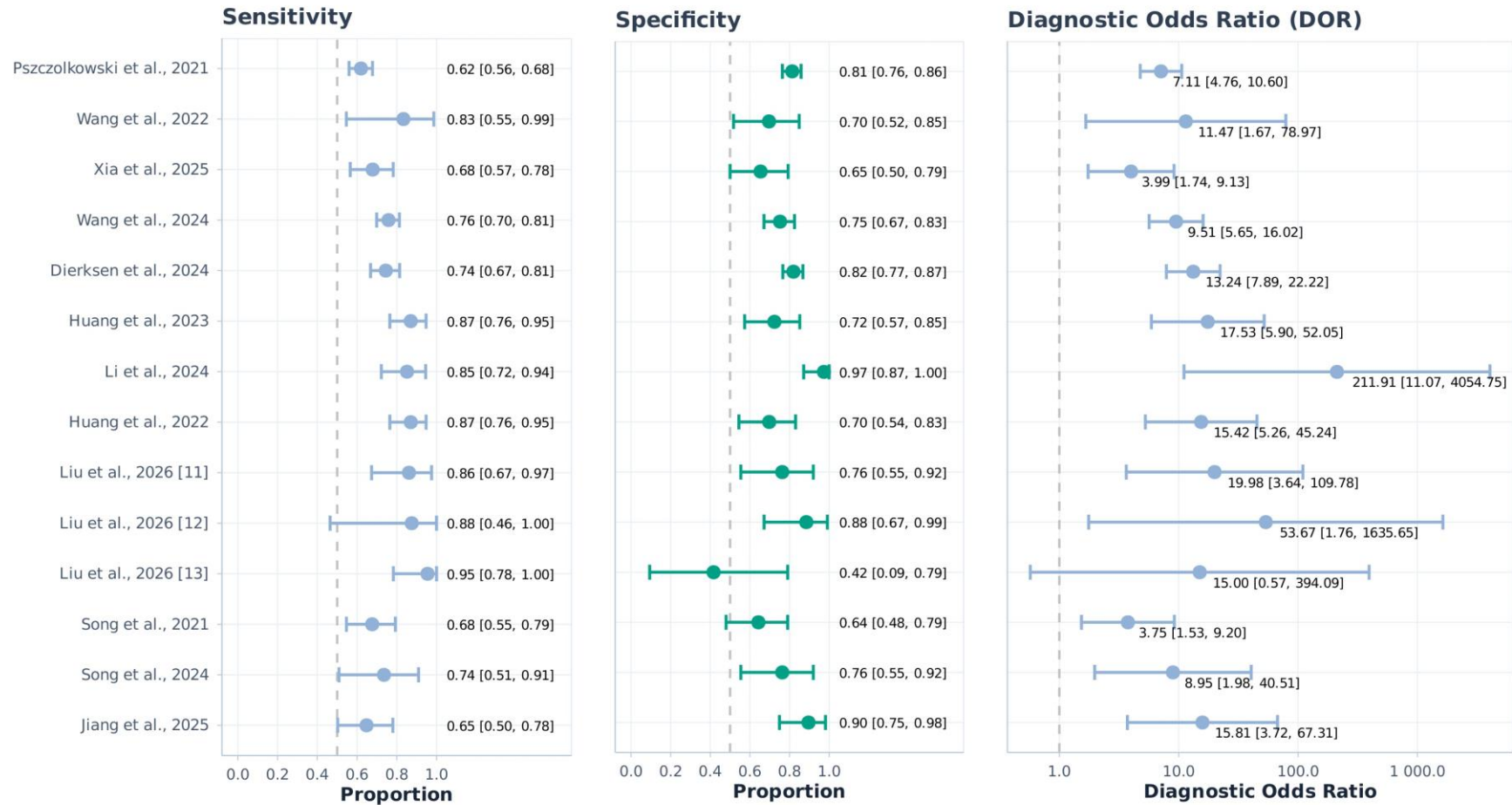


Figure S6A. Forest plots of individual studies for the conventional clinical-CT models (90-day analysis).

Note: Forest plots of per-study sensitivity, specificity, and diagnostic odds ratio (DOR) for the conventional clinical-CT models, with 95% confidence intervals (90-day analysis).

Abbreviations: CI, confidence interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

CT-based radiomics-only ML models - per-study estimates & 95% CI

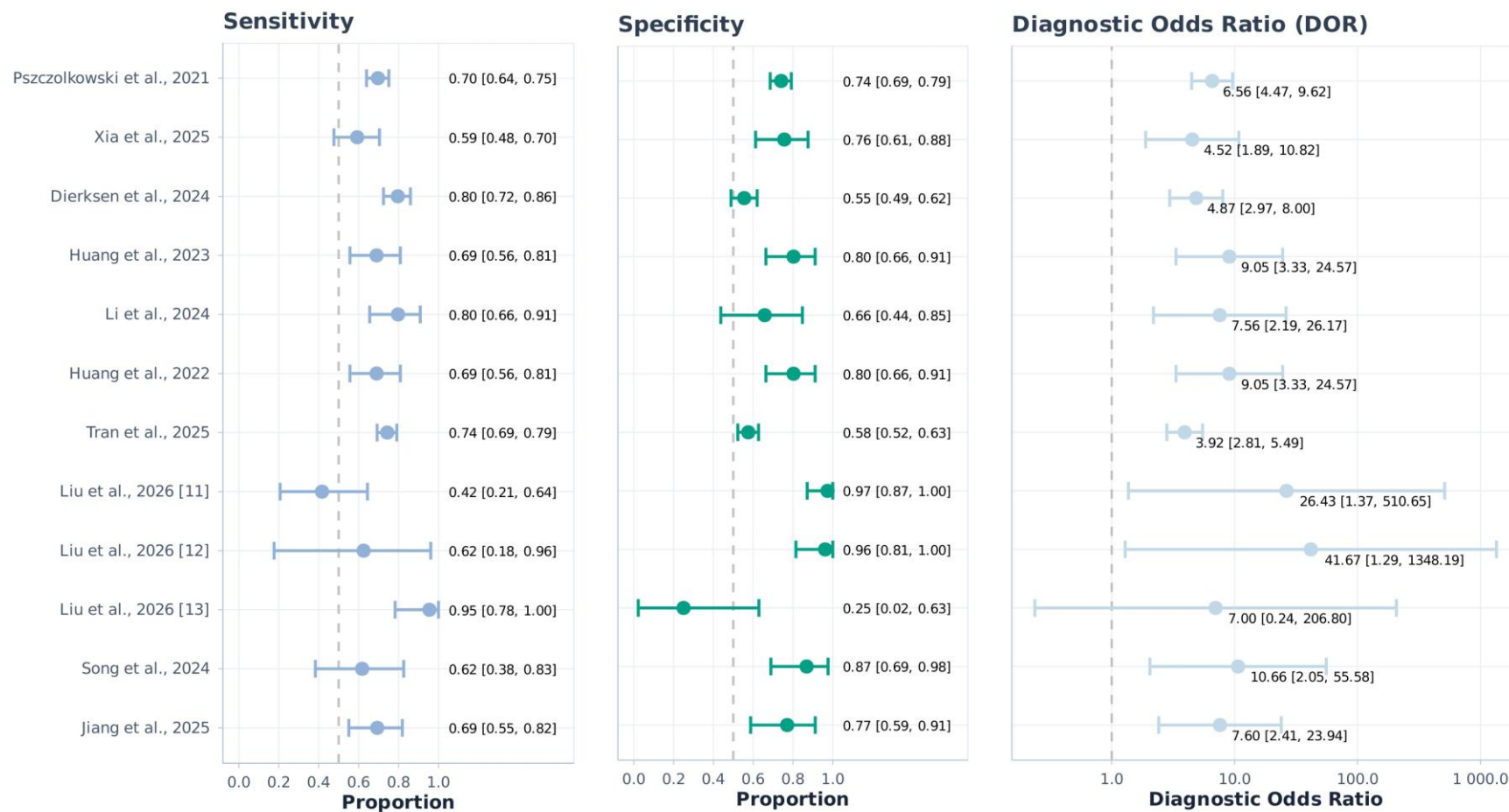


Figure S6B. Forest plots of individual studies for the CT-based radiomics-only ML models (90-day analysis).

Note: Forest plots of per-study sensitivity, specificity, and diagnostic odds ratio (DOR) for the CT-based radiomics-only ML models, with 95% confidence intervals (90-day analysis).

Abbreviations: CI, confidence interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

Clinical-CT-radiomics ML models - per-study estimates & 95% CI

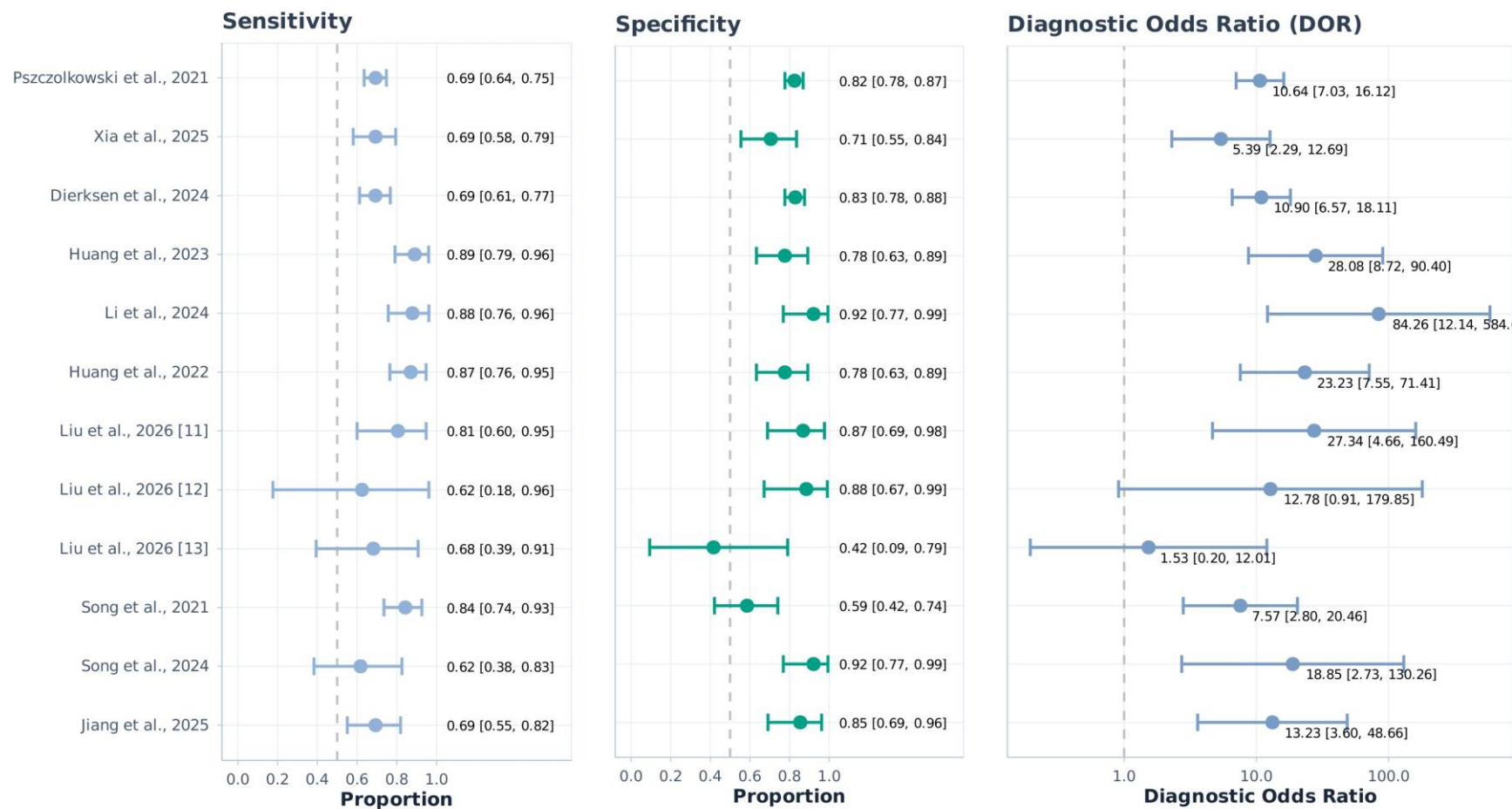


Figure S6C. Forest plots of individual studies for the clinical-CT-radiomics ML models (90-day analysis).

Note: Forest plots of per-study sensitivity, specificity, and diagnostic odds ratio (DOR) for the clinical-CT-radiomics ML models, with 95% confidence intervals (90-day analysis).

Abbreviations: CI, confidence interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

CT-based DL-only models - per-study estimates & 95% CI

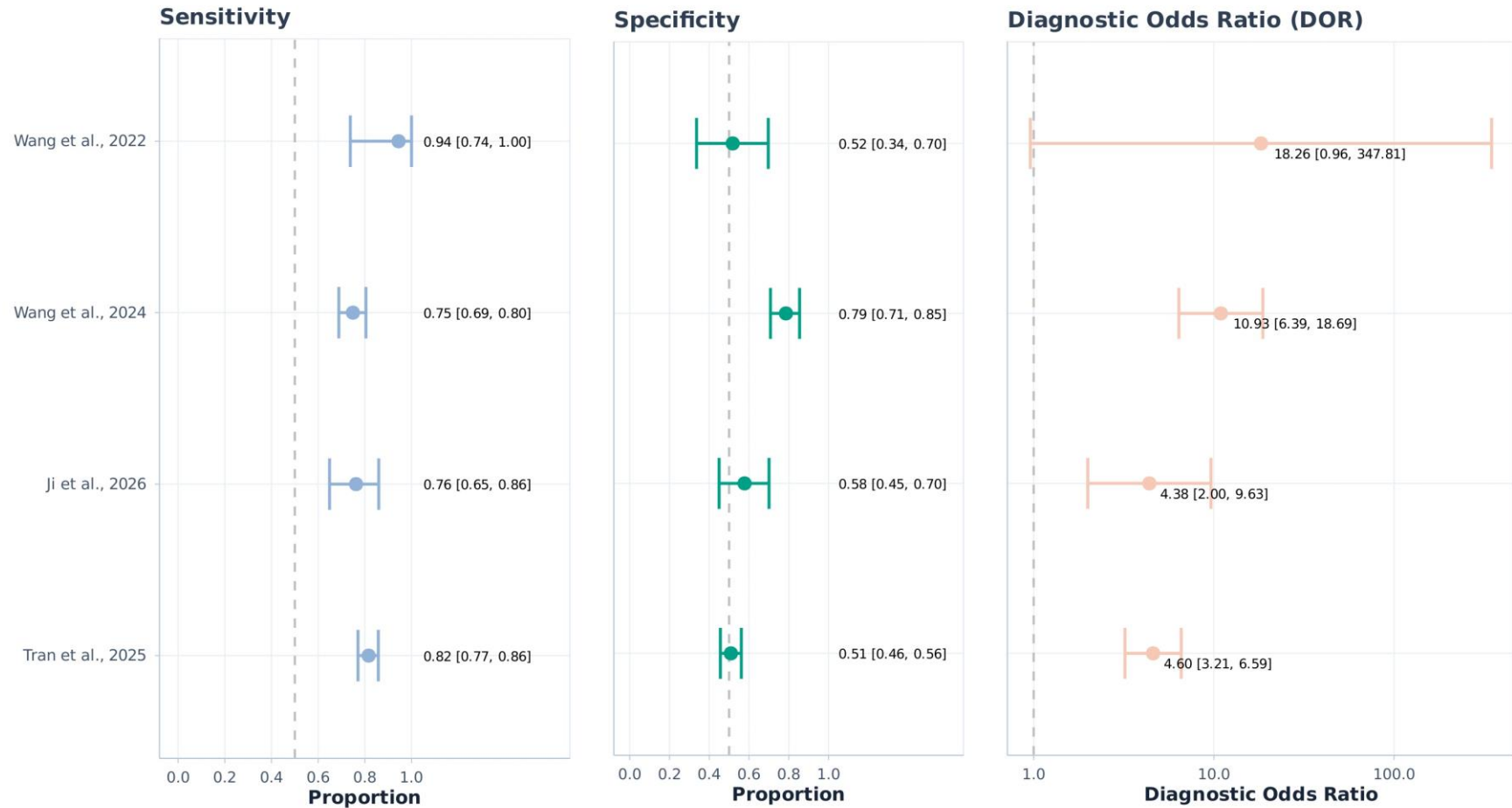


Figure S6D. Forest plots of individual studies for the CT-based DL-only models (90-day analysis).

Note: Forest plots of per-study sensitivity, specificity, and diagnostic odds ratio (DOR) for the CT-based DL-only models, with 95% confidence intervals (90-day analysis).

Abbreviations: CI, confidence interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

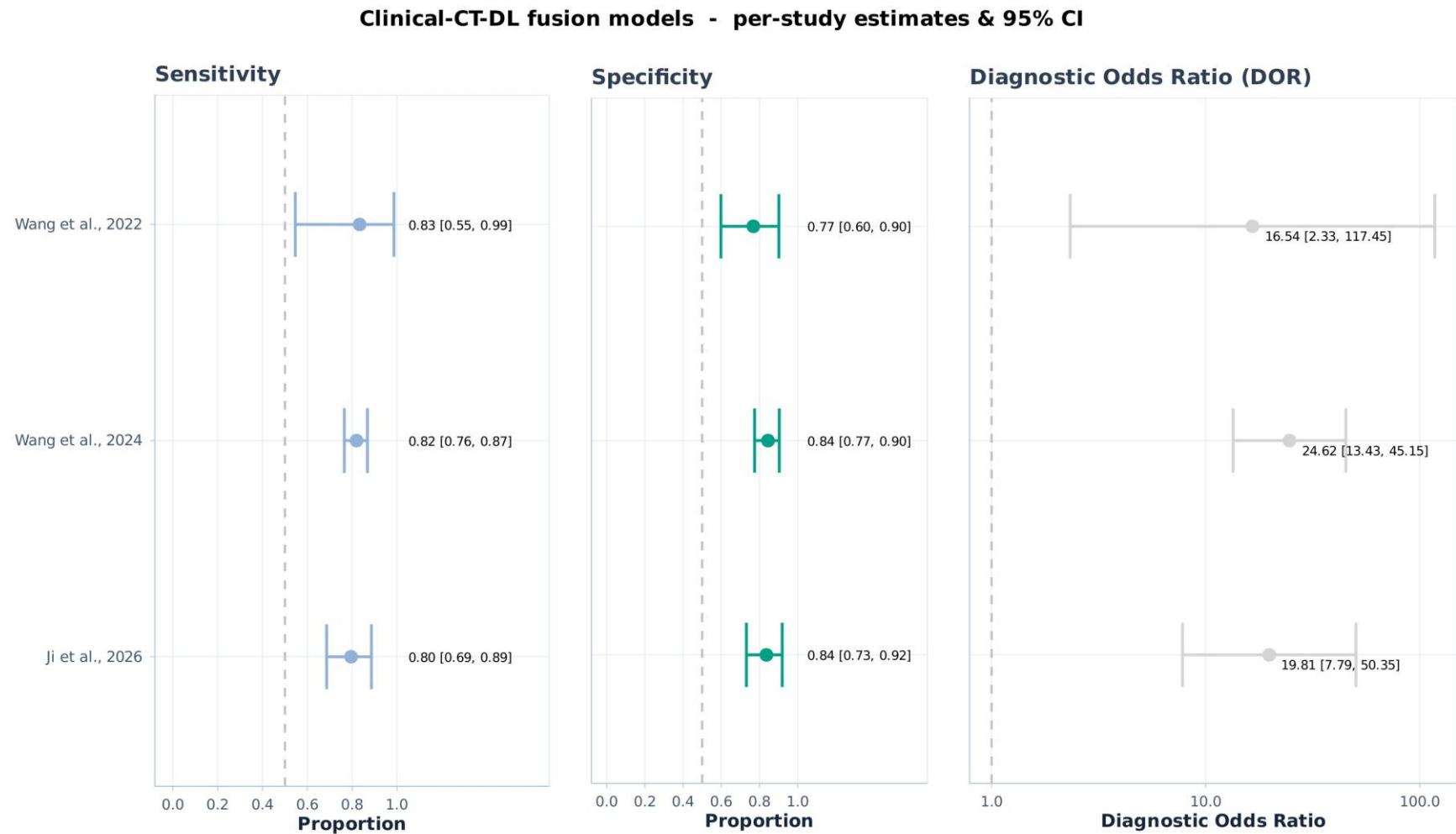


Figure S6E. Forest plots of individual studies for the clinical-CT-DL fusion models (90-day analysis).

Note: Forest plots of per-study sensitivity, specificity, and diagnostic odds ratio (DOR) for the clinical-CT-DL fusion models, with 95% confidence intervals (90-day analysis).

Abbreviations: CI, confidence interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

Direct Evidence Contribution

Bar = n studies; n = total sample size; +indirect = indirect path available

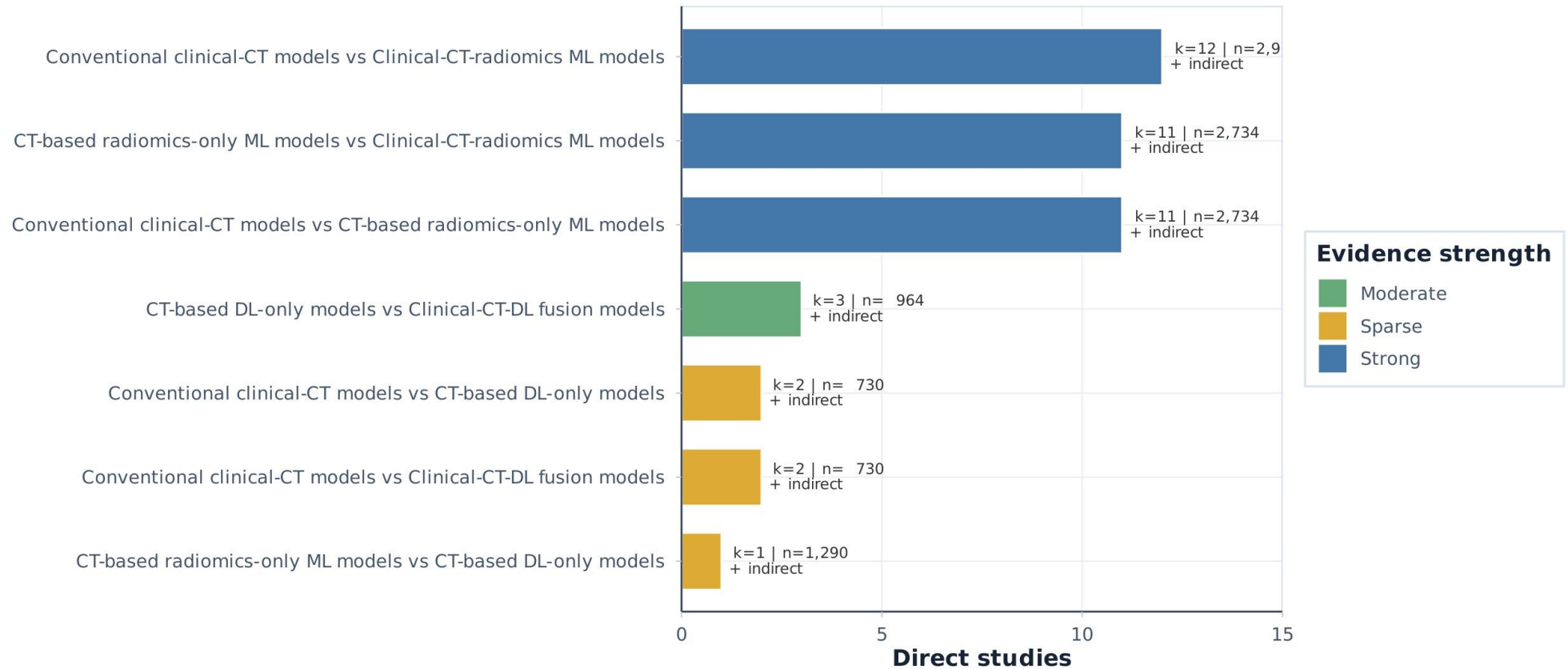


Figure S7. Direct evidence contribution of the network meta-analysis (90-day analysis).

Note: Direct evidence contribution of the network meta-analysis. Bars show the number of direct studies (k) per pairwise comparison; colors indicate evidence strength (strong, moderate, or sparse); '+' indicates an available indirect path.

Abbreviations: CT, computed tomography; DL, deep learning; ML, machine learning.

Pairwise Meta-Analysis: Direct Comparisons

B - A > 0 = B superior; red = 95% CrI excludes 0

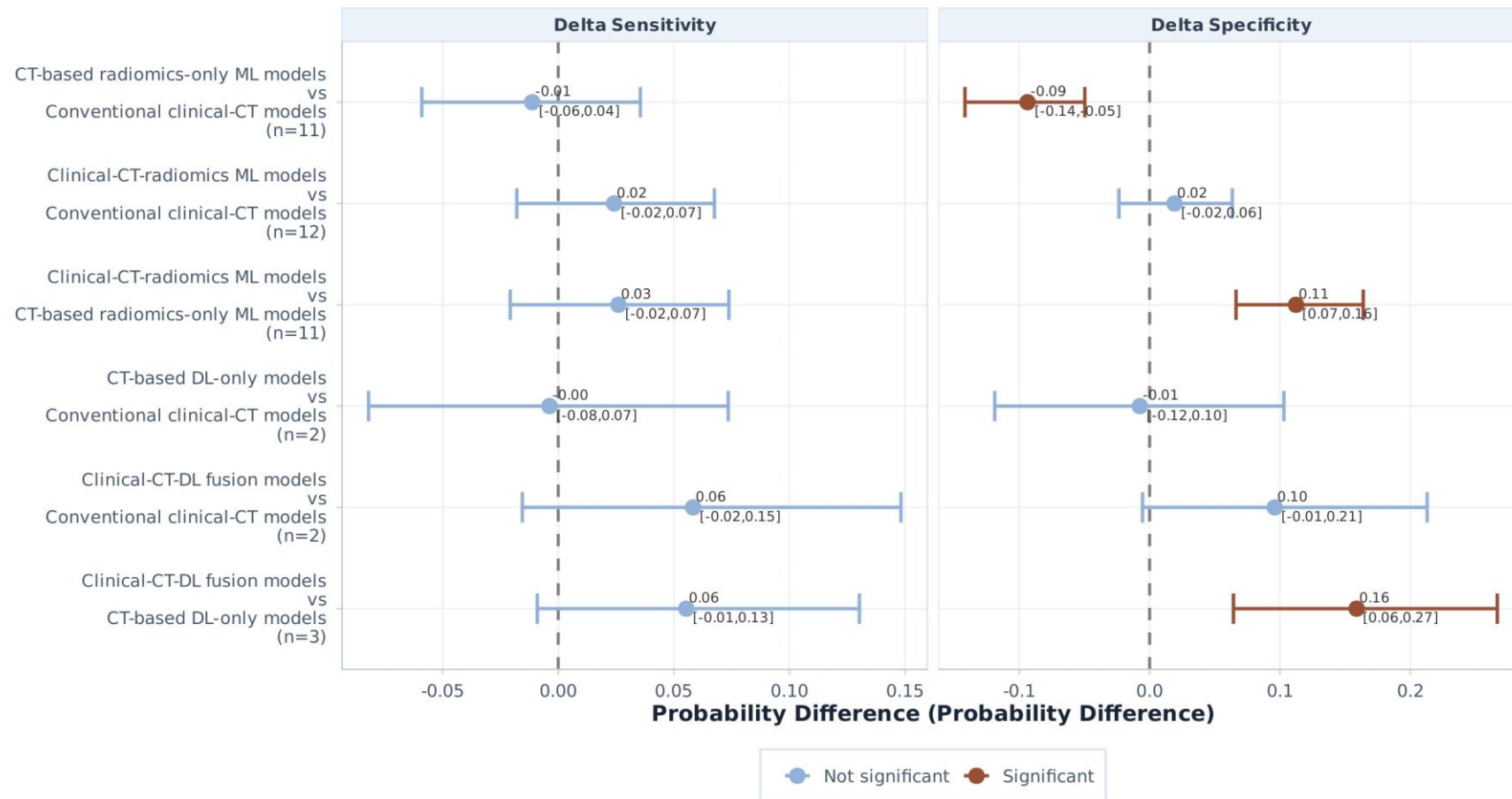


Figure S8. Pairwise meta-analysis of direct comparisons for sensitivity and specificity (90-day analysis).

Note: Pairwise meta-analysis of direct comparisons. Panels show probability differences (Δ) in sensitivity and specificity between paired model categories; red intervals indicate 95% credible intervals excluding zero (B - A > 0 indicates that B is superior to A).

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; ML, machine learning.

Study Outlier Diagnostics (Scale Mixture Weights)

$w < 0.75$ = potential outlier

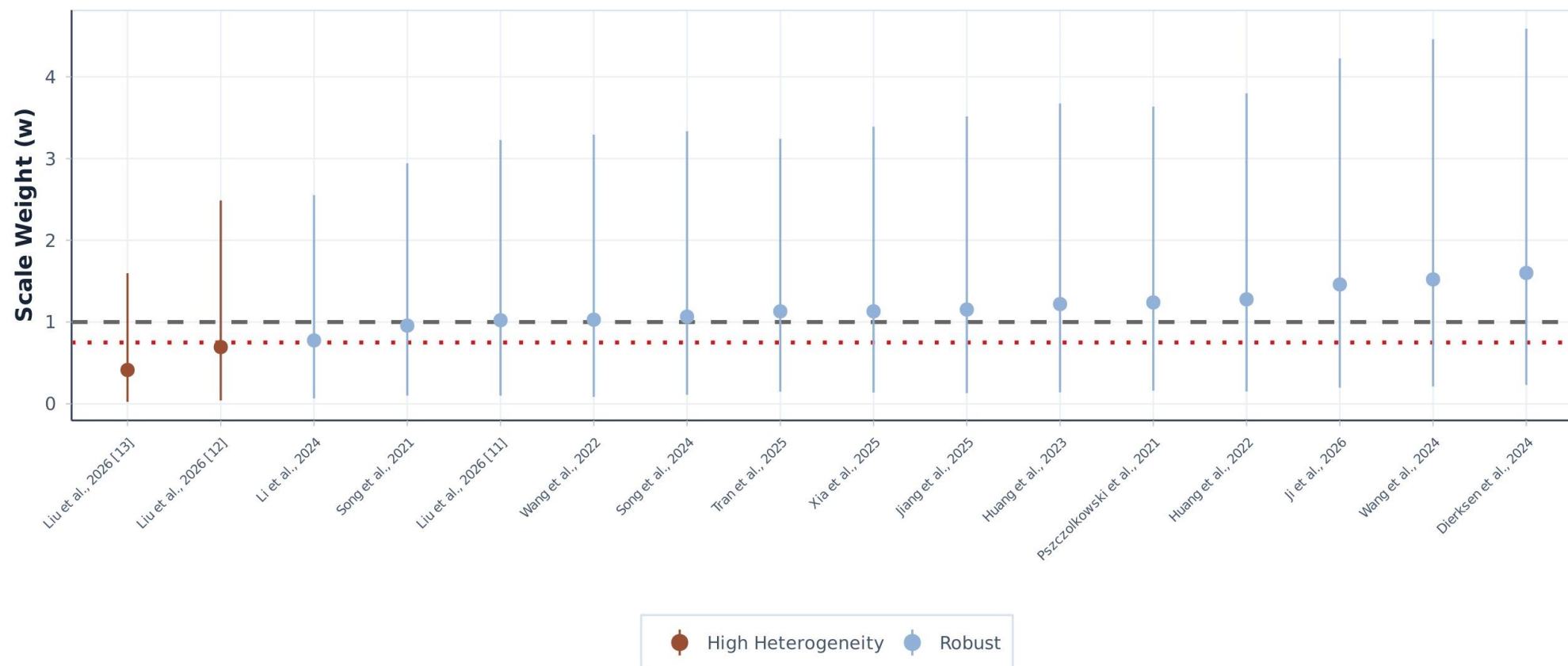


Figure S9. Outlier diagnostics assessment for the included studies (90-day analysis).

Note: Study outlier diagnostics based on scale-mixture weights. Studies with a scale weight $w < 0.75$ were considered potential outliers (orange = high heterogeneity; blue = robust).

Abbreviations: CT, computed tomography; DL, deep learning; ML, machine learning.

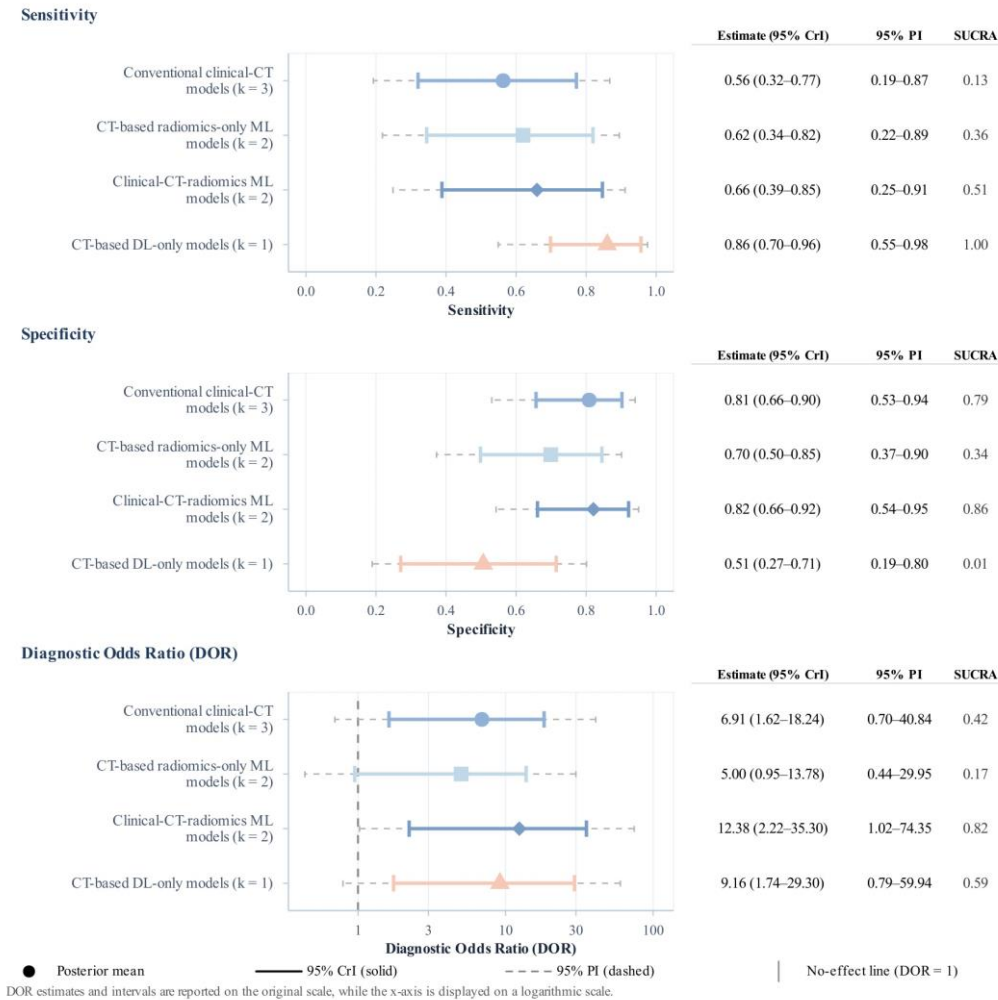


Figure S10A. Forest plots of pooled diagnostic performance for poor functional outcome at 6 months.

Note: Forest plots of pooled sensitivity, specificity, and diagnostic odds ratio (DOR) for poor functional outcome at 6 months. Points show posterior means with 95% credible intervals and prediction intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; PI, prediction interval; SUCRA, surface under the cumulative ranking curve.

SUCRA Evidence Profile

Darker = higher SUCRA; interpret with rank probabilities and GRADE certainty

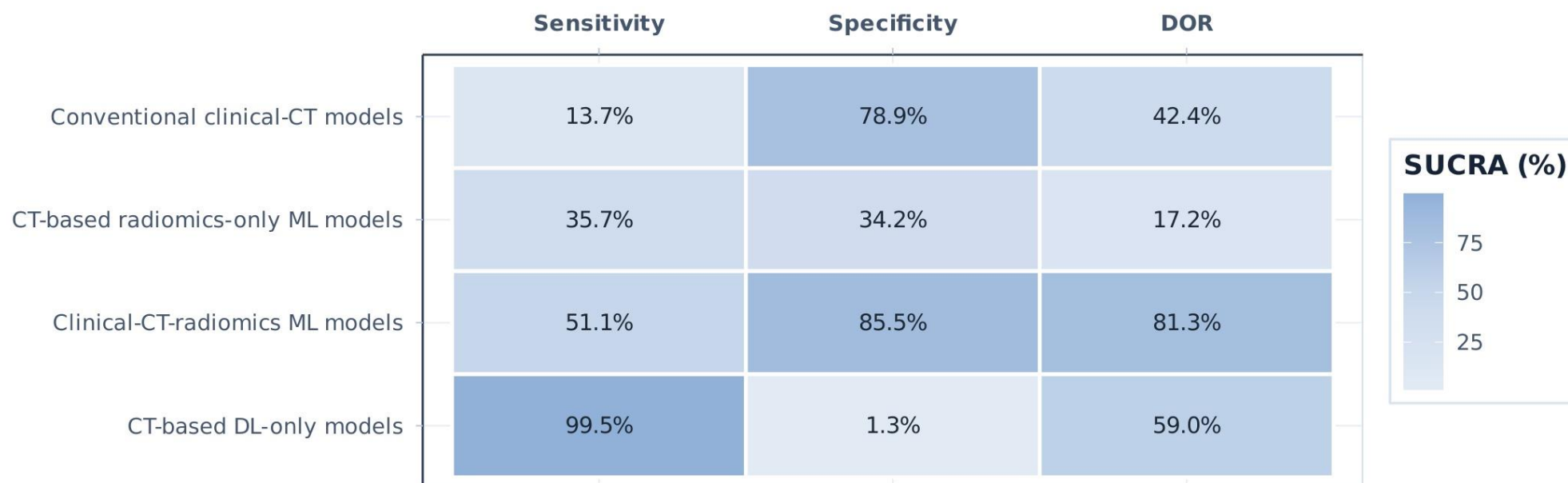


Figure S10B. SUCRA heatmap for poor functional outcome at 6 months.

Note: SUCRA heatmap for poor functional outcome at 6 months. Darker cells indicate higher SUCRA values; interpret alongside rank probabilities and GRADE certainty.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; SUCRA, surface under the cumulative ranking curve.

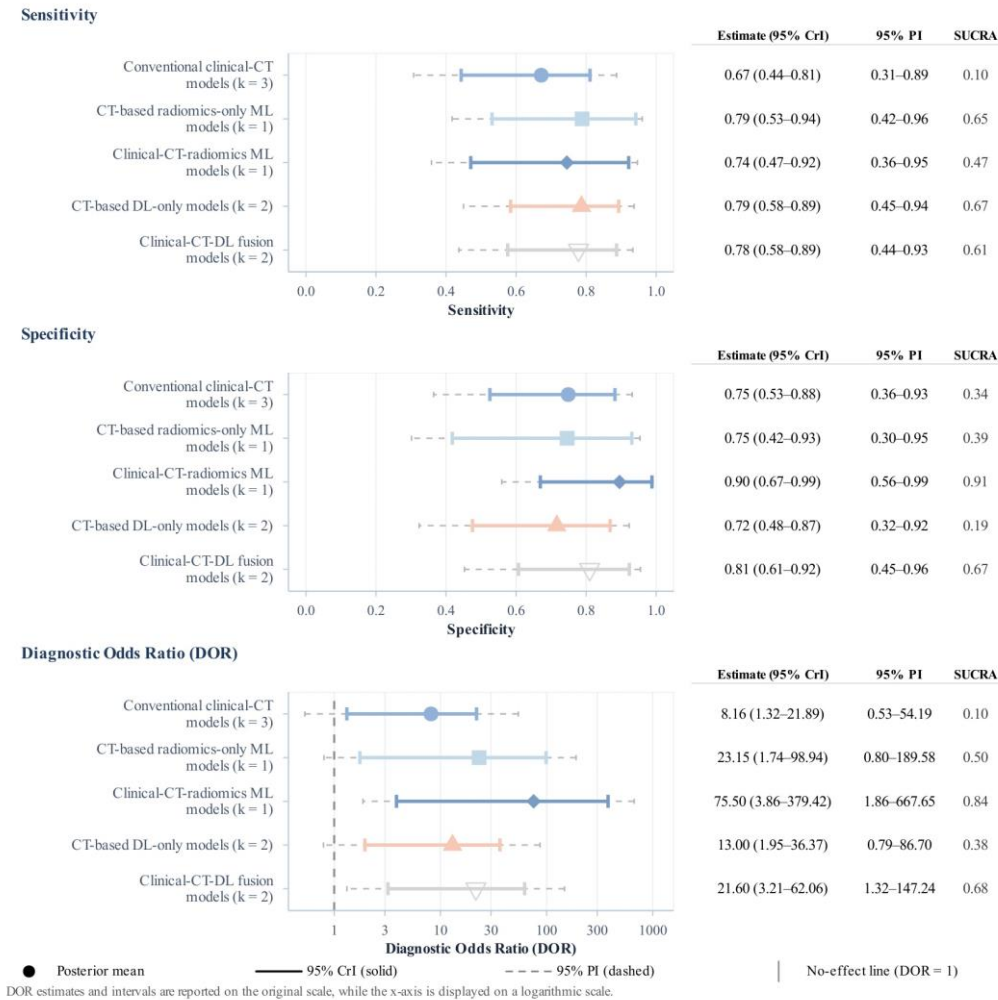


Figure S11A. Forest plots of pooled diagnostic performance for poor functional outcome at discharge.

Note: Forest plots of pooled sensitivity, specificity, and diagnostic odds ratio (DOR) for poor functional outcome at discharge. Points show posterior means with 95% credible intervals and prediction intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; PI, prediction interval; SUCRA, surface under the cumulative ranking curve.

SUCRA Evidence Profile

Darker = higher SUCRA; interpret with rank probabilities and GRADE certainty

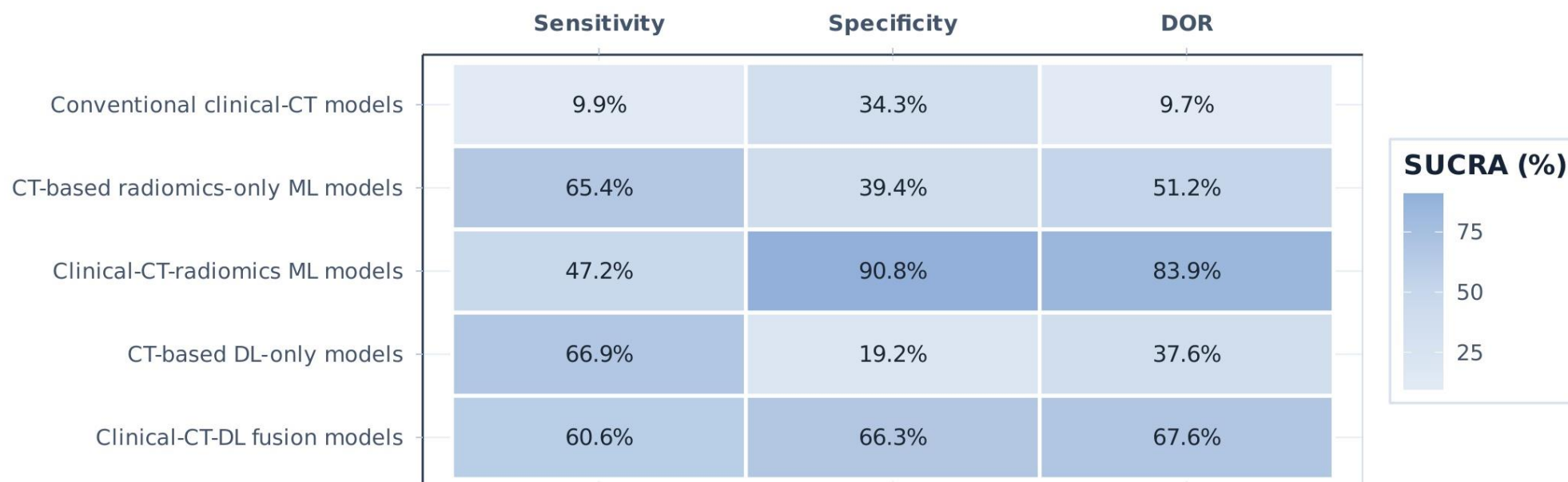


Figure S11B. SUCRA heatmap for poor functional outcome at discharge.

Note: SUCRA heatmap for poor functional outcome at discharge. Darker cells indicate higher SUCRA values; interpret alongside rank probabilities and GRADE certainty.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; SUCRA, surface under the cumulative ranking curve.

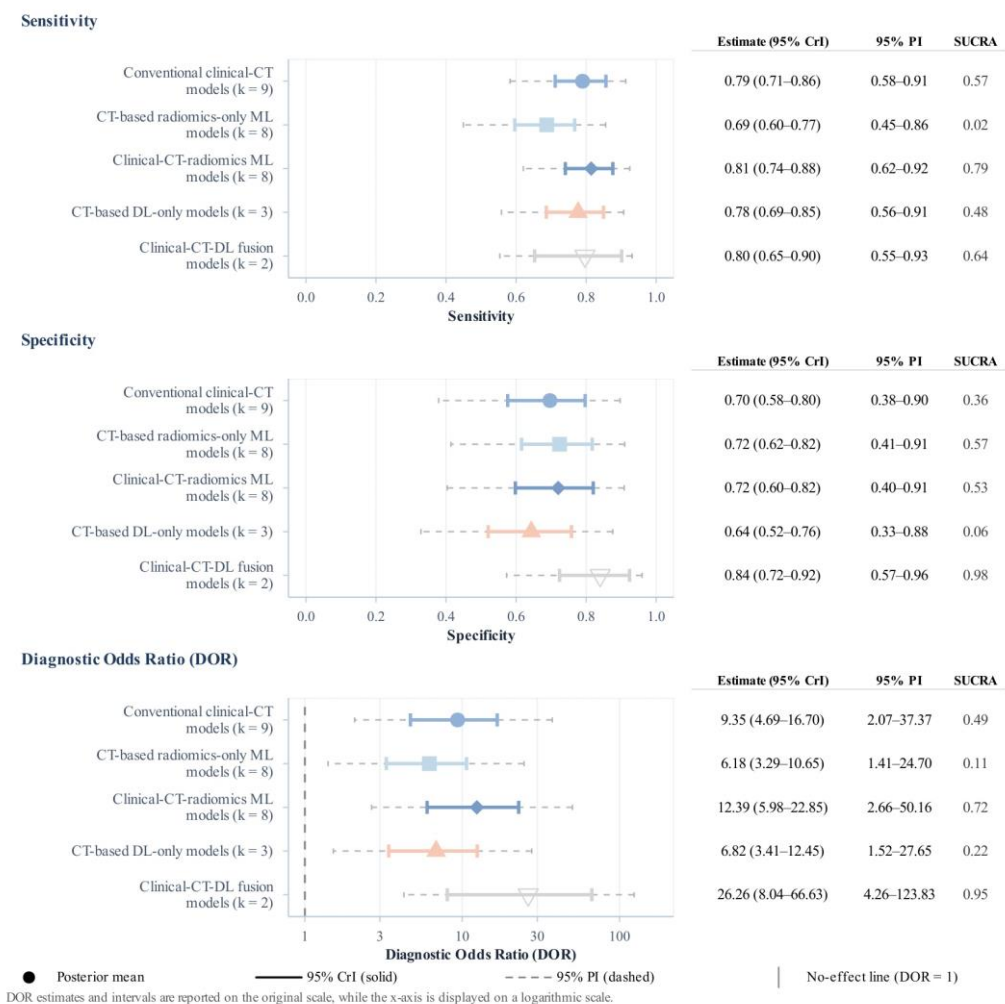


Figure S12A. Sensitivity analysis of pooled diagnostic performance: external validation studies only.

Note: Forest plots of pooled diagnostic performance in the sensitivity analysis restricted to external validation studies. Points show posterior means with 95% credible intervals and prediction intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; PI, prediction interval.

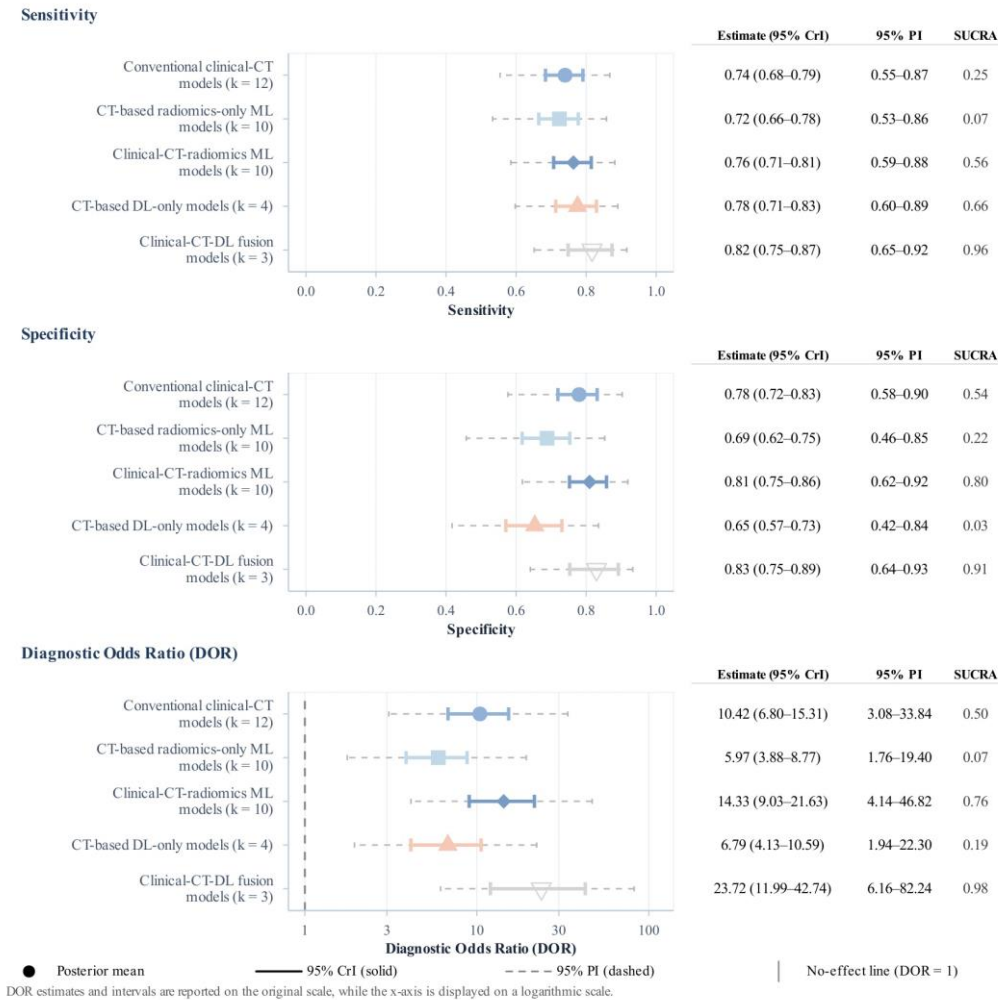


Figure S12B. Sensitivity analysis of pooled diagnostic performance: excluding outlier studies.

Note: Forest plots of pooled diagnostic performance in the sensitivity analysis excluding outlier studies. Points show posterior means with 95% credible intervals and prediction intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; PI, prediction interval.

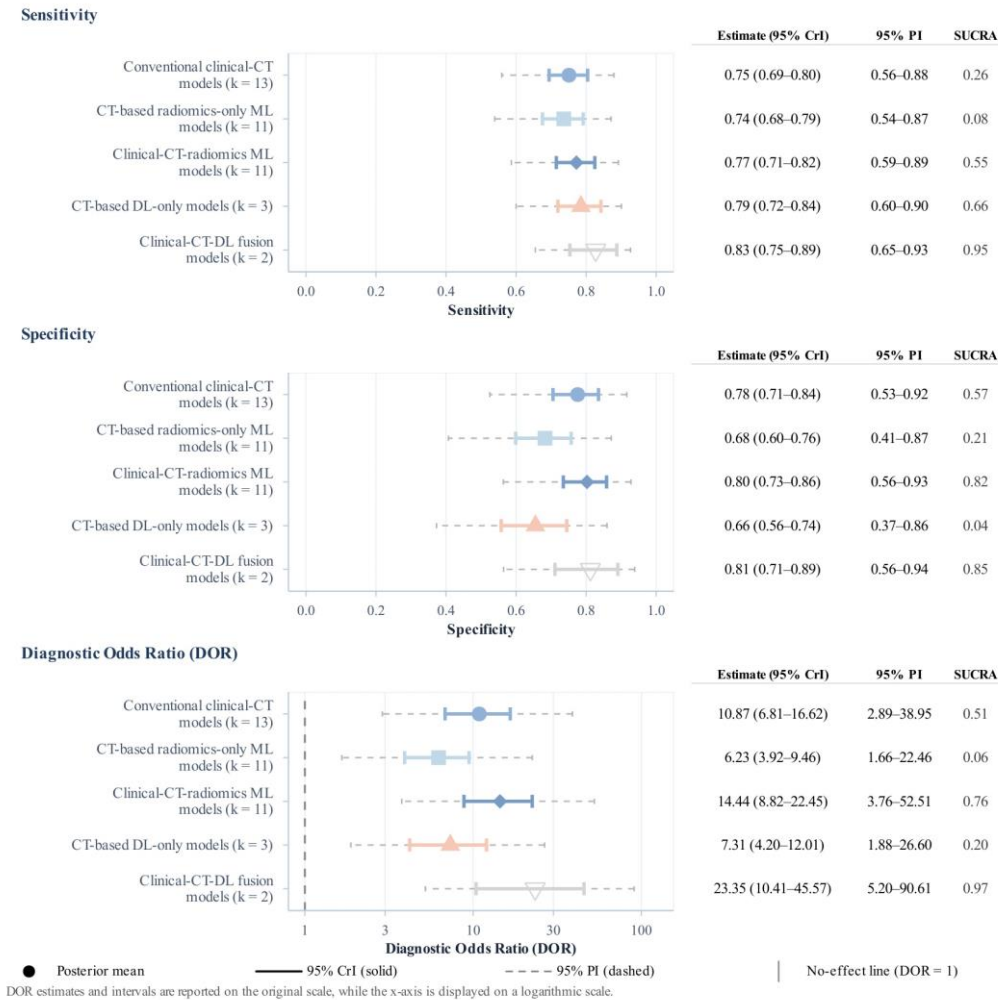


Figure S12C. Sensitivity analysis of pooled diagnostic performance: studies using a mRS >3 threshold.

Note: Forest plots of pooled diagnostic performance in the sensitivity analysis restricted to studies using an mRS >3 threshold. Points show posterior means with 95% credible intervals and prediction intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; mRS, modified Rankin Scale; PI, prediction interval.

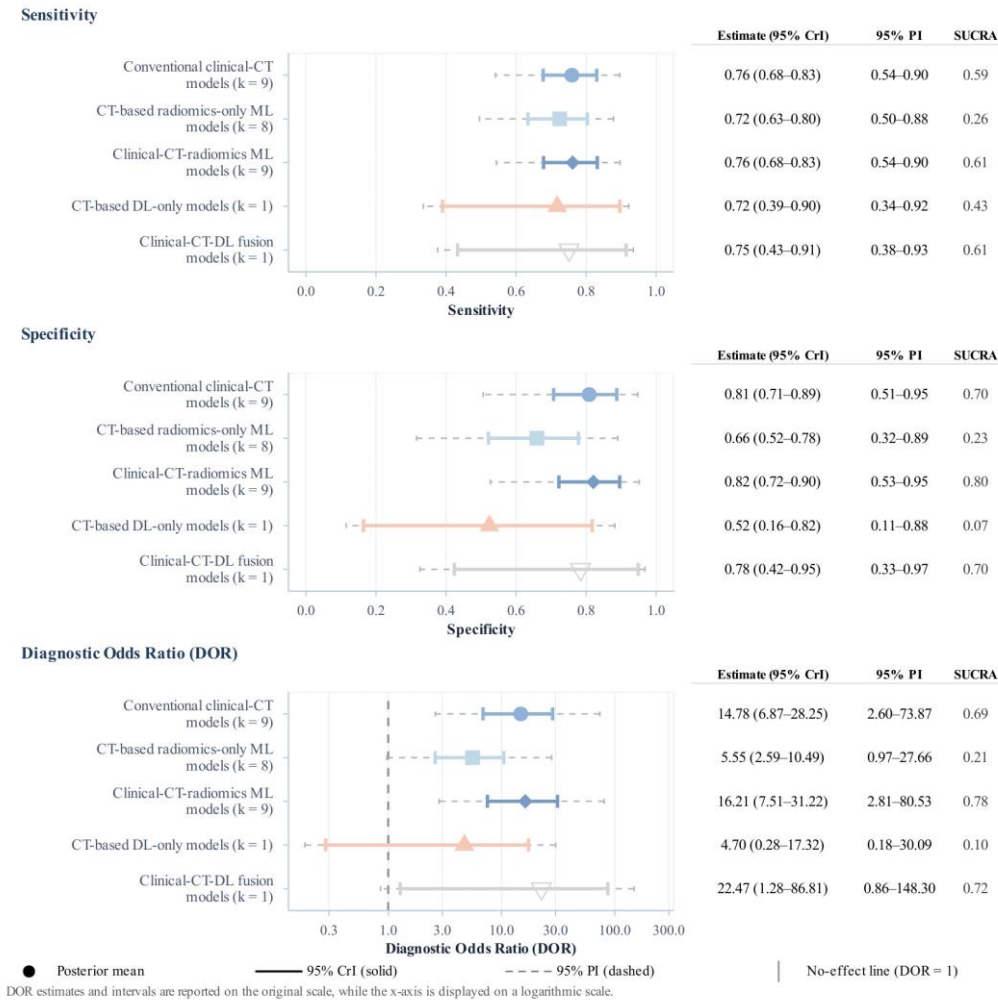


Figure S12D. Sensitivity analysis of pooled diagnostic performance: excluding high-risk studies.

Note: Forest plots of pooled diagnostic performance in the sensitivity analysis excluding studies at overall high risk of bias (PROBAST+AI). Points show posterior means with 95% credible intervals and prediction intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; PI, prediction interval; PROBAST+AI, Prediction model Risk of Bias Assessment Tool for artificial intelligence.

SUCRA Evidence Profile

Darker = higher SUCRA; interpret with rank probabilities and GRADE certainty

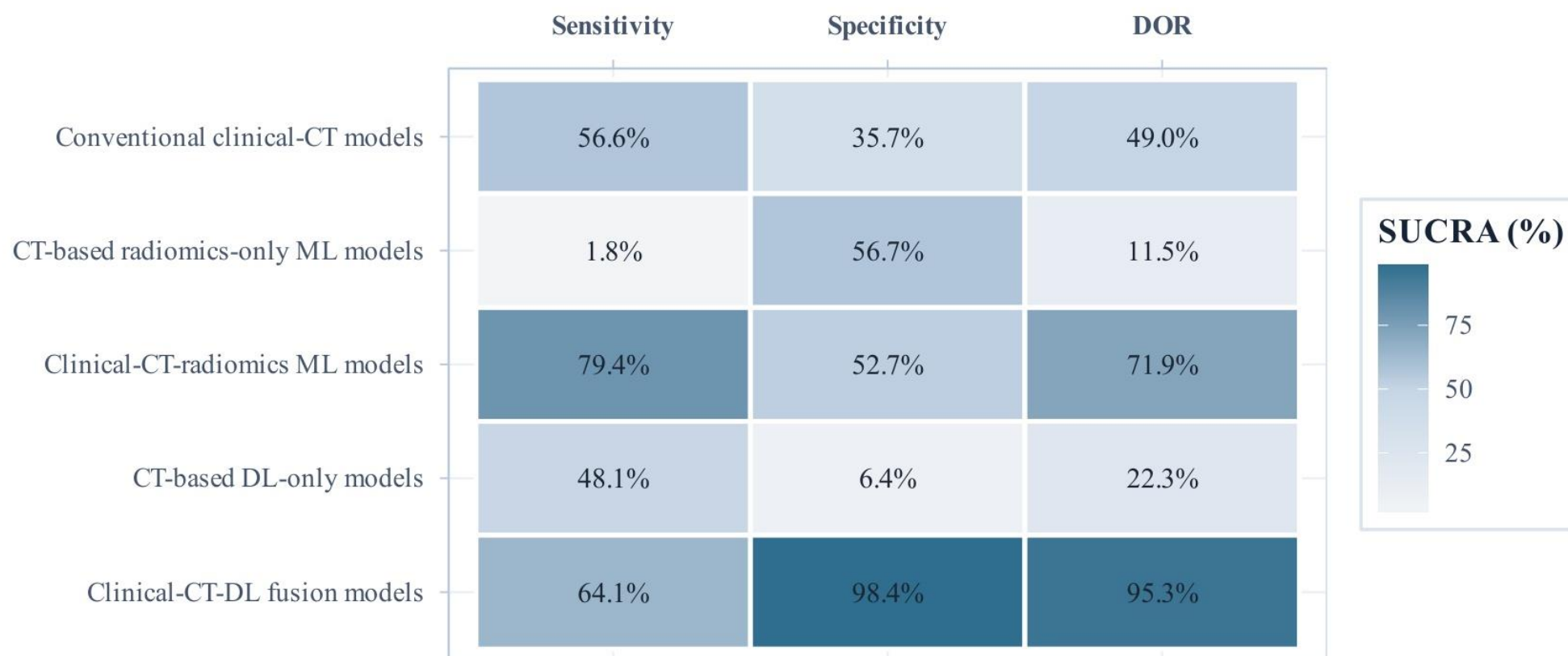


Figure S13A. SUCRA heatmap for the sensitivity analysis scenario in Figure S12A (external validation studies only).

Note: SUCRA heatmap for the sensitivity analysis restricted to external validation studies.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; SUCRA, surface under the cumulative ranking curve.

SUCRA Evidence Profile

Darker = higher SUCRA; interpret with rank probabilities and GRADE certainty

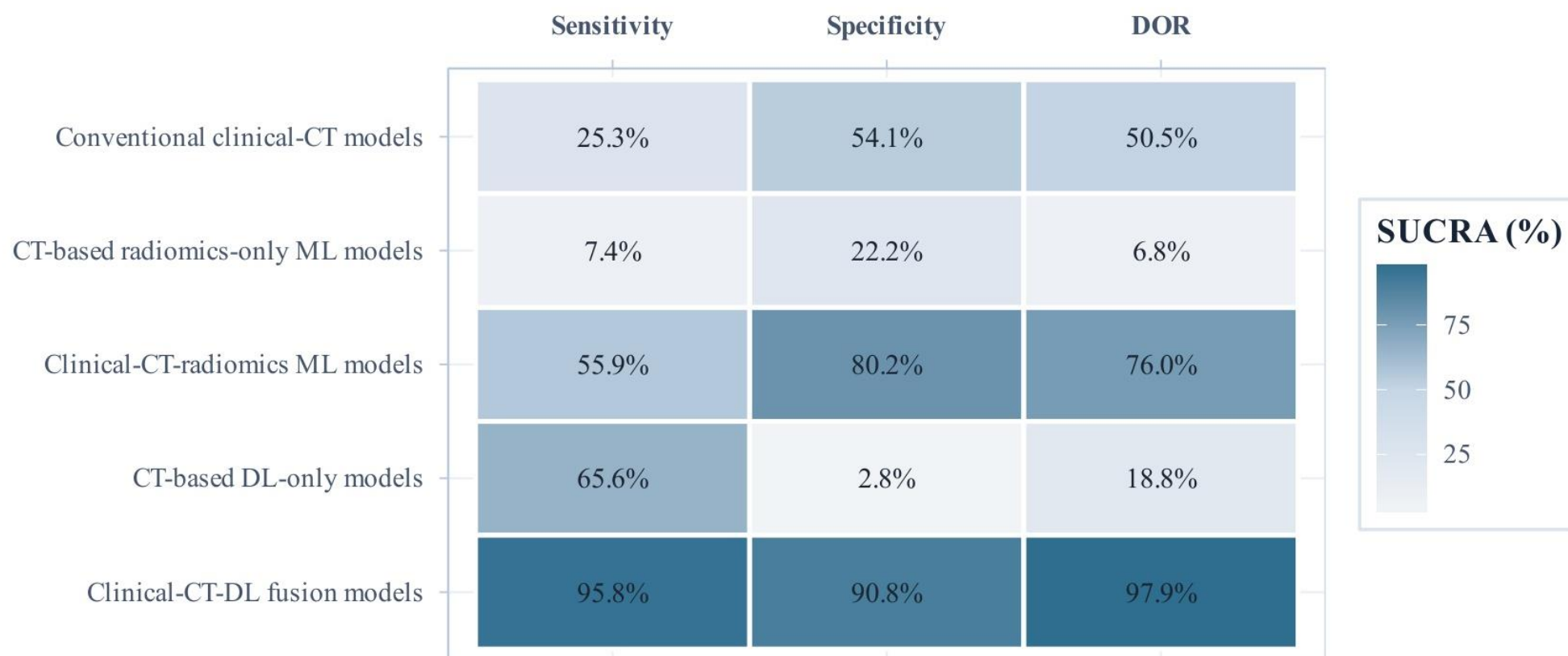


Figure S13B. SUCRA heatmap for the sensitivity analysis scenario in Figure S12B (excluding outlier studies).

Note: SUCRA heatmap for the sensitivity analysis excluding outlier studies.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; SUCRA, surface under the cumulative ranking curve.

SUCRA Evidence Profile

Darker = higher SUCRA; interpret with rank probabilities and GRADE certainty

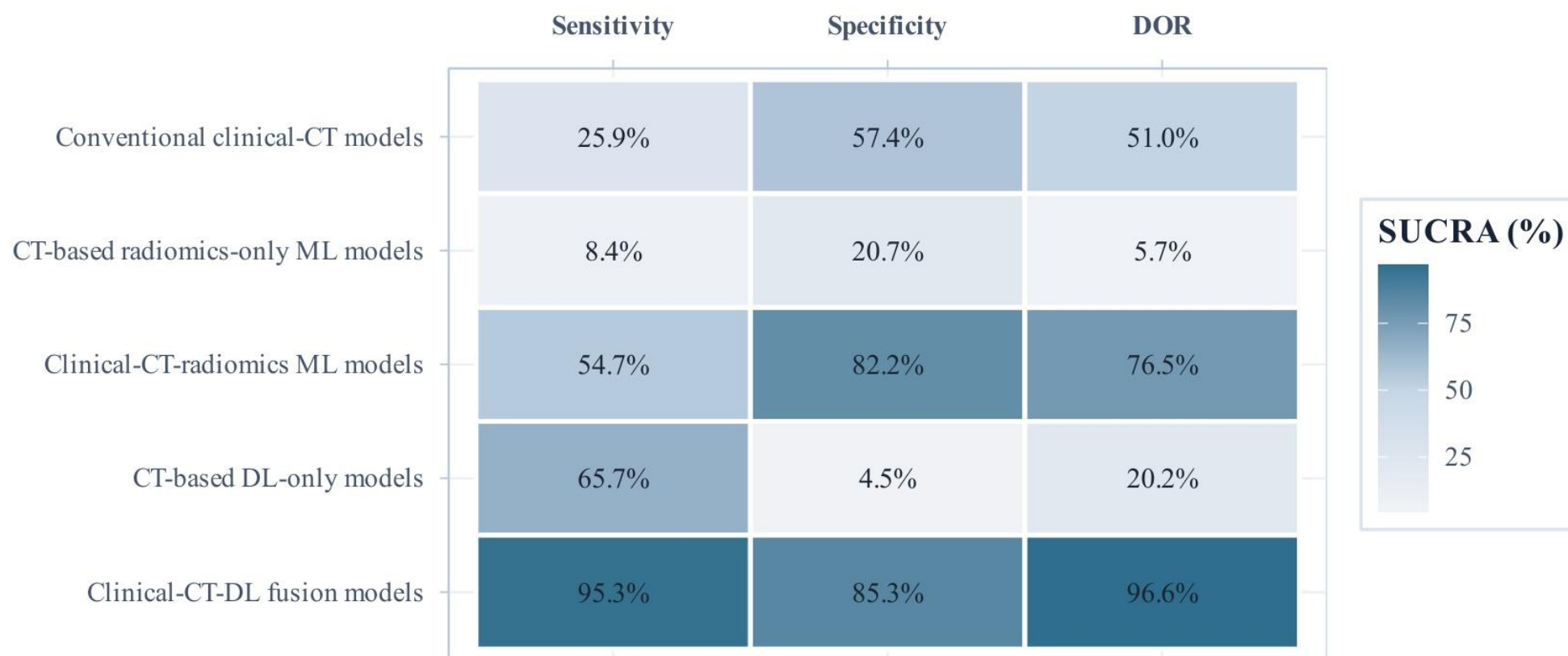


Figure S13C. SUCRA heatmap for the sensitivity analysis scenario in Figure S12C (mRS >3 threshold).

Note: SUCRA heatmap for the sensitivity analysis restricted to studies using an mRS >3 threshold.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; mRS, modified Rankin Scale; SUCRA, surface under the cumulative ranking curve.

SUCRA Evidence Profile

Darker = higher SUCRA; interpret with rank probabilities and GRADE certainty



Figure S13D. SUCRA heatmap for the sensitivity analysis scenario in Figure S12D (excluding high-risk studies).

Note: SUCRA heatmap for the sensitivity analysis excluding studies at overall high risk of bias (PROBAST+AI).

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; PROBAST+AI, Prediction model Risk of Bias Assessment Tool for artificial intelligence; SUCRA, surface under the cumulative ranking curve.

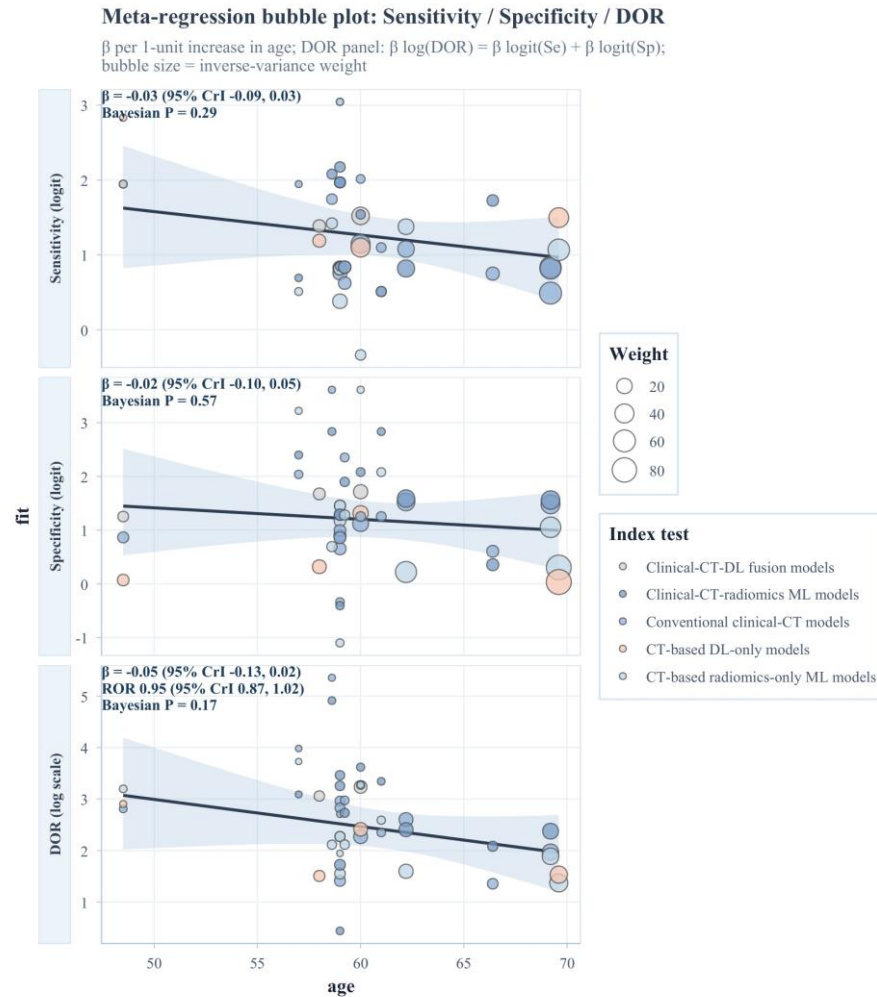


Figure S14A. Univariable meta-regression bubble plot for age.

Note: Univariable meta-regression bubble plot for age. Panels show the fitted association for sensitivity (logit), specificity (logit), and diagnostic odds ratio (log scale); bubble size reflects inverse-variance weight and shaded bands represent 95% credible intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

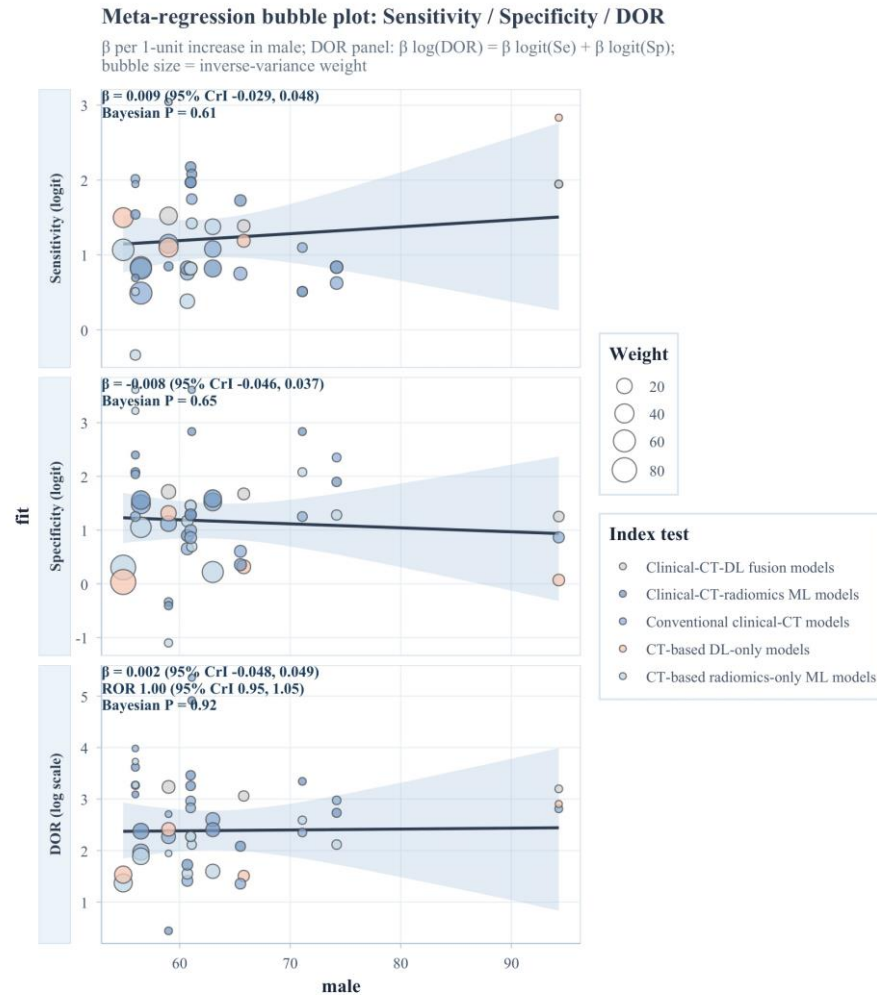


Figure S14B. Univariable meta-regression bubble plot for male proportion.

Note: Univariable meta-regression bubble plot for male proportion. Panels show the fitted association for sensitivity (logit), specificity (logit), and diagnostic odds ratio (log scale); bubble size reflects inverse-variance weight and shaded bands represent 95% credible intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

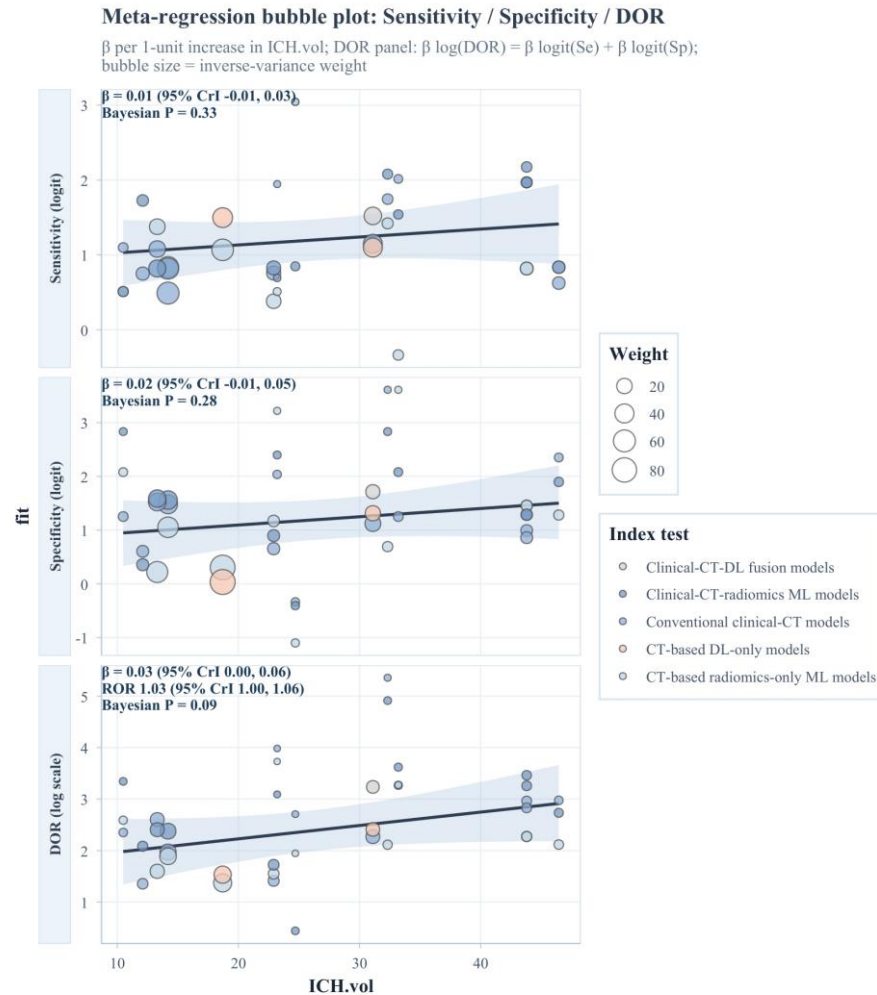


Figure S14C. Univariable meta-regression bubble plot for ICH volume.

Note: Univariable meta-regression bubble plot for ICH volume. Panels show the fitted association for sensitivity (logit), specificity (logit), and diagnostic odds ratio (log scale); bubble size reflects inverse-variance weight and shaded bands represent 95% credible intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ICH, intracerebral hemorrhage; ML, machine learning.

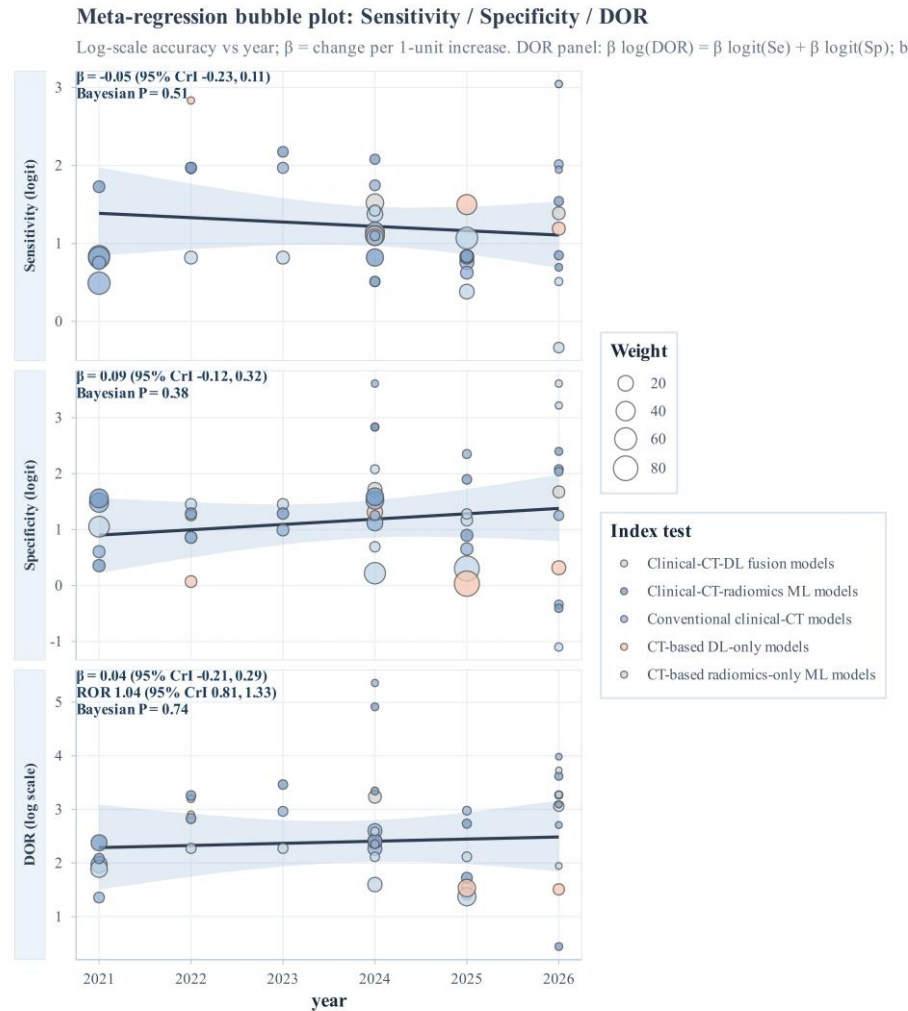


Figure S14D. Univariable meta-regression bubble plot for publication year.

Note: Univariable meta-regression bubble plot for publication year. Panels show the fitted association for sensitivity (logit), specificity (logit), and diagnostic odds ratio (log scale) against publication year; bubble size reflects inverse-variance weight and shaded bands represent 95% credible intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning; ROR, relative diagnostic odds ratio.

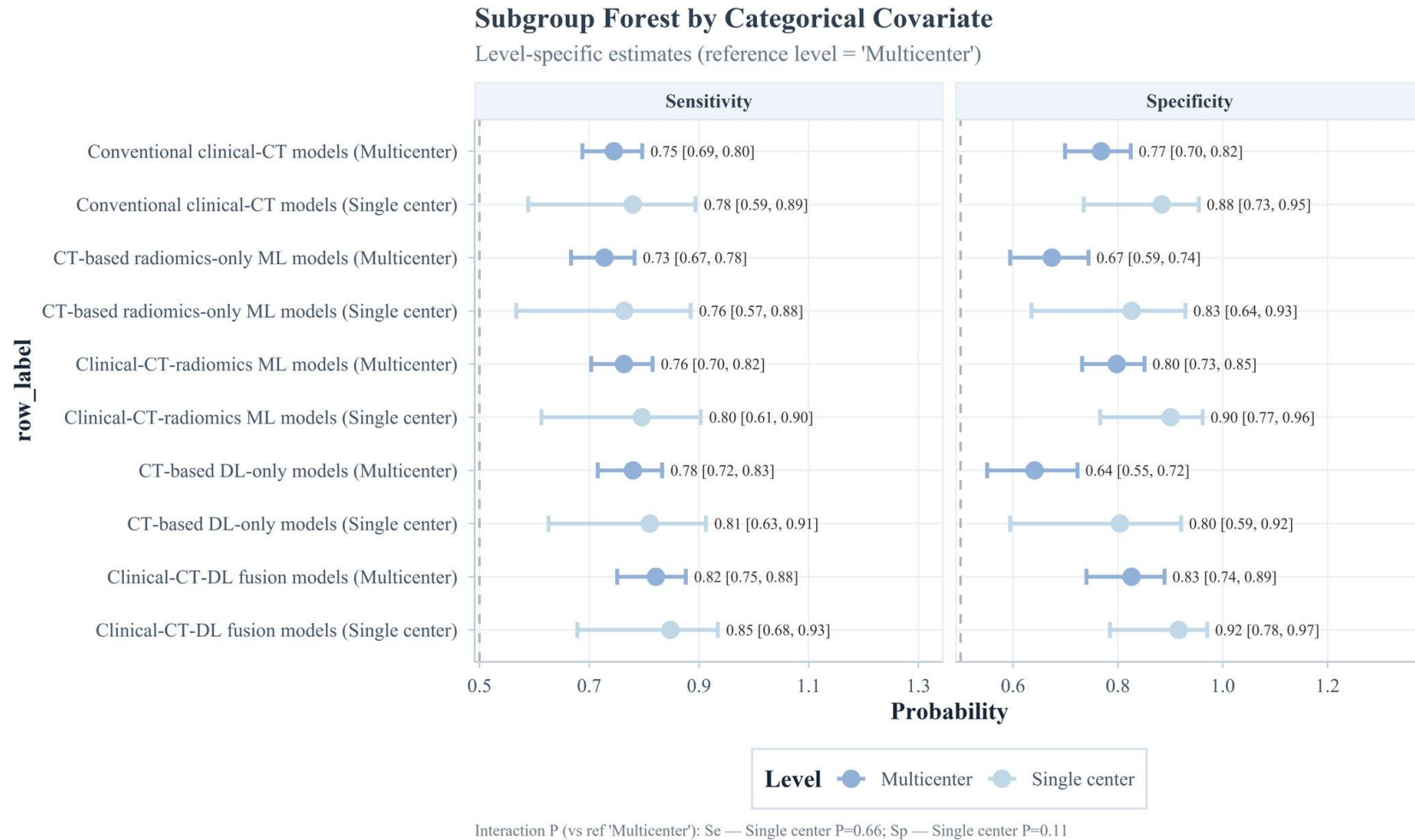


Figure S15A. Subgroup analysis by center.

Note: Subgroup analysis by center (multicenter vs single center). Panels show subgroup-specific sensitivity and specificity with 95% credible intervals; P values indicate differences versus the reference subgroup.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; ML, machine learning.

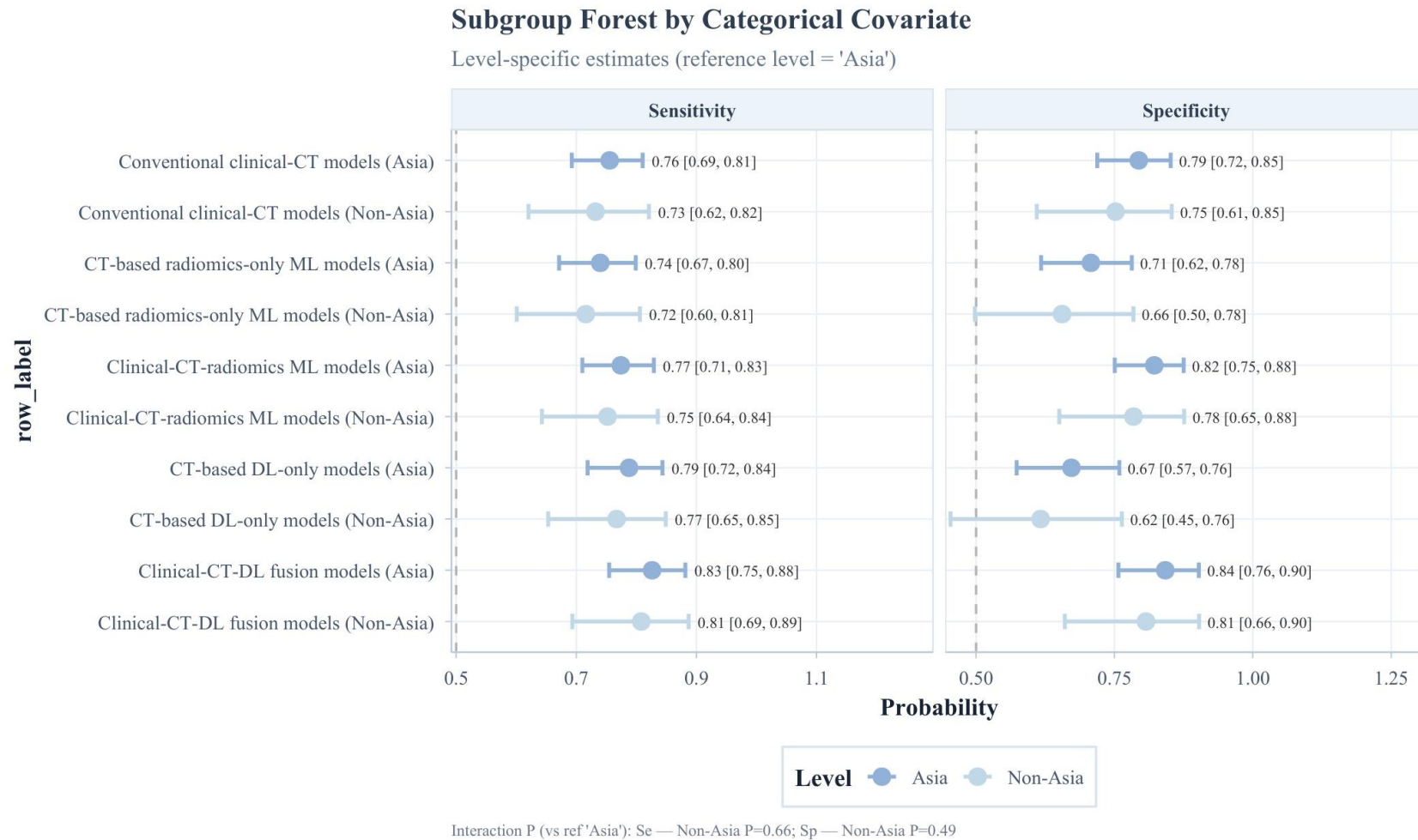


Figure S15B. Subgroup analysis by region.

Note: Subgroup analysis by region. Panels show subgroup-specific sensitivity and specificity with 95% credible intervals; P values indicate differences versus the reference subgroup.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; ML, machine learning.

Subgroup Forest by Categorical Covariate

Level-specific estimates (reference level = 'External validation')

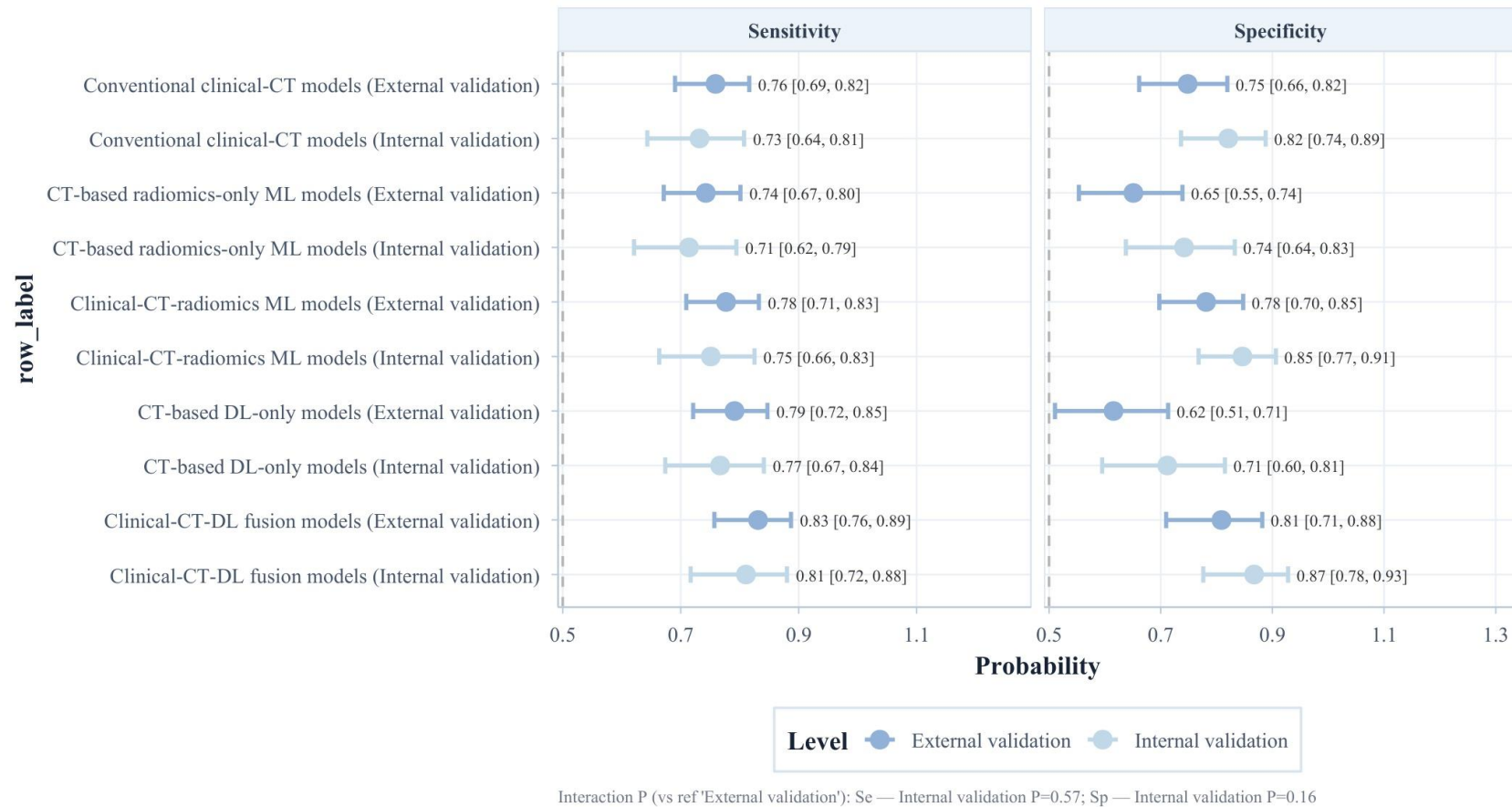


Figure S16A. Subgroup analysis by validation type.

Note: Subgroup analysis by validation type (internal vs external validation). Panels show subgroup-specific sensitivity and specificity with 95% credible intervals; P values indicate differences versus the reference subgroup.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; ML, machine learning.

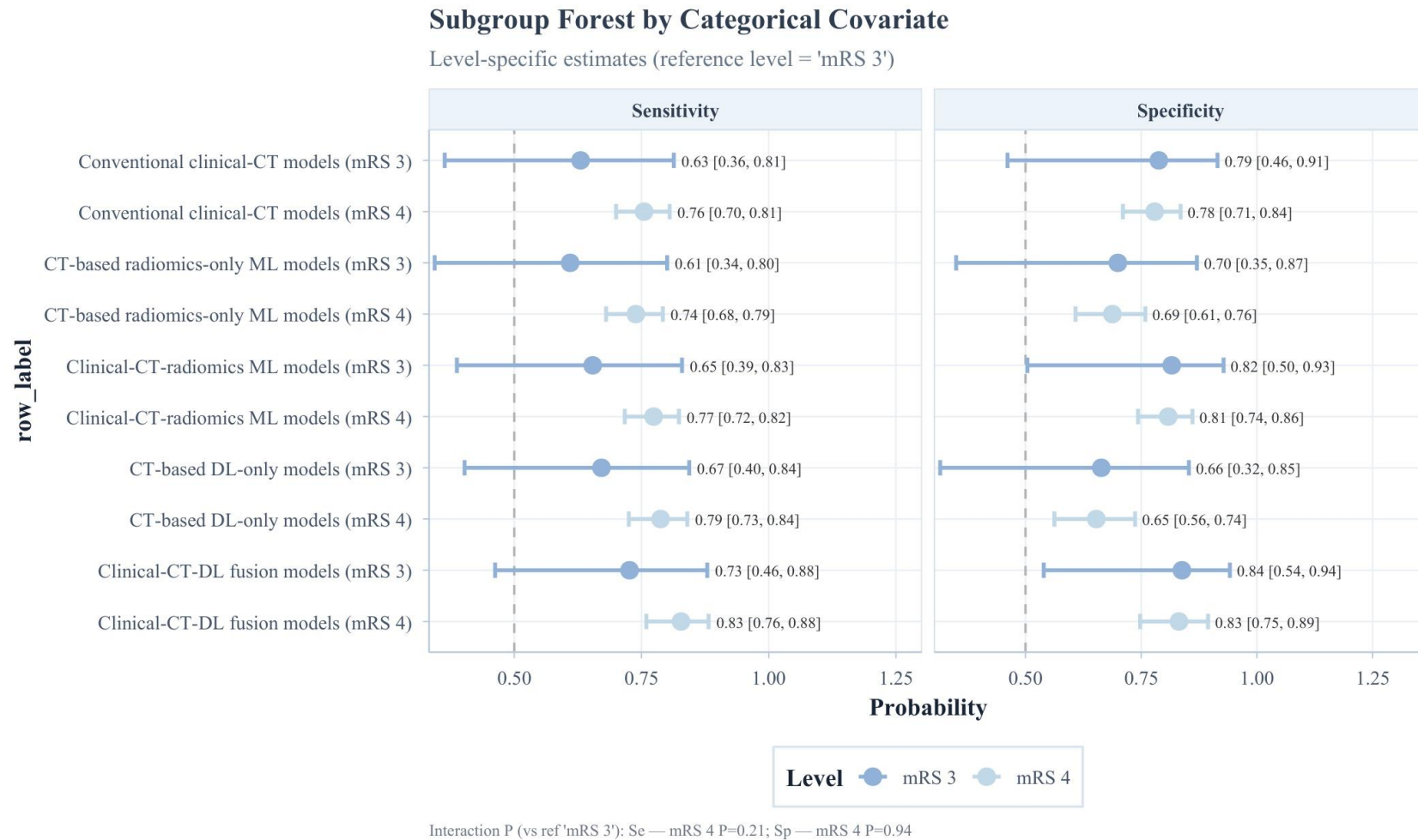


Figure S16B. Subgroup analysis by reference standard.

Note: Subgroup analysis by reference standard. Panels show subgroup-specific sensitivity and specificity with 95% credible intervals; P values indicate differences versus the reference subgroup.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; ML, machine learning.

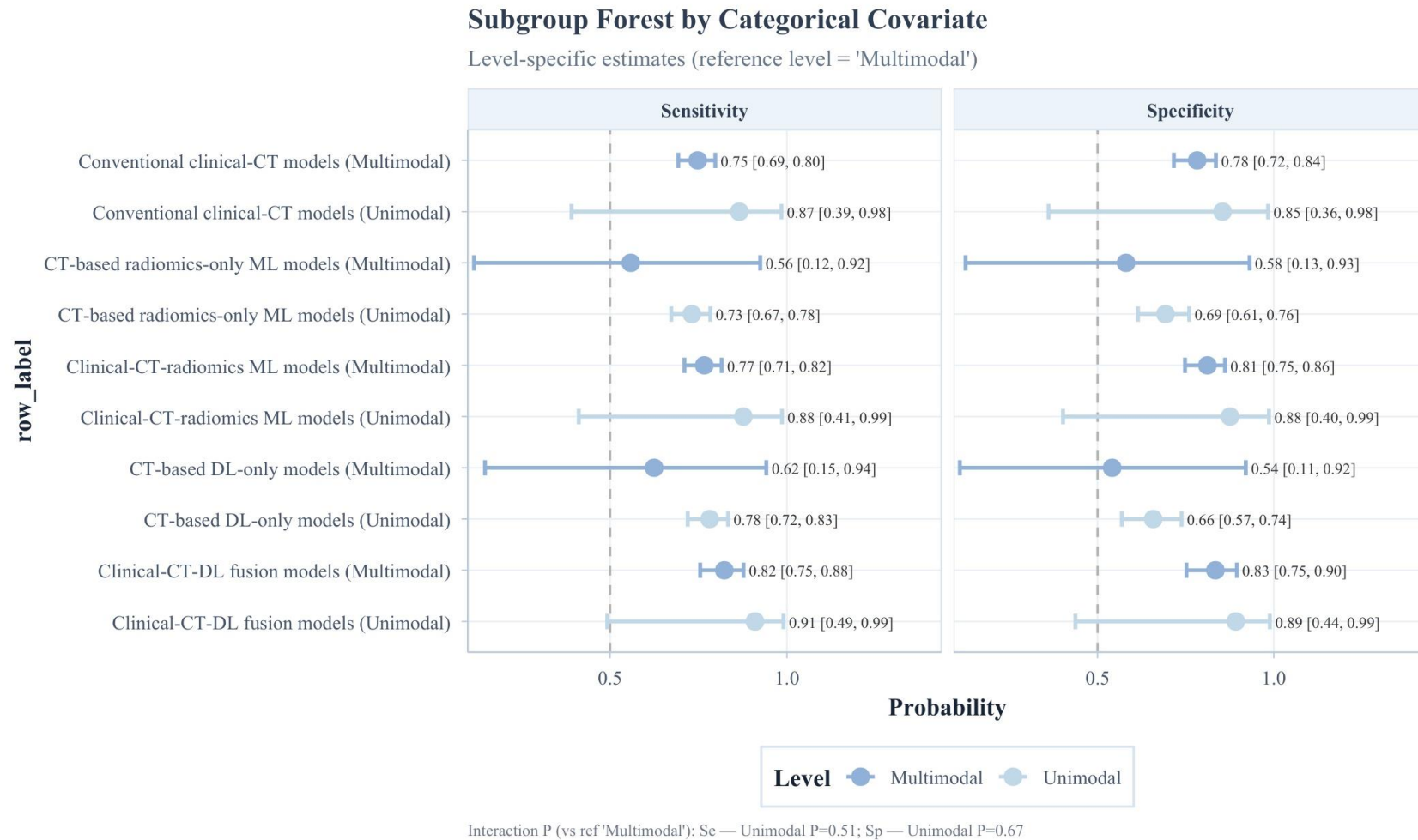


Figure S16C. Subgroup analysis by input modality.

Note: Subgroup analysis by input modality. Panels show subgroup-specific sensitivity and specificity with 95% credible intervals; P values indicate differences versus the reference subgroup.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; ML, machine learning.

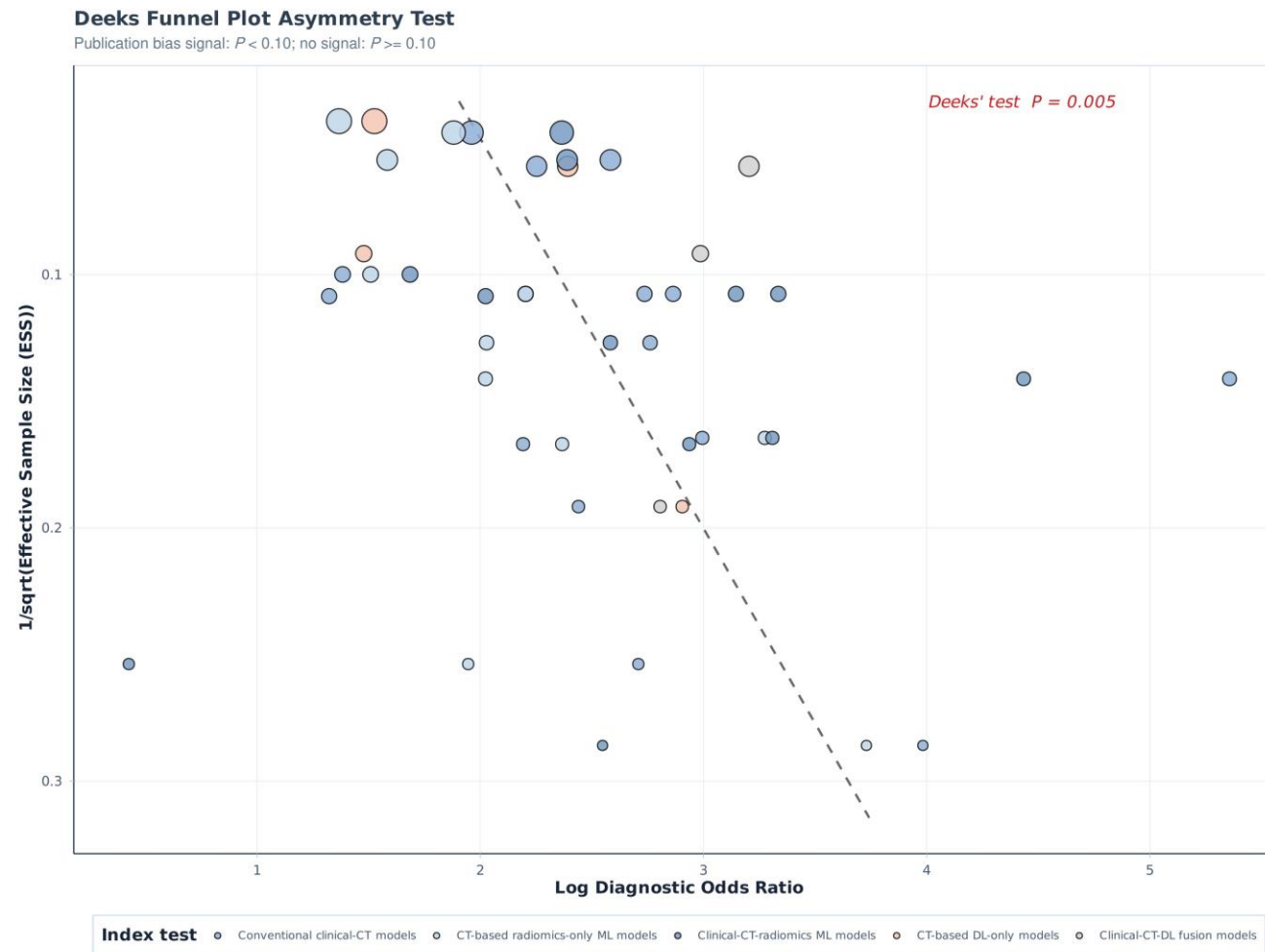


Figure S17A. Deeks' funnel plot asymmetry test for the overall network.

Note: Deeks' funnel plot asymmetry test for the overall network. The dashed line shows the weighted regression; $P < 0.10$ was considered suggestive of small-study effects.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

Deeks Funnel - Conventional clinical-CT models

No significant asymmetry ($P \geq 0.10$)

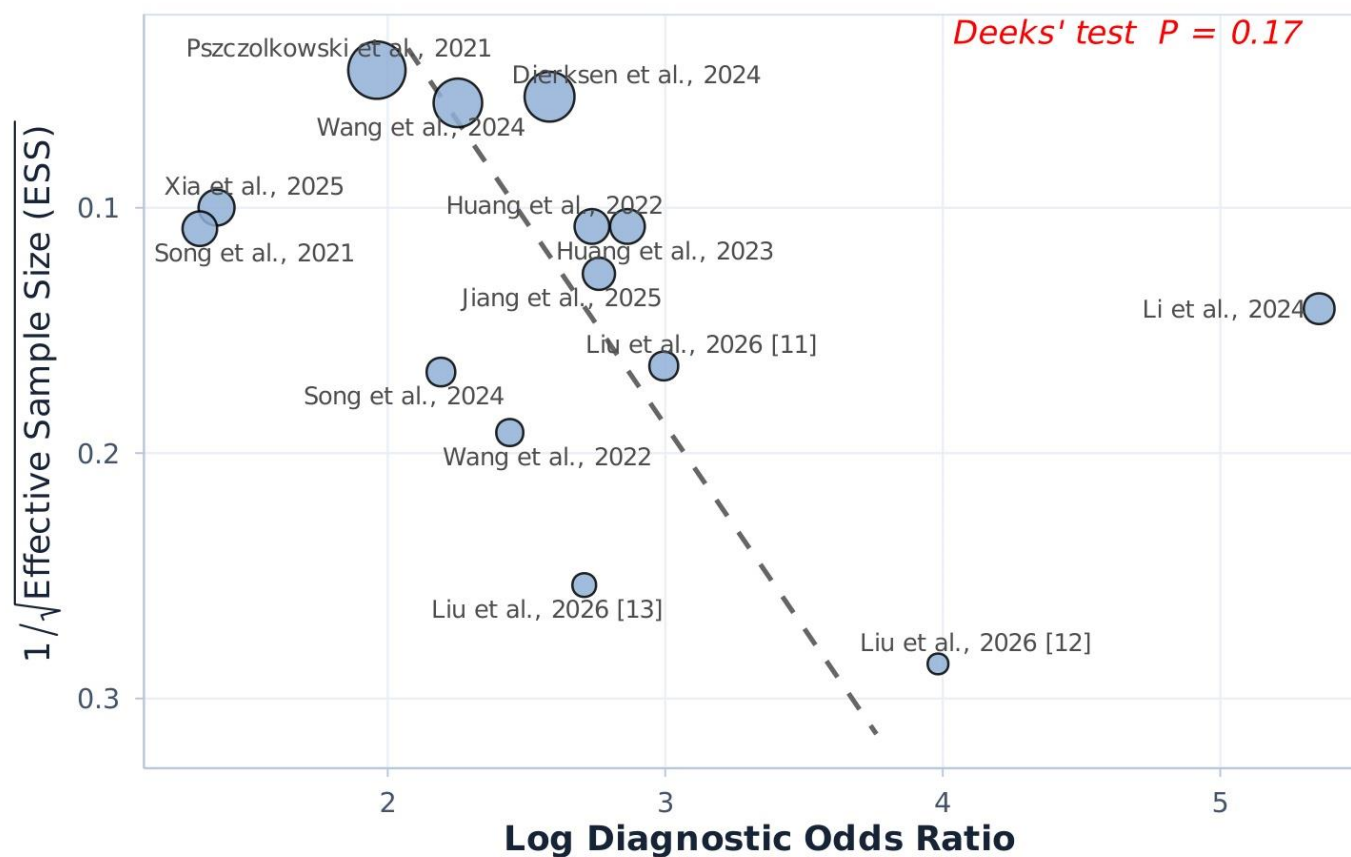


Figure S17B. Deeks' funnel plot asymmetry test for the conventional clinical-CT models.

Note: Deeks' funnel plot asymmetry test for the conventional clinical-CT models. The dashed line shows the weighted regression; $P < 0.10$ was considered suggestive of small-study effects.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

Deeks Funnel - CT-based radiomics-only ML models

Significant asymmetry ($P < 0.10$) - possible publication bias

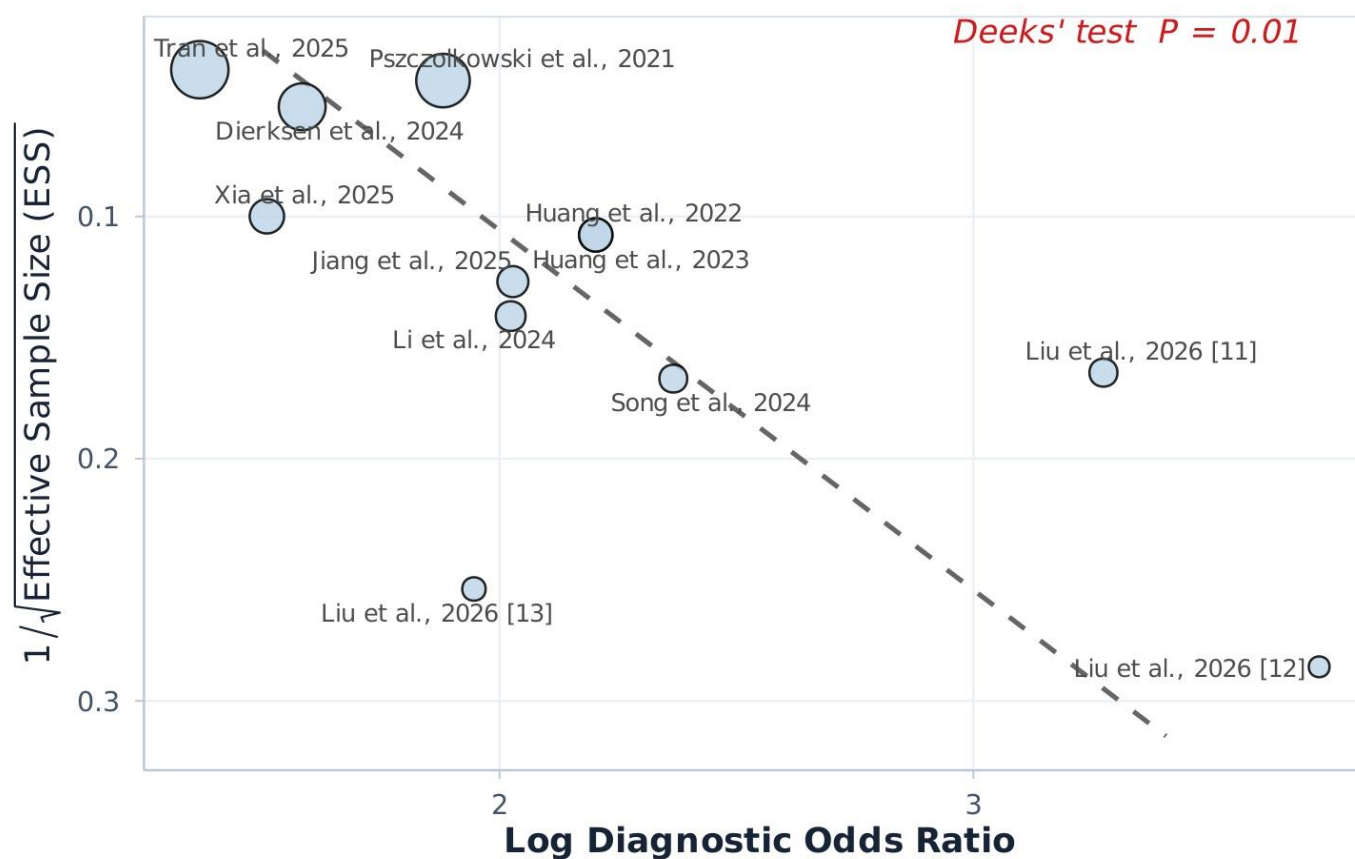


Figure S17C. Deeks' funnel plot asymmetry test for the CT-based radiomics-only ML models.

Note: Deeks' funnel plot asymmetry test for the CT-based radiomics-only ML models. The dashed line shows the weighted regression; $P < 0.10$ was considered suggestive of small-study effects.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

Deeks Funnel - Clinical-CT-radiomics ML models

No significant asymmetry ($p \geq 0.10$)

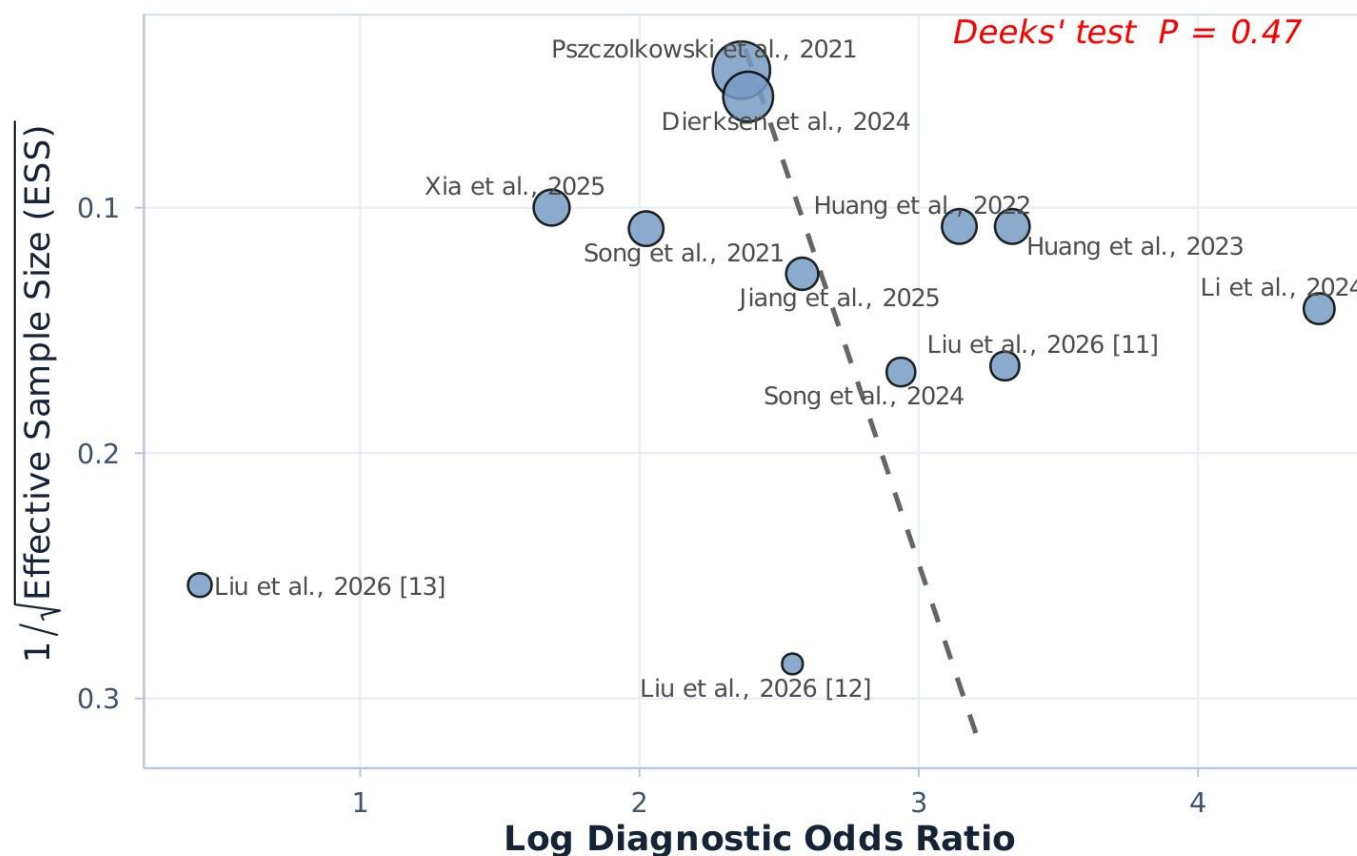


Figure S17D. Deeks' funnel plot asymmetry test for the clinical-CT-radiomics ML models.

Note: Deeks' funnel plot asymmetry test for the clinical-CT-radiomics ML models. The dashed line shows the weighted regression; $P < 0.10$ was considered suggestive of small-study effects.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

Deeks Funnel - CT-based DL-only models

No significant asymmetry ($P \geq 0.10$)

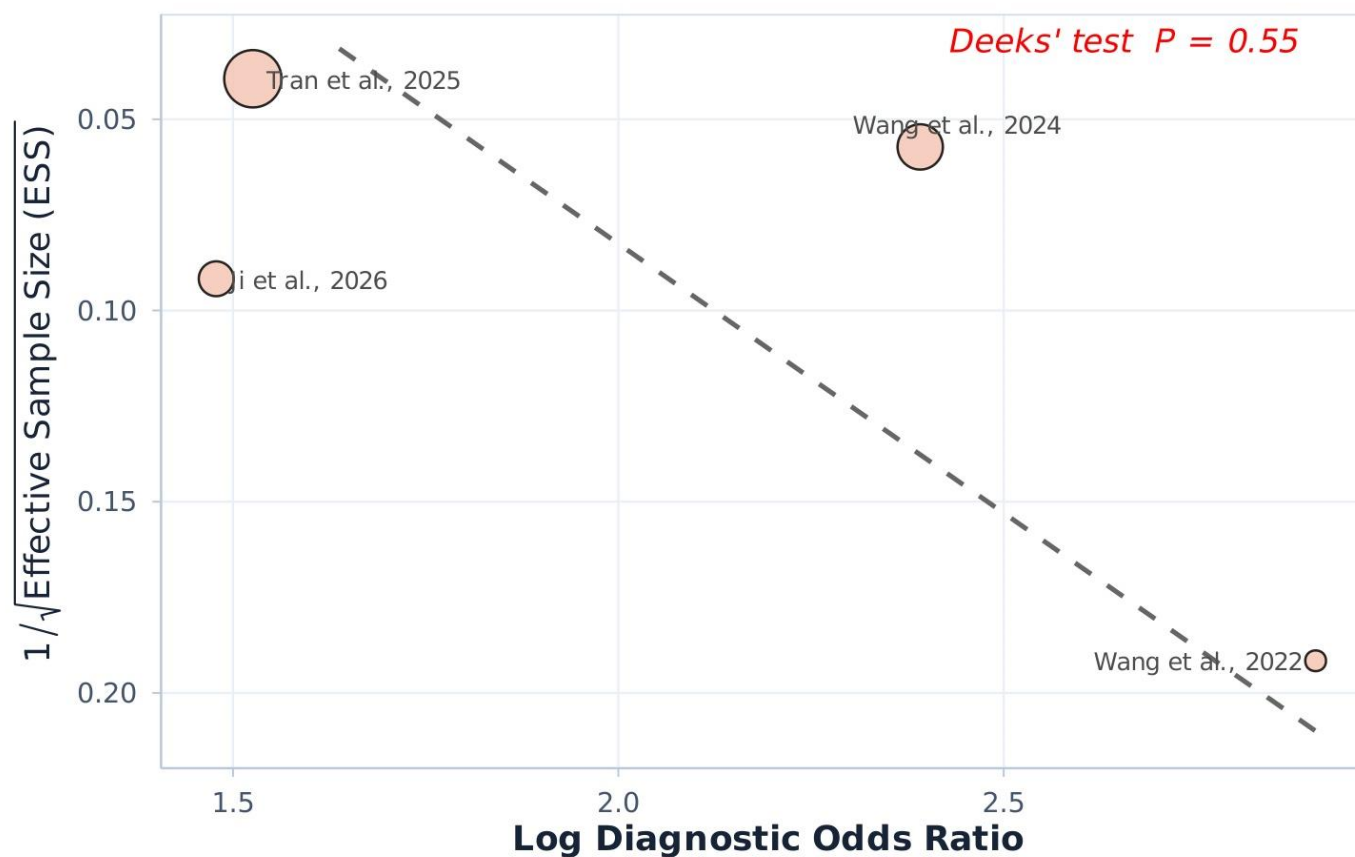


Figure S17E. Deeks' funnel plot asymmetry test for the CT-based DL-only models.

Note: Deeks' funnel plot asymmetry test for the CT-based DL-only models. The dashed line shows the weighted regression; $P < 0.10$ was considered suggestive of small-study effects.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

Deeks Funnel - Clinical-CT-DL fusion models

No significant asymmetry ($P \geq 0.10$)

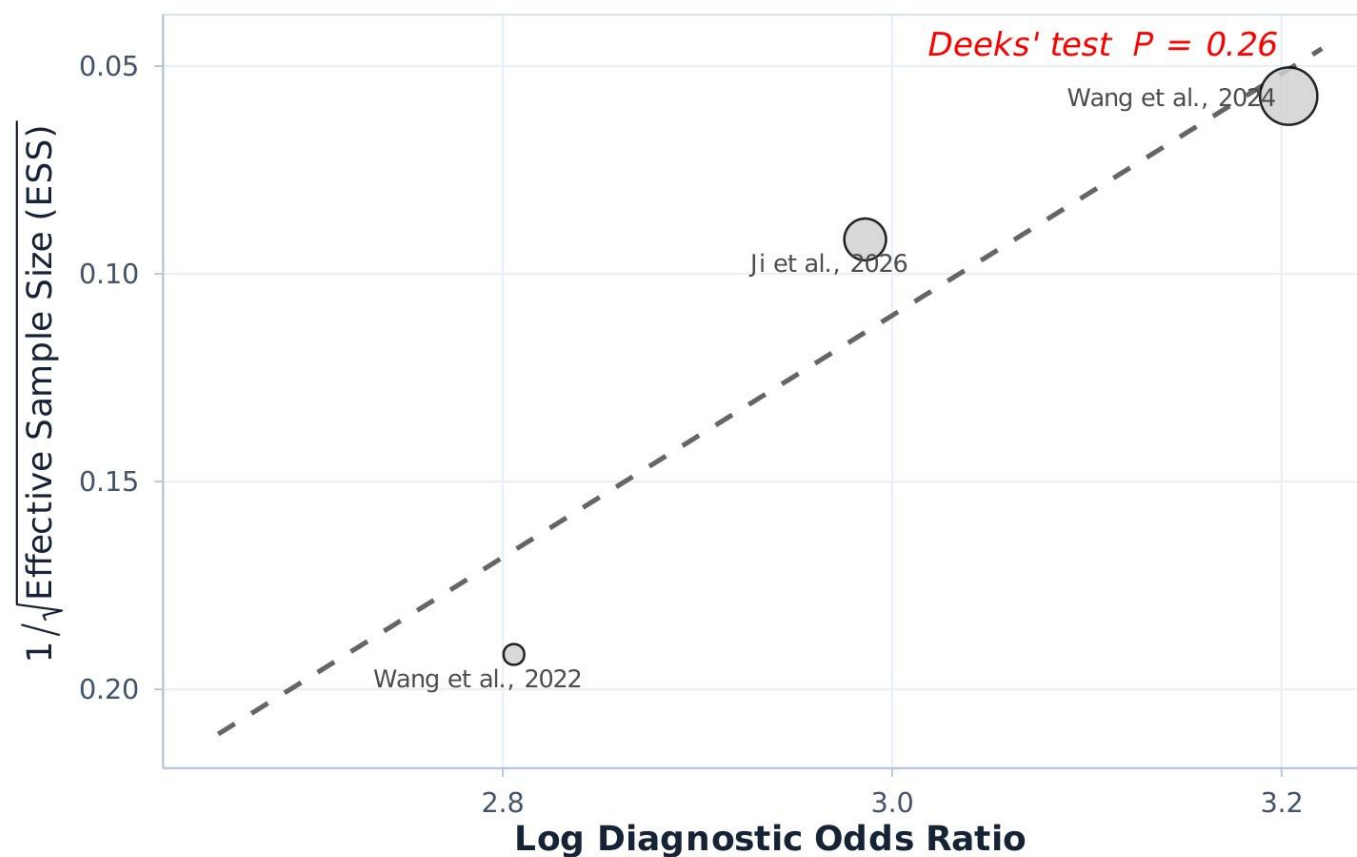


Figure S17F. Deeks' funnel plot asymmetry test for the clinical-CT-DL fusion models.

Note: Deeks' funnel plot asymmetry test for the clinical-CT-DL fusion models. The dashed line shows the weighted regression; $P < 0.10$ was considered suggestive of small-study effects.

Abbreviations: CT, computed tomography; DL, deep learning; DOR, diagnostic odds ratio; ML, machine learning.

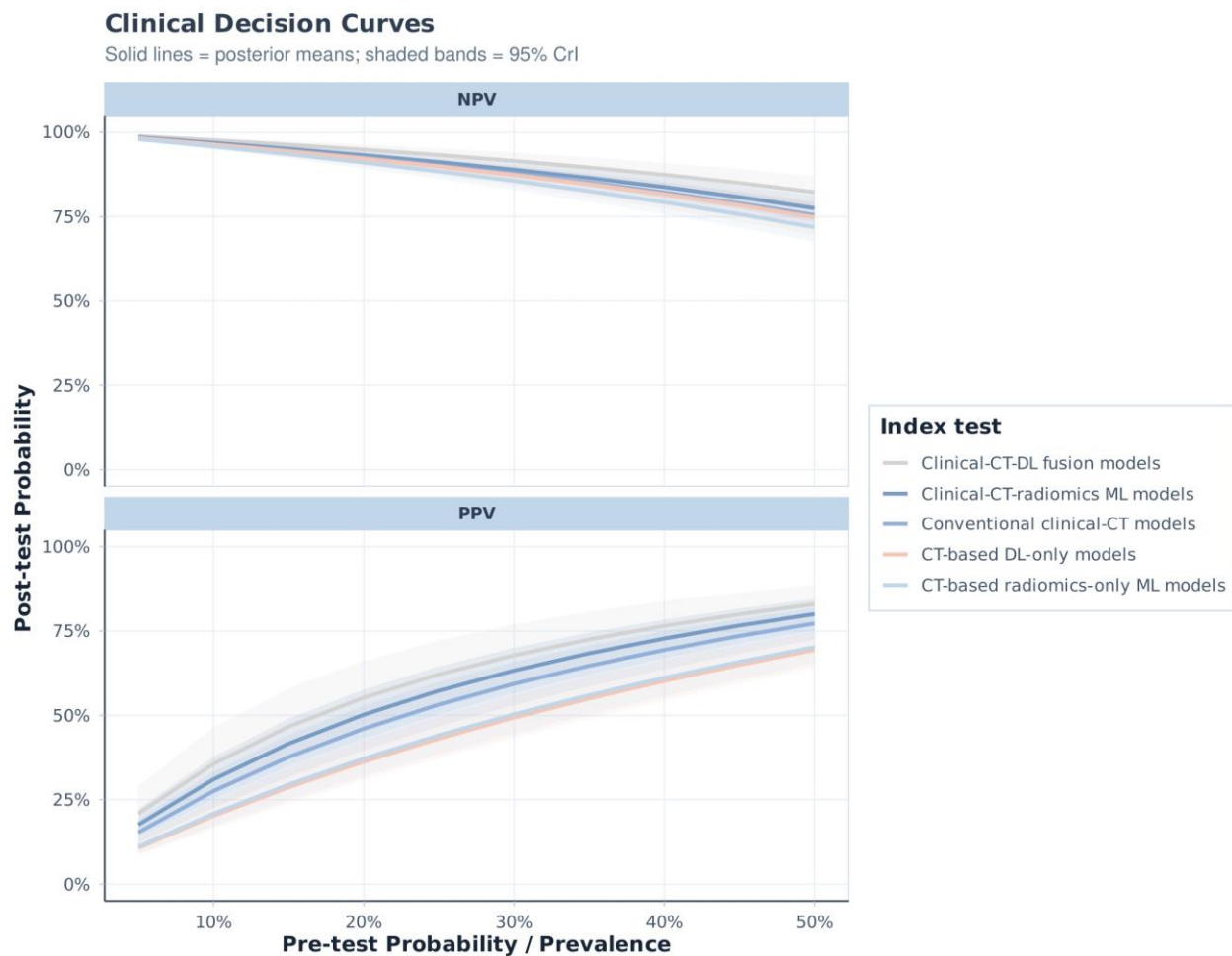


Figure S18. Clinical decision curves of positive and negative predictive values across assumed prevalence scenarios.

Note: Decision curves of pooled positive and negative predictive values across assumed prevalence scenarios. Solid lines represent posterior means; shaded bands represent 95% credible intervals.

Abbreviations: CrI, credible interval; CT, computed tomography; DL, deep learning; ML, machine learning; NPV, negative predictive value; PPV, positive predictive value.